



JOHN HOARAU

Measuring and Closing the Gap Between Declared and Real Urgency in a Supply Chain Digital Twin

*Cognitive adequacy scoring, forecast-assisted declaration,
and an opening toward satellite-observed capacity*

MSc in Computational and Software Techniques in Engineering

Digital Engineering Design

ESTIA Institute of Technology, Bidart

Academic Year: 2025–2026

Academic Supervisor (Cranfield): Dr Peter Sherar

Academic Supervisor (ESTIA): Prof. David Gomez

Industrial Supervisor: Sébastien Perthuisot, ALTEN-DIN

August 2026



Research carried out at the Direction Innovation Numérique, ALTEN Group

© Cranfield University 2026. All rights reserved.

Company Confidential

Confidentiality and Use of Software Tools

Confidentiality. This thesis contains commercially sensitive information belonging to the ALTEN Group, including internal scoring parameters and the architecture of an unreleased software product. It is submitted for examination on a confidential basis and is marked *Company Confidential* throughout.

Use of software tools. Numerical results were produced with the Python codebase described in Section 3.3, using NumPy, SciPy, Matplotlib, Shapely, PyProj and Ultralytics. Large language models were used in two distinct and clearly separated capacities, both declared in full. First, as an *object of study*: the eight role-playing agents of the HÉLIOS campaign are language models, and this is an experimental design choice reported and discussed as such in Sections 3.4.3 and 4.8. Second, as a *writing aid* for language correction and $\text{\LaTeX}$ formatting. All scientific claims, all numerical results and all interpretations are my own, and every figure in this thesis was produced either by the project’s own code or by hand.

Abstract

In mutualised logistics networks the urgency an operator declares often diverges from the urgency computable from the operational state. Over-declaration wastes shared capacity; under-declaration hides rupture risk. No existing digital twin measures that divergence or propagates it across a supplier network.

This thesis builds and tests one. Real urgency is aggregated at each node from six KPI blocks through a weighted probabilistic OR and propagated upstream across a directed acyclic supply graph; declared urgency is elicited through a consistency-gated pairwise questionnaire and propagated downstream; their mismatch is scored on $[0, 100]$ with an asymmetric Prospect-Theory penalty. The system was evaluated through a pre-registered campaign replaying the 2020 to 2022 semiconductor shortage over nineteen turns and eight nodes, with a blind control arm and a forecast-assisted treatment arm on a frozen scenario.

The results separate two claims that are easily conflated. The gap carries information about which supplier will fail: over the six turns preceding the public onset of the crisis, the nodes it flags are, without a single false alarm, nodes that go on to miss a delivery commitment, three of the four that eventually do, and the worst-scoring among them is flagged eight turns before its miss while its own manager declares an urgency of 0.001. The signal then weakens for a structural reason rather than a defect, since declared urgency converges on computed urgency once the crisis is visible and a measure defined as their difference closes by construction. This bounds the instrument to early warning, which is where a warning is worth most.

Converting that signal into a calibrated probability attached to a named week is a further step, and it failed on the first attempt, diagnostically and precisely. The forecast layer showed no discrimination above chance under the production outcome definition and was severely miscalibrated, at a Brier skill between -3.4 and -6.9 against a base-rate forecaster. Forecasts below 0.1 were well calibrated, while twenty-one announcing a mean probability of 0.948 were followed by the predicted outcome exactly zero times. One block explains it: the time block read a lead-time distribution that no shock ever reached. In the treatment arm, on the turn the forecast first became visible, all eight declarants revised their urgency downward or held it and none revised upward, two turns before the scenario's critical event. Both arms carry identical forecasts, so the effect belongs to the display. The outcome definition itself moves the answer by more than any modelling decision taken here: a naive definition improves the apparent skill fivefold while leaving discrimination unchanged, and judging a milestone against the deadline first committed to rather than the deadline it currently carries raises the four-week AUC from 0.485 to 0.654.

The diagnosis was then acted on and re-measured on the identical frozen forecasts. The obvious repair, having the events write a lead-time impact, proved wrong under measurement: lost time must be added to the completion date rather than multiplied into it, or a shock attenuates to nothing on the milestones nearest their deadline. After repair the Brier deficit roughly halves at every horizon, from -0.553 to -0.216 at four weeks, and remains negative. A closed-loop campaign on the repaired system, in which the same agents act on the chain, meets five milestones of eleven on their original date, exactly as many as the frozen run: it changes which milestone is met, not how many.

What bounds every figure above is the sample rather than the construction: eight nodes of

which four are positive, eleven observable milestones, one scripted scenario, and language-model declarants. The instrument and its diagnostic method transfer to any supply chain; the rates measured here do not yet transfer to anything, and widening that base is the work this thesis hands on.

Keywords: supply chain digital twin, urgency adequacy, cognitive bias, multi-criteria decision analysis, forecast calibration, automation bias, serious game.

Résumé

Dans les réseaux logistiques mutualisés, l'urgence qu'un opérateur déclare diverge souvent de celle que l'on peut calculer à partir de l'état opérationnel. La sur-déclaration gaspille de la capacité partagée ; la sous-déclaration dissimule un risque de rupture. Aucun jumeau numérique existant ne mesure cette divergence ni ne la propage à l'échelle d'un réseau de fournisseurs.

Ce mémoire en construit un et le met à l'épreuve. L'urgence réelle est agrégée à chaque nœud à partir de six blocs de KPI par un OU probabiliste pondéré, puis propagée vers l'aval sur un graphe orienté acyclique ; l'urgence déclarée est élicitée par un questionnaire de comparaisons par paires soumis à un contrôle de cohérence, puis propagée vers l'amont ; leur écart est noté sur $[0, 100]$ par une pénalité asymétrique issue de la théorie des perspectives. Le système a été évalué par une campagne pré-enregistrée jouant la pénurie de semi-conducteurs de 2020 à 2022 sur dix-neuf tours et huit nœuds, avec un bras témoin en aveugle et un bras de traitement assisté par prévision, sur un scénario gelé.

Les résultats séparent deux affirmations qu'il serait facile de confondre. L'écart porte bien de l'information sur le fournisseur qui va défaillir : sur les six tours précédant l'irruption publique de la crise, les nœuds qu'il signale sont, sans une seule fausse alerte, des nœuds qui rateront un engagement de livraison, trois des quatre qui finiront par le faire, et le plus mal noté est signalé huit tours avant son échec alors que son propre responsable déclare une urgence de 0,001. Le signal s'affaiblit ensuite pour une raison structurelle et non par défaut : quand la crise devient visible, l'urgence déclarée converge vers l'urgence calculée et une mesure définie comme leur différence se referme par construction. L'instrument est donc borné à l'alerte précoce, là où une alerte vaut le plus.

Convertir ce signal en probabilité calibrée attachée à une semaine nommée est une étape de plus, et elle a échoué au premier essai, de façon précise et diagnosticable. La couche de prévision ne discriminait pas mieux que le hasard sous la définition d'issue de production et se révélait fortement mal calibrée, avec un *skill* de Brier compris entre $-3,4$ et $-6,9$ face à un prévisionniste annonçant le taux de base. Les prévisions inférieures à 0,1 étaient bien calibrées, tandis que vingt-et-une annonçant une probabilité moyenne de 0,948 n'ont été suivies de l'issue prédite exactement zéro fois. Un seul bloc l'expliquait : le bloc temporel lisait une distribution de délai qu'aucun choc n'atteignait jamais. Dans le bras de traitement, au tour où la prévision devient visible pour la première fois, les huit déclarants ont tous revu leur urgence à la baisse ou l'ont maintenue, aucun ne l'a revue à la hausse, deux tours avant l'événement critique du scénario. Les deux bras portent des prévisions identiques : l'effet tient donc à l'affichage. La définition de l'issue déplace la réponse davantage qu'aucune décision de modélisation prise ici : une définition naïve multiplie par cinq le *skill* apparent sans modifier la discrimination, et juger un jalon sur l'échéance initialement engagée plutôt que sur celle qu'il porte aujourd'hui fait passer l'aire sous la courbe ROC à quatre semaines de 0,485 à 0,654.

Ce diagnostic a ensuite été traité et re-mesuré sur les mêmes prévisions gelées. Le correctif évident, faire écrire aux événements un impact sur le délai d'approvisionnement, s'est révélé faux à la mesure : le temps perdu doit s'ajouter à la date d'achèvement et non s'y multiplier, faute de quoi le choc s'atténue jusqu'à disparaître sur les jalons les plus proches de leur échéance. Après correction, le déficit de Brier est divisé par deux à chaque horizon, de $-0,553$ à $-0,216$ à quatre semaines, et demeure négatif. Une campagne en boucle fermée sur le

système réparé, où les mêmes agents agissent sur la chaîne, tient cinq jalons sur onze à leur date initiale, exactement autant que la campagne gelée : elle change quel jalon est tenu, pas combien.

Ce qui borne chacun de ces chiffres est l'échantillon et non la construction : huit nœuds dont quatre positifs, onze jalons observables, un scénario scripté, et des déclarants qui sont des modèles de langage. L'instrument et sa méthode de diagnostic se transposent à n'importe quelle chaîne ; les taux mesurés ici ne se transposent encore nulle part, et élargir cette base est le travail que ce mémoire transmet.

Mots-clés : jumeau numérique de chaîne d'approvisionnement, adéquation d'urgence, biais cognitif, aide multicritère à la décision, calibration de prévisions, biais d'automatisation, jeu sérieux.

Acknowledgements

I would like to thank everyone who supported this research project at the Direction Innovation Numérique of the ALTEN Group.

My particular thanks go to **Émilienne Lardy**, **Selma Ferhat** and **Sébastien Perthuisot**. Émilienne and Selma supervised the scientific side of this work throughout, and their demands, their availability and the quality of their reviews did more than anything else to raise the rigour of what follows. Sébastien, Research Programme Director and Quality & Supply Chain Expert at the DIN, provided the industrial supervision and the confidence to let the scientific direction of the subject develop as it did.

I thank my academic supervisors, **Dr Peter Sherar** of the Centre for Computational Engineering Sciences at Cranfield University, and **Prof. David Gomez** of the ESTIA Institute of Technology, for their methodological guidance and for the academic framing of this thesis.

Finally, I thank the whole Direction Innovation Numérique team, and the other interns in the department, for the welcome they gave me and for the discussions that fed this work from beginning to end.

Contents

Confidentiality and Use of Software Tools	i
Abstract	ii
Résumé	iv
Acknowledgements	vi
List of Figures	xi
List of Tables	xii
1 Introduction	1
1.1 Industrial and Technical Context	1
1.2 The DISCO Digital Twin and Prior Research at ALTEN-DIN	1
1.3 Problem Statement	3
1.4 Aim, Objectives and Research Questions	4
1.5 Research Hypotheses	5
1.6 Scope and Delimitations	7
1.7 Thesis Structure	7
2 Literature Review	8
2.1 Introduction and Key Definitions	8
2.2 Risk Propagation and Deadline-Sensitive Value	8
2.3 Cognitive Bias in Urgency Declaration	10
2.4 Indicators of Delivery Rupture Risk	11
2.5 Closest Antecedents: Network Risk Twins and MCDM Supplier Scoring	12
2.6 Enabling Method: Bayesian and Probabilistic Risk Updating	13
2.7 Enabling Method: Digital Twins and Real-Time Decision Support	13
2.8 Enabling Methods: Multi-Criteria Decision Methods	14
2.9 Forecast-Assisted Decision and Automation Bias	14
2.10 Validation Methodology: Serious Games	15
2.11 Probabilistic Forecast Verification	16
2.12 Analogical and Contextual Literature	16
2.13 Source Review Grid	17
2.14 Identified Gaps	18
3 Methods	19
3.1 Methodological Framework	19

3.1.1	The Proposed Structure	19
3.2	Mathematical Model	20
3.2.1	Local Real Urgency	20
3.2.2	Why These Six Blocks	23
3.2.3	Local Declared Urgency	24
3.2.4	Network Propagation	24
3.2.5	Network Criticality Index	27
3.2.6	Asymmetric Cognitive Adequacy	27
3.3	System Implementation	29
3.3.1	Architecture	29
3.3.2	AHP Elicitation	29
3.3.3	Fuzzy BWM and PROMETHEE II Network Outranking	30
3.3.4	Weekly Review Cycle and Calibrated Event Engine	32
3.3.5	The Forecast Layer	33
3.3.6	Explainability, Export and Operational Hardening	34
3.4	Experimental Design: The HÉLIOS Campaign	34
3.4.1	Scenario and Network	34
3.4.2	Data Provenance	36
3.4.3	The Three Arms	37
3.4.4	Scoring Protocol	38
3.4.5	Protocol Integrity and Controls	39
3.5	Limitations and Simplifications of the Method	39
3.6	Reproducibility	40
3.7	Ethical Considerations	40
3.8	Conclusion	41
4	Results and Discussion	42
4.1	Introduction	42
4.2	The Delivered System	42
4.3	Mechanical Validation: The Synthetic Dry Run	43
4.4	Arm A: Blind Declaration by Eight Isolated Agents	43
4.5	What the Adequacy Signal Gets Right: Node-Level Targeting	47
4.5.1	Protocol	47
4.5.2	Result	47
4.5.3	Why the Signal Degrades, and Why That Is Not a Defect	48
4.5.4	What the Repair Changes Here	48

4.5.5	Statistical Standing	48
4.6	Arm A Rescored: Discrimination Without Calibration	48
4.6.1	Discrimination	48
4.6.2	Calibration	49
4.6.3	Ruling Out Right-Censoring	50
4.6.4	The Mechanism	51
4.7	The Target-Definition Trap	52
4.8	Arm B: What Happens When the Forecast Is Shown	53
4.9	The Repair and Its Re-measurement	55
4.9.1	The Repair That Was Recommended Does Not Work	55
4.9.2	Two Further Defects Exposed by the First	56
4.9.3	The Re-measurement	56
4.9.4	A Third Arm, and What It Does Not Show	57
4.9.5	A Hypothesis That Became Measurable Again	58
4.10	Discussion	58
4.11	Opening: Grounding Real Urgency in Observed Capacity	60
4.11.1	Motivation	60
4.11.2	Method and Calibrated Constants	60
4.11.3	The Physical Limit and the Reframing	62
4.11.4	Approaches Tested and Rejected	63
4.11.5	Learned Dock-Zone Segmentation	64
4.11.6	Annotation Integrity	65
4.11.7	What This Can and Cannot Feed	66
4.12	Synthesis Against the Research Questions	68
5	Conclusion	70
5.1	Contributions	70
5.2	Status of the Research Hypotheses	72
5.3	Limitations	72
5.4	Future Work	74
5.5	Personal Conclusion	76
5.5.1	Technical Skills Developed	76
5.5.2	What I Learned	76
	References	78
A	Ethics Documentation	86

B	Source Review Grid	86
C	HÉLIOS Campaign Protocol Extract	91
D	Full Scoring Tables	93
E	Satellite Capacity Estimation: Protocol and Data	98

List of Figures

1	The DISCO serious game network	2
2	Declared and real urgency, and the two mismatch pathologies	4
3	Map of the literature streams	9
4	The layered adequacy architecture	20
5	The test-and-learn iteration loop	21
6	Causal structure behind the six-block selection	25
7	Bidirectional urgency propagation on the DAG	26
8	One iteration of the shock simulation	28
9	The SupplyScore architecture	30
10	The four AHP criteria and their computed counterparts	31
11	The AHP questionnaire page	31
12	From a declared event to a calibrated KPI update	33
13	Per-node explainability decomposition	35
14	The HÉLIOS network and its per-arc coefficients	36
15	The weekly review page	43
16	Network criticality ranking	44
17	Real against synthetic declared urgency, dry run	45
18	Reliability of the four-week forecast, arm A	50
19	ROC under both outcome definitions	53
20	Arm B: influence of the displayed forecast, by turn	55
21	The fill factor against the auditable corpus	61
22	The same equipped facade at 20 cm and at 8 cm	62
23	Equipped facades on which no trailer was detected	63
24	Validation mask mAP50-95 across the four runs	65
25	Predicted dock zones on held-out imagery	66
26	End-to-end output on one deployment site	67
27	Where satellite-observed capacity would enter the real-urgency model, and which blocks it does not reach.	68

List of Tables

1	DISCO technology stack	2
2	Urgency pathologies and their consequences	4
3	Each hypothesis, its research question, and the cost of failure	6
4	Aggregation and growth models found in the literature	10
5	Temporal Motivational Theory parameters	11
6	The six real urgency blocks	22
7	Selection filter applied to each candidate block	24
8	Terms of the propagation equations	26
9	Terms and parameters of the adequacy equations	28
10	The eight HÉLIOS campaign nodes	35
11	The twelve calibrated engine events	37
12	Arm A verdicts on the six pre-registered hypotheses	46
13	Final per-node snapshot of the arm A project	46
14	Nodes flagged by hidden risk against nodes that later miss a commitment, arm A	47
15	The arm A forecast layer, scored by horizon	49
16	Reliability of the arm A four-week forecasts	49
17	The time block at turn 6	51
18	The same forecasts under three outcome definitions	52
19	Arm B: influence of the displayed forecast, by turn	54
20	The frozen campaign rescored after repair, original-commitment target	57
21	Calibrated constants of the capacity model	60
22	Approaches tested and rejected	64
23	Segmentation training progression	64
24	Status of the six hypotheses across the executed runs	72
25	Documented limitations and their implications	73
26	Source review grid	87
27	Complete arm A scoring, production outcome definition	94
28	Complete arm A scoring, naive outcome definition	94
29	Every arm A forecast at or above 0.5 at four weeks	95
30	Arm B at turn 4, node by node	95
31	The eleven milestones observable within the campaign, judged against the deadline first committed to	96
32	Arm A before repair, original-commitment target	97
33	Arm A after the structural repair alone	97
34	Arm A after the structural repair and the progress-uncertainty term	97
35	Degeneracy of the milestone-miss probability, 120 forecasts	98
36	Data sources of the capacity pipeline	98

1 Introduction

1.1 Industrial and Technical Context

Supply chain planning has become a multi-dimensional assessment problem. An operator deciding which of two late shipments to expedite is simultaneously weighing cost, deadline pressure, supplier reliability and, increasingly, carbon footprint, while most monitoring and planning tools still process one criterion at a time (Pinedo, 2012; Chakraborty et al., 2020). That mismatch between the dimensionality of the decision and the dimensionality of the tooling is precisely why practitioners fall back on judgment-based prioritisation, and why the quality of that judgment has become an engineering concern rather than a purely human one.

This thesis was produced during an R&D placement in the Direction Innovation Numérique (DIN) of the ALTEN Group, which conducts applied research where digital twin technology, logistics optimisation and sustainable transport meet. Its Smart Green Supply Chain programme (Ben Zakour, 2024) pursues three aims at once. The lean aim attacks waste, meaning cost, delay and over-stock, through pull-flow logic and real-time scheduling. The green aim attacks the carbon footprint of each transport leg, working to the Global Logistics Emissions Council (GLEC) Framework v3.2 (Smart Freight Centre, 2023) and ISO 14083:2023 (International Organization for Standardization, 2023). The predictive aim is the one this thesis sits inside: anticipating disruption through probabilistic modelling, in the manner of a digital supply chain twin (Ivanov; Dolgui, 2020).

Behind all three is Industry 4.0 (Kagermann et al., 2013), and its promise that cyber-physical systems, the Internet of Things (IoT) and digital twins can turn a reactive supply chain into a self-organising one. The Physical Internet (Montreuil, 2011) supplies the setting for the simulation environment used here, since it imagines logistics networks that route shipments through shared infrastructure the way a network routes packets.

1.2 The DISCO Digital Twin and Prior Research at ALTEN-DIN

DISCO, the Supply Chain Illustrator, is a serious game and digital twin developed by the DIN to simulate a complete supply chain for a fictitious satellite product. It involves 22 component suppliers across structural assemblies, optical systems and embedded electronics (Baxas, 2025). Players take the role of supply chain buyers and decide on supplier selection, transport routes and factory allocation. Figure 1 shows the supplier network they work with and the three dimensions their decisions are scored on, and Table 1 summarises the technology stack underneath it.

Its scoring engine applies Min-Max normalisation over the values of the current game session, illustrated here on the carbon dimension:

$$\text{ScoreCO}_2 = 1 + 9 \times \frac{\text{realCO}_2 - \text{CO}_{2,\min}}{\text{CO}_{2,\max} - \text{CO}_{2,\min}}, \quad (1)$$

with analogous formulae for cost and risk, on a scale from 1 for the best achievable performance to 10 for the worst.

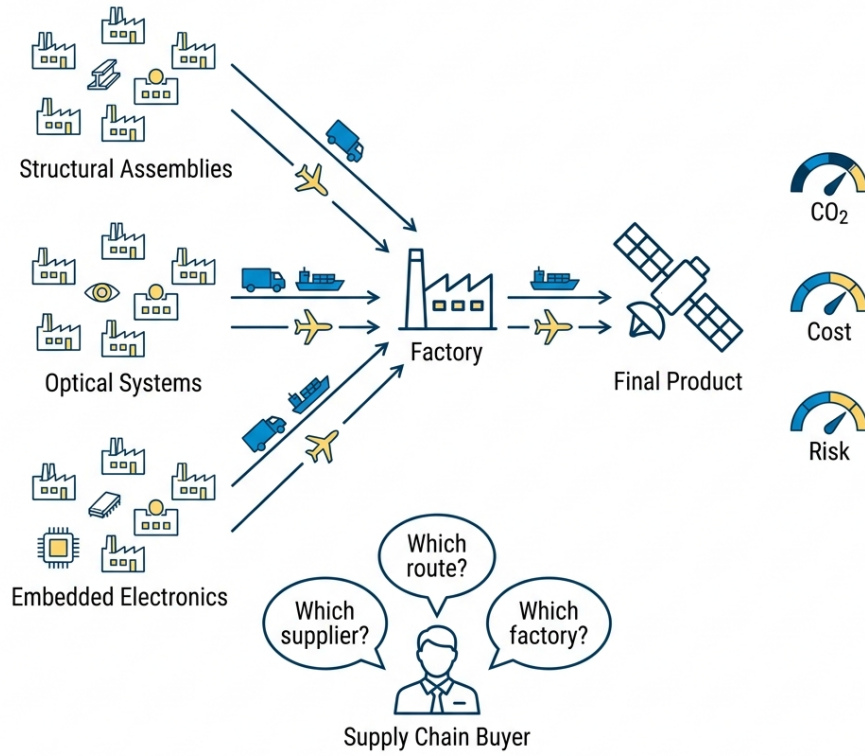


Figure 1: The DISCO serious game network

Table 1: DISCO technology stack

Component	Technology / Value
Backend API	Spring Boot + Java 21
Database	PostgreSQL (Azure)
Frontend	Angular
Authentication	Player key / Azure AD SSO OAuth2
Scores computed	CO ₂ , Cost, Risk per supplier and itinerary, on $[1, 10]$

This work sits within a body of prior DIN research. Baxas (2025) defines the DISCO software architecture including the Time To Service (TTS) and Time To Recovery (TTR) formalisation. Baulain et al. (2024) developed the Physical Internet routing module, providing transit times through the HERE Maps API with Vincenty distances and mandatory Heavy Goods Vehicle (HGV) pause compliance, together with the Itinerary Performance Index (Itinerary Performance Index (IPG)) normalised on $[0, 1]$. Kabouche (2025) developed the supplier risk scorer combining delay history, geographic vulnerability and financial fragility. Cordova (2024) developed the per-route CO₂ quantification module using the GLEC and VECTO methodology. Mondelice (2025) developed the Transport Management System interface providing real-time transit-time extraction.

These works form the organisational and thematic context of the present research. The system delivered here, named SupplyScore and described in Section 3.3, is a standalone Python application that runs alongside these modules rather than inside them. Their coupling is a stated perspective of the programme, not a delivered capability, and this thesis makes no

claim to the contrary.

1.3 Problem Statement

This thesis addresses the layer that comes *before* any planning or scheduling decision, and which the classical literature has not fully investigated: the urgency signal itself is systematically biased at the source, and must be measured and corrected before it can safely inform any downstream decision (M. Zhu et al., 2018; Rusou et al., 2020).

The consequences are worse where capacity is shared. In an interconnected network, a biased urgency signal does not stay local: it propagates, and misallocation grows as it travels. Over-triage, declaring more urgency than the situation warrants, takes capacity from whoever needed it more. Under-triage does the opposite and is more dangerous, because a risk nobody has declared is a risk nobody is watching (Li et al., 2020; Ivanov; Dolgui, 2020).

Digital twin architectures and real-time decision support systems have demonstrated strong capabilities in operational monitoring, disruption detection, resource reallocation and order synchronisation (Ding et al., 2023; Leung et al., 2022; Hrušovský et al., 2021; Karam et al., 2022; X. Zhu et al., 2025). They do so primarily by computing the physical operational state of the system. To the best of the review presented in Section 2, existing systems do not explicitly model the gap between a declared urgency and a computed operational urgency.

Figure 2 shows what that gap looks like over the life of a task. Two curves are plotted against time. The step line is declared urgency, which moves only when the operator submits a review, and the smooth curve is real urgency, which moves as the operational state does. Reading left to right, the figure passes through four phases: the two signals agree, then a first adequacy defect opens, then a second defect of the opposite sign, then agreement returns. Shading tells the two defects apart. Where the step line sits above the curve the operator is claiming more urgency than the state justifies, and the enclosed area is false urgency. Where the curve sits above the step line the state is worse than anyone has declared, and the enclosed area is hidden risk. Leader lines pointing at a curve label that signal; leader lines pointing into a shaded area label the gap between them.

Formally, let $U_d \in [0, 1]$ denote the declared urgency and $U_r(t) \in [0, 1]$ the real urgency computed from operational data in the digital twin. The mismatch between the two signals is

$$\Delta U(t) = U_d - U_r(t) \in [-1, 1]. \quad (2)$$

A tolerance band $\varepsilon \in [0, 1]$ is introduced so that small deviations are not treated as operationally meaningful. Over-declaration then corresponds to $\Delta U(t) > \varepsilon$, under-declaration to $\Delta U(t) < -\varepsilon$, and approximate agreement to $|\Delta U(t)| \leq \varepsilon$. Table 2 gives the operational reading of each condition.

The two conditions need not cost the same. Weighting them asymmetrically, with $\lambda_{\text{under}} \geq \lambda_{\text{over}}$, prices under-prioritisation above over-prioritisation. Two bodies of work support doing so: operations research showing that decision errors carry unequal downstream consequences (Kim et al., 2020; DiwasSingh et al., 2020), and scheduling work that models differentiated tardiness penalties instead of a uniform delay cost (Baker et al., 1982; Chauhan et al., 2022). How much asymmetry is right is a calibration question, answerable only in a specific setting.

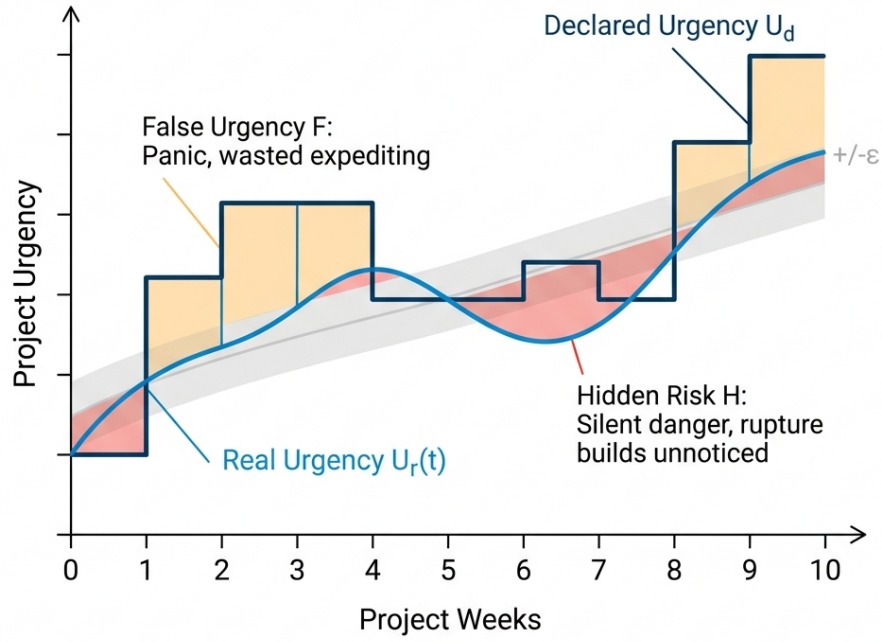


Figure 2: Declared and real urgency, and the two mismatch pathologies

Table 2: Urgency pathologies and their consequences

Condition	Interpretation	Potential consequence
$\Delta U(t) > \varepsilon$	Over-declaration of urgency, or false urgency	Excess expediting, avoidable cost, possible delay for other flows
$\Delta U(t) < -\varepsilon$	Under-declaration of urgency, or hidden risk	Hidden lateness or stockout risk, delayed intervention, downstream disruption
$ \Delta U(t) \leq \varepsilon$	Acceptable alignment	No material urgency mismatch

1.4 Aim, Objectives and Research Questions

The aim of this research is to establish whether the divergence between declared and computed urgency is measurable to a standard that would justify acting on it.

That wording is deliberate on two counts. It commits to a falsifiable target rather than to building a system: a score that cannot beat a trivial baseline against recorded outcomes has not met the standard, however well engineered the system producing it. And it makes measurability the question rather than assuming it, because the two signals being compared are not independent by construction, and an architecture that lets the same person supply both sides is measuring its own input.

Three objectives follow from that aim:

- O1.** Define a multi-block, DAG-propagated real urgency $U_r(t)$ and a cognitive adequacy score $A(t) \in [0, 100]$ measuring the gap between the planner's declared urgency U_d and the digital twin's operational state.
- O2.** Develop and implement a network propagation and criticality-driven prioritisation layer,

combining PROMETHEE II outranking with systematic shock simulation, to identify structurally critical nodes.

- O3.** Validate the resulting signal empirically, against recorded outcomes, and measure what happens to human declaration behaviour when the system's own forecast is placed in front of the declarant.

Four research questions structure the scientific approach:

- RQ1.** *Multi-block real urgency.* How can a physically and operationally grounded real urgency $U_r(t)$ be computed dynamically in a multi-tenant digital twin from heterogeneous KPI blocks, covering time, capacity, performance, risk, cost and environmental dimensions, and propagated coherently across a multi-tier supply chain graph?
- RQ2.** *Cognitive adequacy formulation.* How should an adequacy score $A(t)$ be formulated to measure the alignment between declared urgency U_d and computed real urgency $U_r(t)$ while asymmetrically penalising under-triage against over-triage, and does that score demonstrate predictive validity against recorded operational outcomes?
- RQ3.** *Network prioritisation and criticality.* How can pairwise elicitation methods, namely AHP and the Fuzzy Best-Worst Method, be combined with outranking aggregation through PROMETHEE II and systematic shock simulation to identify which nodes of a supply network are structurally critical?
- RQ4.** *Operational validation and the effect of showing the forecast.* How can this framework be operationalised and empirically validated within a persistent digital twin exercised as a weekly-cadence serious game, and what happens to declared urgency once the declarant is shown the system's own prediction?

The second half of RQ4 is not an afterthought. The campaign produced a result about the declarant rather than only about the model, and reporting the model's scores without it would be misleading (Section 4.8).

1.5 Research Hypotheses

These six hypotheses were pre-registered. They were written, with their numeric thresholds, and frozen before the campaign of Section 3.4 produced any data, which is why they appear here rather than next to the results they decide. They are reproduced unchanged. Where the campaign later showed a threshold to be badly chosen, or a hypothesis to be untestable on the scenario as scripted, that is reported as a finding in Section 5.2 and not repaired by rewriting the hypothesis.

Each states an observable outcome with a threshold the delivered system either meets or does not, and each serves one of the research questions above.

H1 (Cognitive latency): Declared urgency follows computed real urgency with a delay of at least one weekly cycle on the majority of nodes. Test: the lagged correlation between U_d and U_r peaks at a lag of one week or more for at least five of the eight campaign nodes.

H2 (Early detection of hidden risk): The hidden-risk signal $H = \max(U_r - U_d, 0)$ exceeds 0.3 at the crisis-origin node several weeks before the rupture becomes publicly visible.

H3 (Structural criticality): The systematic shock simulation ranks the true crisis origin first, by impact on the final customer, on at least 80% of the informative, non-saturated

weekly rounds.

H4 (Predictive validity): At threshold $H \geq 0.5$ with a four-week horizon, the hidden-risk signal predicts recorded negative outcomes, meaning a missed milestone, a critical event or a node abandonment, with precision and recall both above 0.5.

H5 (Urgency contagion): On weeks when crisis news becomes public, operators of nodes that are not physically affected still raise their declared urgency, reproducing the mere-urgency effect (M. Zhu et al., 2018) at network scale.

H6 (Perception hysteresis): Once the physical situation improves, declared urgency decays at less than half the rate of real urgency.

Table 3 maps each hypothesis to the question it serves and states what its failure would mean, so that a negative result carries a consequence rather than just a verdict.

Table 3: Each hypothesis, its research question, and the cost of failure

Hypothesis	Serves	If it fails
H1 cognitive latency	RQ2	Declared urgency tracks the physical signal without delay, so there is no lag to correct and the adequacy layer has nothing to add.
H2 early detection	RQ2	The hidden-risk signal arrives no earlier than the rupture it is meant to anticipate, which makes it a description rather than a warning.
H3 structural criticality	RQ3	The network ranking cannot be trusted to identify the bottleneck, so criticality-driven prioritisation is unsafe to act on.
H4 predictive validity	RQ2	The signal does not predict recorded outcomes, and the score is a diagnostic curiosity rather than a decision aid. This is the hypothesis the aim of Section 1.4 stands or falls on.
H5 urgency contagion	RQ1, RQ4	Operators do not import urgency from context, so declaration bias is individual rather than systemic and does not propagate.
H6 perception hysteresis	RQ2	Over-declaration corrects itself once the situation eases, so the asymmetric penalty is mis-specified and λ_{under} needs recalibrating.

One omission from this register is worth flagging in advance, because Section 4.7 measures its cost. The thresholds were pre-registered but the definition of an adverse outcome was not, and that definition turns out to move the measured result by more than the thresholds do.

Distinct from these testable hypotheses, the framework rests on stated modelling assumptions, each verifiable in the delivered code: the network is a directed acyclic graph, so there are no feedback loops and a linear topological sort is valid; reviews follow a strict ISO weekly cycle; declared urgency is damped as it descends toward suppliers while real risk is amplified as it ascends toward the customer, each scaled by a per-arc coefficient; the six KPI blocks are treated as independent so that a probabilistic aggregation is admissible; task statuses drive explicit override rules; and the cognitive loss is deliberately asymmetric, with under-declaration priced at roughly twice the cost of panic. These are design commitments, not

empirical claims, and the limitations they induce are catalogued in Section 5.3.

1.6 Scope and Delimitations

Four boundaries are set explicitly, because each of them shapes what the results in Section 4 can and cannot support.

The declarants are language-model agents, not human consultants. The campaign reported here was run with eight isolated language-model agents, one per role, each confined to its own role card and turn briefings. This is an experimental design choice with real consequences for external validity, discussed in Sections 3.4.3 and 4.10. The human campaign for which the protocol was originally frozen has not been run.

The validation is a single scenario. All results come from one pre-registered scenario, a semiconductor shortage propagating through an eight-node satellite supply chain over nineteen weekly turns. Statements about how the framework behaves are statements about its behaviour on that scenario.

SupplyScore is standalone. It does not integrate with DISCO, with an ERP, or with the routing and carbon modules of Section 1.2. Every KPI reaching the model arrives either through a manual weekly review or through a scripted campaign event.

The satellite observation work is an opening, not a second contribution. Section 4.11 reports a separate system that estimates warehouse capacity from public satellite imagery. It is included because it addresses the freshness limitation that the rest of this thesis documents about itself, and it is reported at the depth that claim requires, without any assertion that the two systems have been connected.

1.7 Thesis Structure

Four sections follow. Section 2 reviews the streams that bear on the problem and states the gap each one leaves. Section 3 keeps three things apart that are easily confused: the mathematical model, the choices made in building it into software, and the experiment designed to test it. It closes with the method's own limits, its reproducibility provisions and its ethics. Section 4 reports what was measured, then diagnoses it. Section 5 states what the work contributes, what each hypothesis did, where the work stops, and what it opens.

2 Literature Review

2.1 Introduction and Key Definitions

This review answers a single question: which existing methods can be assembled to measure, correct and propagate a biased urgency signal, and which part of that assembly is missing from the literature? The relevant work is broad but fragmented across operations research, behavioural science and digital twin engineering. Read individually, each stream solves one piece of the problem. Read together, they leave a specific slot empty. No retrieved work combines, within a single supply chain decision architecture, the five elements this thesis requires: a multi-block operational urgency function, a structured client-side urgency input, an adequacy layer comparing stated and operational urgency, network propagation of that urgency across a multi-tier DAG, and empirical validation of the adequacy signal against recorded outcomes (Sawik, 2013; Sawik, 2014; Sawik, 2016; Cohen et al., 2019; Hosseini; Ivanov, 2020).

The review is organised by the role each stream plays in that argument rather than by chronology or method family. Figure 3 maps the streams onto that organisation and shows where the empty slot falls. Each section closes with a statement of the gap it leaves, and Section 2.14 collects those gaps.

Six terms are used throughout with the following meanings.

Urgency is how far a task needs prompt action to avoid a material loss of operational performance. Two neighbouring words mean something else. *Importance* is about strategic value; *priority* is about current execution order. A task can be important and not urgent, and it can be first in the queue for reasons that have nothing to do with either (J. Chen et al., 2025; M. Zhu et al., 2018).

Real-time scheduling refers to the class of algorithms that assign execution order to tasks given temporal constraints, resource constraints and optimality criteria (Pinedo, 2012). This literature is drawn on only as historical grounding for deadline-sensitive value functions; no scheduling algorithm is delivered here.

Digital twin is a synchronised digital representation of a physical system supporting monitoring, simulation and decision-making from real-time and historical data (Ivanov; Dolgui, 2020).

Multi-criteria decision-making (Multi-Criteria Decision-Making (MCDM)) covers methods that combine multiple, potentially conflicting performance dimensions into a decision (Saaty, 1980; Brans; Vincke, et al., 1986).

Physical Internet (Physical Internet (PI)) is the model of open, mutualised logistics networks routing shipments through shared infrastructure using standardised modular containers (Montreuil, 2011).

Serious game is a simulation environment designed to train decision-makers through realistic scenarios with immediate feedback and scoring (Sterman, 1989; Prensky, 2001).

2.2 Risk Propagation and Deadline-Sensitive Value

Within real-time systems, deadline-sensitive scheduling theory models task value as a function of completion or response time, an approach known as time-value or utility-accrual

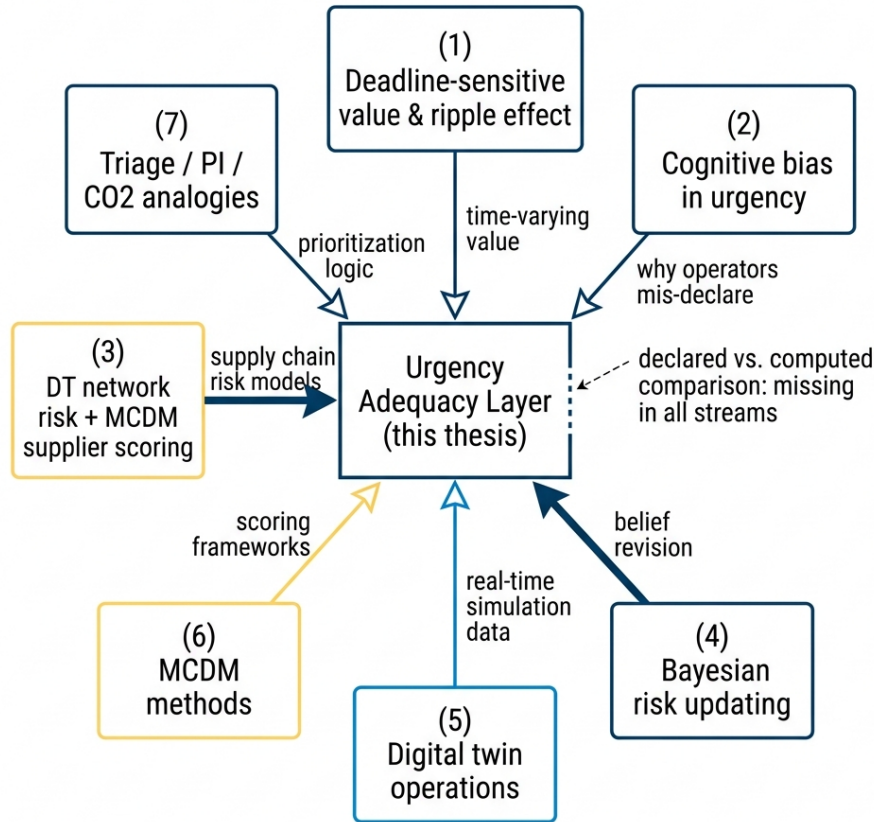


Figure 3: Map of the literature streams

scheduling (K. Chen et al., 1996; J. Chen et al., 2025). Chen and Mühlethaler (K. Chen et al., 1996) show that task utility is typically constant up to a deadline and then non-increasing with delay, and a broader survey confirms that deadline-sensitive literature models task value as a function of response time with bounded, structured degradation after critical thresholds (J. Chen et al., 2025). This establishes the precedent for treating urgency as time-varying, but it stops at a single task in isolation.

A complementary line of evidence comes from network risk propagation. Ivanov and colleagues formalise the ripple effect, whereby a local disruption at one supplier cascades through downstream tiers and amplifies as it travels (Ivanov; Dolgui, 2020; Ivanov; Dolgui, 2019; Li et al., 2020). Probabilistic risk inference for supply chain systems (Zheng et al., 2020a) further shows that aggregating heterogeneous risk indicators into a single node-level score is itself an open modelling question, distinct from how that score should then be propagated.

Table 4 compares four candidate aggregation and growth models found across these two lines.

The two block-level families differ in what they hide. A weighted average is fully compensatory, so a catastrophic reading on one indicator can be offset by acceptable readings elsewhere. A noisy-OR is not, because any factor approaching its maximum drives the aggregate toward its own maximum regardless of the others. Which behaviour is desirable depends on whether

Table 4: Aggregation and growth models found in the literature

Model	Formula	Key property	Where it appears
Linear in time	$U_0 + \lambda t$	Constant pressure	Deadline-sensitive scheduling (K. Chen et al., 1996)
Hyperbolic in time	$K/(t_c - t)$, bounded	Finite-time singularity, discounting	Temporal discounting (Steel et al., 2006); crash precursors (Johansen et al., 2001)
Weighted average of blocks	$\sum_m \omega_m u_m$	Fully compensatory: one weak block is hidden by strong ones	Composite indicators (Greco et al., 2018)
Noisy-OR over blocks	$1 - \prod_m (1 - u_m)^{\omega_m}$	Non-compensatory: one extreme block dominates	Combining independent causal evidence (Pearl, 1988)

the aggregate is meant to describe average condition or exposure to the worst active cause, and the composite-indicator literature treats that as a modelling decision rather than a solved question (Greco et al., 2018; Becker et al., 2017).

The hyperbolic family carries a further caveat. A structurally similar acceleration appears in Log-Periodic Power Law (Log-Periodic Power Law (LPPL)) models of financial crash precursors (Johansen et al., 2001; Johansen et al., 2000), but empirical analyses find LPPL-family models highly parameter-sensitive and unsuited to use as production risk models (Feigenbaum, 2001; Brée et al., 2010).

Gaps in risk propagation and deadline-sensitive value. Neither stream specifies how to aggregate heterogeneous, block-structured operational risk indicators at a single node before propagating them. Neither addresses declared urgency, or the mismatch between what is stated and what is computable. And the aggregation literature stops at the observation that the choice matters, without establishing whether equal values on structurally different indicators carry equal risk meaning, which is the question any multi-block aggregate has to answer before it can be interpreted.

2.3 Cognitive Bias in Urgency Declaration

Behavioural operations research suggests that human decision-makers systematically misrepresent urgency, a finding with direct implications for any system relying on operator-declared urgency as an input.

Two convergent lines show this from different angles. M. Zhu et al. (2018) report the Mere-Urgency Effect in five controlled experiments: participants consistently over-prioritised imminent low-value tasks over distant high-value tasks, and the bias persisted even when participants were informed of its existence. Steel et al. (2006) provides a complementary formal model through Temporal Motivational Theory (Temporal Motivational Theory (TMT)):

$$V(t) = \frac{U \times E}{1 + \Gamma \times D(t)^k}, \quad (3)$$

where U is objective utility, E is the expectancy of success, $D(t)$ is the delay to deadline, and Γ and k are individual discounting parameters. As the delay to deadline shrinks toward zero, the motivational value $V(t)$ converges to the full product of utility and expectancy, so immediacy becomes more salient as the deadline approaches. Near-term tasks can therefore attract disproportionate attention even when their broader objective value is lower (Steel et al., 2006). Table 5 gives the operational role of each parameter.

Table 5: Temporal Motivational Theory parameters

Symbol	Concept	Operational role
$V(t)$	Temporal motivational value	Dynamic task salience
U	Objective utility	Task strategic importance
E	Expectancy of success	Probability of task completion
$D(t)$	Delay to deadline	Remaining safe execution window
Γ	Impulsiveness parameter	Operator sensitivity to delay
k	Sensitivity scaling parameter	Non-linear scaling of delay discounting

These laboratory findings are supported by evidence from operational settings. Rusou et al. (2020) show that operators under workload preferentially complete small, easy tasks, the operational analogue of under-triage for large-scope supply chain activities. DiwasSingh et al. (2020) confirm in a large field study that completing easy tasks first significantly hurts overall performance. Hospital admission control under occupancy uncertainty (Kim et al., 2020) produces over- and under-triage patterns analogous to those seen in logistics operators facing uncertain lead times. Human-centred decision support systems such as that of Rosenthal et al. (2020) use visual feedback to show the mismatch between human plans and system temporal constraints, helping users calibrate their schedules.

Gaps in cognitive bias research. Behavioural research diagnoses the problem and quantifies it in controlled settings, but the magnitude of the bias in industrial supply chain contexts remains a matter of calibration, and this literature does not prescribe a real-time operational correction.

2.4 Indicators of Delivery Rupture Risk

If urgency is to be aggregated from several indicators, the literature has to supply the indicators. This section reviews the body of work that identifies which observable quantities precede a delivery rupture, meaning a failure to deliver at the agreed time, place, quantity and quality. It is the source from which Section 3.2.2 later selects.

Heckmann, Comes and Nickel establish rupture as the terminal event a supply risk measure should anticipate, and criticise the field for measuring exposure without connecting it to that event (Heckmann et al., 2015). Indicator-based supplier monitoring proceeds by identifying observable precursors rather than modelling the event directly (Mokhtar et al., 2019).

Five families of precursor recur across that work. Slack, meaning the margin between a realistic lead time and the committed deadline, is the one indicator whose exhaustion is irreversible, since a task with no remaining slack cannot be recovered (Cohen et al., 2019; Bojko et al., 2021; Sun et al., 2020; Sawik, 2013). Capacity saturation adds waiting time that a nominal lead time does not show, so a loaded system carries a rupture risk its announced schedule conceals (Tomlin, 2006; Hosseini; Tajik, et al., 2019; Taghavi et al., 2024). Resource performance drift is monitored through composite proxies, most often Overall Equipment Effectiveness, because direct performance data is rarely available on the supplier side (Mokhtar et al., 2019; Hlioui et al., 2017). Adverse-event exposure is treated probabilistically, as the product of a failure probability, a recovery duration and an impact severity (Hosseini; Ivanov; Dolgui, 2020; Liu et al., 2021; Hosseini; Ivanov, 2020). Cost signals such as expediting premiums are read as symptoms of strain rather than causes of rupture: paying for air freight to protect a deadline is evidence the system is already under stress, and it also bounds which corrective actions stay affordable (Shen et al., 2019; Tomlin, 2006). Emissions constraints enter the same way, as a restriction on the admissible set of responses rather than as a rupture mechanism, and become decisive mainly in the trade-off between slow sea freight and fast air freight (Hoen et al., 2014; Smart Freight Centre, 2023; International Organization for Standardization, 2023).

Gaps in rupture-risk indicators. This literature names precursors and monitors them individually. It does not establish a selection criterion for deciding which precursors belong in a single composite, nor does it address what happens when several are combined: whether the indicators are independent enough to be multiplied, and whether a given numerical level on one carries the same risk meaning as the same level on another. Those two questions are left open, and any multi-block urgency model inherits them.

2.5 Closest Antecedents: Network Risk Twins and MCDM Supplier Scoring

The works most structurally similar to this thesis combine multi-criteria supplier scoring with disruption-aware network monitoring. Sawik (2013) develops mixed-integer programming models for resilient supplier order assignment integrating cost, service level and disruption risk, extended in later work to joint and integrated formulations (Sawik, 2014; Sawik, 2016). Cohen et al. (2019) and Afrasiabi et al. (2022) extend this to MCDM supplier prioritisation under disruption, combining criteria weighting with allocation optimisation. Chakraborty et al. (2020) confirm that rule-based heuristics remain competitive against exact methods in dynamic industrial environments. These works are cited for the closeness of their subject matter, not because this thesis reuses their optimisation machinery, which addresses allocation and sequencing rather than urgency measurement.

Gaps in the closest antecedents. Two gaps distinguish this stream from the present work. First, none introduces a network-propagated urgency signal computed independently at every node and combined across tiers; risk is typically evaluated once, at the point of decision. Second, none compares a client-stated urgency against a model-derived urgency: where

the declared signal exists at all, it is used directly, without validation against live operational proxies.

2.6 Enabling Method: Bayesian and Probabilistic Risk Updating

Bayesian methods are a well-supported approach for revising risk estimates from new evidence in supply chain research (Pearl, 1988; Koller et al., 2009). Rather than a full Bayesian network with an explicit conditional-probability-table structure, this study draws on the narrower and more tractable case of Beta-Bernoulli conjugate updating: a failure probability p is treated as a Beta-distributed belief with an equivalent prior sample size n_0 , revised from a small number of newly observed equivalent trials (k, n) each time a calibrated operational event is declared.

Three convergent findings support this choice. Hosseini; Ivanov (2020) survey Bayesian methods applied to supply chain risk and resilience, finding that probabilistic belief-revision methods are often preferred to deterministic scoring when sparse new evidence must update an existing estimate. Huang et al. (2020) and Deng et al. (2020) develop Bayesian models for disruption and delay probability estimation. Lockamy et al. (2011) and Qazi et al. (2017) support Bayesian revision as a practical tool under sparse, event-driven evidence, consistent with the small-sample setting of a weekly-cadence digital twin. Plain frequency counts, by contrast, are unstable when only a handful of events are declared per node per project (Zheng et al., 2020b).

Gaps in Bayesian risk updating. Bayesian belief revision quantifies disruption exposure but does not, by itself, compare it against a client-declared urgency, and it says nothing about whether the resulting probabilities are well calibrated once they reach a human reader.

2.7 Enabling Method: Digital Twins and Real-Time Decision Support

The digital twin concept was formalised by Grieves (2014) as a virtualised representation of a physical process. Tao et al. (2018) extended it to manufacturing, and Fuller et al. (2020) surveyed the enabling technologies. In supply chain management, Ivanov; Dolgui, et al. (2021) define the Supply Chain (SC) Digital Twin (DT) as integrating simulation, optimisation and machine learning. Ivanov; Dolgui (2020) address digital supply chain twins for disruption risk and resilience management. Wang et al. (2020) demonstrate twin-driven planning in a manufacturing context analogous to DISCO's satellite component network. In the Industry 5.0 setting, these systems have evolved toward human-in-the-loop Cognitive Digital Twins, where semantic models and reasoning methods capture human feedback and synchronise mental models with simulated physical operations (Nilloofar et al., 2023).

Real-time freight planning work shows that operational urgency proxies, meaning slack, lateness risk and shortage exposure, are already present within these systems (Hrušovský et al., 2021; Perez et al., 2022). Model-predictive control for vehicle routing (Karam et al., 2022) and twin-based disruption response (Ding et al., 2023; Leung et al., 2022; X. Zhu et al., 2025) confirm the state of the art in live operational replanning.

Gaps in digital twin decision support. In the retrieved literature, no system combines a declared urgency input, a computed operational urgency, and their propagated mismatch

within one architecture. A second gap matters for this thesis specifically: these systems are evaluated on the accuracy of their physical state estimation, and their inputs are assumed to arrive. Where an input depends on the same operator whose judgment the system exists to support, that circularity is not usually treated as a measurement problem.

2.8 Enabling Methods: Multi-Criteria Decision Methods

This research builds on two distinct domains within multi-criteria decision-making: pairwise comparison methods for eliciting weights, and outranking aggregation for turning those weights into a defensible ranking of nodes.

Pairwise elicitation. To construct a structured representation of declared urgency U_d , criteria weights must be systematically elicited. The Analytic Hierarchy Process (Analytic Hierarchy Process (AHP)) (Saaty, 1980) derives weights as the principal eigenvector of a pairwise comparison matrix, validating logical consistency through the Consistency Ratio ($ConsistencyRatio(CR) \leq 0.10$). While widely applied in logistics, AHP suffers from rank reversal, where introducing a new alternative alters the relative ranking of existing ones, and is sensitive to cognitive noise during elicitation (Dyer, 1990; Triantaphyllou; Yanase, 2024). To reduce operator load and improve consistency, the Best-Worst Method (Best-Worst Method (BWM)) (Rezaei, 2015) reduces the number of comparisons from $n(n-1)/2$ to $2(n-1)$ by anchoring around chosen best and worst criteria. The Fuzzy Best-Worst Method (Fuzzy Best-Worst Method (F-BWM)) (S. Guo et al., 2017; Yucesan et al., 2019; Amiri et al., 2020) extends this with Triangular Fuzzy Numbers (Triangular Fuzzy Numbers (TFNs)) to absorb judgment uncertainty, and consistency conditions for the fuzzy variant have been established (Liang et al., 2020).

Outranking aggregation and the composite indicator critique. For prioritising which nodes need attention, direct weighted averages risk fully compensatory aggregation, where an excellent score on one criterion completely masks a critical failure on another. Outranking methods such as PROMETHEE II (Brans; Vincke, et al., 1986; Brans; Mareschal, 2005) establish pairwise preference flows ϕ^+ and ϕ^- , allowing a non-compensatory ordering. A cross-industry survey (Behzadian et al., 2010) confirms its suitability for supplier and node evaluation. Composite-index literature nonetheless warns that nominal weights often diverge from effective importance because of correlation, normalisation and aggregation dynamics (Greco et al., 2018; Becker et al., 2017; Borgonovo et al., 2021), and the choice of normalisation is itself non-neutral, since it can produce rank reversal (Triantaphyllou, 2000).

Gaps in multi-criteria decision methods. These methods weight criteria and rank alternatives, but they are static elicitation and aggregation tools. None is routinely compared against a computed operational counterpart. And the composite-indicator critique is generally raised as a caveat at the end of a paper rather than turned into something a running system can check on its own data.

2.9 Forecast-Assisted Decision and Automation Bias

The streams above concern how to compute a signal. This one concerns what happens when the signal is shown to the person whose judgment it was built to support, and it is included

because the campaign reported in Section 4.8 produced a result in exactly that register.

Parasuraman et al. (1997) set out the canonical taxonomy of misuse, disuse and abuse of automation, establishing that the failure modes of automated aids are as much about human reliance as about machine accuracy. Skitka et al. (1999) demonstrate automation bias directly: participants supported by an automated aid committed both omission errors, failing to react to a problem the aid did not flag, and commission errors, following the aid against contradicting evidence. Cummings (2004) extends the finding to time-critical decision support, the family to which a weekly supply chain review belongs. Goddard et al. (2012) provide a systematic review in clinical decision support, and report that automation bias is frequent, that it is mediated by the confidence with which advice is presented, and that it is only partially mitigated by training.

The direction of the effect is not uniform. Dietvorst et al. (2015) document algorithm aversion, where people abandon an algorithm after seeing it err even when it outperforms them, while Logg et al. (2019) document algorithm appreciation, where people weight algorithmic advice more heavily than human advice. Bansal et al. (2021) show that adding explanations to a model's output can increase acceptance of its recommendations whether or not those recommendations are correct, so that explanation improves persuasion more reliably than it improves joint accuracy.

Two consequences follow for the present work. First, an advisory signal changes the behaviour it measures, which means a system that both elicits declared urgency and displays a forecast to the declarant is not observing an independent signal. Second, a confidently presented but poorly calibrated probability is a specific hazard rather than a cosmetic defect, since the confidence with which advice is presented is itself a mediator of reliance (Goddard et al., 2012).

Gaps in forecast-assisted decision research. This literature is dominated by single-decision laboratory tasks and clinical decision support. It does not address a repeated, multi-turn setting in which the same operator is shown a forecast about their own node, week after week, while their declaration feeds back into the model's state. Nor does it address the case where the displayed forecast is systematically miscalibrated in a known direction, which is the situation measured in Section 4.6.

2.10 Validation Methodology: Serious Games

The Beer Game (Sterman, 1989) is the foundational supply chain serious game. Sterman shows that players systematically misperceive the feedback in their own supply line, underestimating orders already placed but not yet received, and over-order with a lag once a shortage appears, amplifying the resulting oscillation as it travels upstream, the mechanism known as the bullwhip effect (Lee et al., 1997). This decision bias is analogous to the false-urgency pathology, where an operator over-reacts to a signal after a delay rather than adjusting smoothly. Prensky (2001) establishes the theoretical basis for digital game-based learning, and Deterding et al. (2011) formalise gamification design principles.

Mohaddesi et al. (2020) introduce the Gamette methodology, a bite-sized approach to capturing decision-making patterns for behavioural modelling, directly applicable to a weekly-review protocol in which operators repeatedly submit urgency assessments. Bellotti et al. (2013)

provide an evaluation framework for serious game learning outcomes. Lafkihi et al. (2025) use a freight transportation game to study carrier behaviour on online spot-market platforms, a setting connected to the Physical Internet’s vision of mutualised capacity allocation.

Gaps in serious game validation. Serious games are established as instruments for observing decision bias, but they are usually scored on learning outcomes or on aggregate system performance. They are rarely used as a controlled comparison in which the same frozen scenario is replayed with and without a treatment applied to the declarant, which is the design adopted in Section 3.4.3.

2.11 Probabilistic Forecast Verification

Because this thesis scores a probabilistic forecast against recorded outcomes, it inherits the vocabulary of forecast verification. Brier (1950) introduced the mean squared error of probabilistic forecasts, and Murphy (1973) decomposed it into reliability, resolution and uncertainty, establishing the distinction this thesis relies on: a forecast can order outcomes correctly while announcing the wrong probabilities, and the two failures require different remedies. Gneiting et al. (2007) formalise strictly proper scoring rules, under which a forecaster minimises expected loss only by reporting their true belief.

The distinction has been rediscovered in machine learning. Niculescu-Mizil et al. (2005) show that several strong classifiers produce distorted probabilities that require post-hoc correction, and C. Guo et al. (2017) show that modern neural networks are systematically over-confident, with accuracy improving while calibration degrades. Both report that the correction is a separate, cheap, post-hoc layer rather than a change to the model.

Gaps in forecast verification practice. The verification literature assumes the outcome being predicted is unambiguously defined. In an operational supply chain setting it is not: whether a node-week counts as an adverse outcome depends on choices about which events qualify and over what horizon. Section 4.7 shows that this choice, and nothing else, moves the measured skill of the same forecasts by a factor of five.

2.12 Analogical and Contextual Literature

Three streams provide vocabulary and context rather than foundations, and are flagged as non-foundational wherever they are used.

Medical triage as a conceptual analogy. Medical triage provides an analogy for prioritising under severe capacity constraints and cognitive pressure. Protocols such as the Manchester Triage System (Manchester Triage System (MTS)) (Mackway-Jones et al., 2013) categorise patients into five urgency levels using flowcharts, whereas the Emergency Severity Index integrates predicted resource consumption. Comparative clinical studies show substantial classification discrepancies between the two systems, at a weighted $\kappa = 0.51$, so even trained professionals disagree on urgency assessment (Kuligowski et al., 2025). The clinical literature also establishes an asymmetry that logistics work rarely states explicitly: under-triage, meaning a patient classified as less urgent than they are, is more dangerous than over-triage, because the first costs a life and the second costs a queue (Kim et al., 2020; Mackway-Jones et al., 2013).

Physical Internet and synchromodality. The Physical Internet is a model of interconnected logistics networks in which routing, storage and transport assets are mutualised across actors (Montreuil, 2011). In such networks urgency stops being a local concern and becomes a network-level quality of service variable, because the capacity one actor claims is capacity another loses.

A related but separate concept is synchromodality. Where the Physical Internet describes who shares the infrastructure, synchromodality describes when the mode is chosen: a shipment can be moved between road, rail and waterway in real time as conditions change, instead of being committed to one mode when it is booked. The two can operate together and neither presupposes the other. Dynamic routing under synchromodality needs priority signals that stay valid across rolling planning windows (W. Guo et al., 2020), and disruptions travel quickly through interconnected networks, producing cascading ripple effects (Li et al., 2020).

Regulatory carbon accounting. Corporate sustainability reporting requirements have made transport emissions a reported quantity rather than an optional one (Cordova, 2024). The GLEC Framework v3.2 (Smart Freight Centre, 2023) and Vehicle Energy Consumption calculation TOol (VECTO) (European Commission, 2017) define emissions baselines for heavy goods vehicles, and emissions factors relate non-linearly to vehicle speed and cargo load, so a carbon figure cannot be recovered from cost or distance alone (ADEME, 2014).

Earth observation of built and logistics assets. The opening reported in Section 4.11 rests on a separate literature. Overhead object detection has a standard benchmark in DOTA (Xia et al., 2018), whose oriented bounding boxes and vehicle classes make it the reference for overhead vehicle detection, while building footprint extraction is benchmarked by Inria (Maggiori et al., 2017) and SpaceNet (Van Etten et al., 2018). Promptable segmentation foundation models (Kirillov et al., 2023) have made it possible to sample candidate sites from imagery alone rather than from a cadastral database. Separately, a body of economics work establishes that satellite observation can serve as an independent proxy for activity that official statistics record slowly or not at all, from night-time lights as a growth proxy (Henderson et al., 2012) to daytime imagery as a poverty predictor (Jean et al., 2016).

Gap in Earth observation for supply chain state. That economics literature measures aggregate activity at regional scale. It does not produce the per-facility quantities a supply chain twin needs, meaning storable volume and transfer capacity at a named site, and the detection literature stops at footprints and vehicles without converting them into an operational capacity figure.

2.13 Source Review Grid

Following the structured review method required by the programme's writing guidelines, each source analysed above has been synthesised in an evaluation grid recording its main argument, methodological approach, conclusions, thematic relevance to this thesis, and a critical appraisal of its limitations. The grid is given in Appendix B. It provides a comparative frame for assessing the contribution of each source and makes explicit which claims in this thesis rest on foundational work and which rest on analogy.

2.14 Identified Gaps

Each stream reviewed above closes on the limit of what it settles. Collected, those limits describe one absence rather than several.

Operational monitoring evaluates one criterion at a time and does not capture the multi-block character of supply chain state. The rupture-risk literature names precursors but supplies no criterion for choosing which belong in one composite, and leaves open whether equal values on structurally different indicators mean equal risk. Pairwise comparison methods elicit weights reliably but are static, and are not compared against a computed counterpart. Out-ranking prevents compensatory aggregation but is not combined with a network-propagated signal. Digital twins and Bayesian revision monitor disruption well, and have no mechanism for comparing an operator's input against the physical state they maintain. The automation bias literature shows that displaying a forecast changes the judgment it supports, without examining a repeated loop in which that judgment feeds back into the model. Forecast verification separates ordering from calibration, and assumes the outcome is unambiguously defined.

The reviewed work therefore covers multi-block risk aggregation, network propagation, twin operations, multi-criteria elicitation, and the human response to advisory signals. No retrieved work combines them with a structured declared-urgency input and an empirically tested adequacy layer. That missing combination is the gap, and closing it is the contribution this thesis claims: a measurement of the divergence between declared and computed urgency, propagated across a supplier network, and scored against recorded outcomes rather than asserted.

Three of the gaps above become testable questions rather than design choices, and they are the ones this thesis carries into Section 3: whether the divergence can be measured at all once both signals come from independent sources, whether the resulting signal predicts anything, and what happens to the declared side when the computed side is shown to the person declaring.

3 Methods

3.1 Methodological Framework

This research is structured as an agile test-and-learn programme rather than a linear sequence, operating through repeated iterations of literature analysis, mathematical specification, software implementation and scenario-based simulation (Figure 5). That structure is mapped onto the IMRaD form of this thesis as follows: the specification and implementation iterations are reported in this section as the method, and the simulation and campaign iterations are reported in Section 4 as the results.

The programme is organised into four operational areas:

- **Literature review and gap analysis**, establishing the psychological and mathematical foundations of multi-block urgency aggregation and cognitive bias (Section 2).
- **Mathematical framework specification**, deriving the multi-block real urgency function, its DAG propagation and the asymmetric adequacy score from first principles, so that the result is dimensionally and probabilistically consistent (Section 3.2).
- **Pairwise comparison elicitation**, formulating crisp AHP and Fuzzy Best-Worst Method weight vectors to build a structured representation of declared urgency (Section 3.3.2).
- **Implementation, simulation and validation**, building a modular, tested Python library exercised through a multi-page Dash web application, running controlled shock and scenario simulations on a persistent digital twin, and confronting the resulting signal with recorded outcomes through a dedicated calibration service and a serious game campaign (Sections 3.3 and 3.4).

The remainder of this section keeps three things apart, because mixing them is what makes a framework of this kind hard to read. Section 3.2 states the mathematical model on its own terms, saying nothing about tooling. Section 3.3 gives the application choices, meaning how each object in that model is elicited, computed and exposed. Section 3.4 gives the use case built to test it. The method's own limitations, its reproducibility provisions and its ethical considerations close the section.

3.1.1 The Proposed Structure

Section 2.14 identified one absence rather than several. The response designed here is a two-layer loop, and stating it before the mathematics makes the equations that follow easier to place.

The first layer is a cognitive adequacy check. It captures the operator's declared urgency U_d through consistency-gated pairwise comparisons, computes a real urgency $U_r(t)$ from live operational state, and scores their mismatch through an asymmetric penalty. The second layer propagates and prioritises. Local urgency travels across the supply network, declared urgency descending toward suppliers and real risk ascending toward the final customer, and the resulting network state feeds an outranking and a shock-simulation criticality index that together indicate which nodes warrant attention. Figure 4 shows how the two layers sit relative to each other and where the declared and computed signals meet.

Two choices inside that design are worth naming immediately, since Section 2.2 left both open. The aggregation over blocks is a noisy-OR rather than a weighted average, because a

complete capacity failure should not be offset by acceptable cost and carbon readings, and non-compensatory behaviour is the property that guarantees it. The penalty over the mismatch is asymmetric, pricing under-declaration above over-declaration, on the clinical analogy of Section 2.12, where under-triage costs more than a queue. Both are design commitments open to challenge, and Section 5.3 records what each one costs.

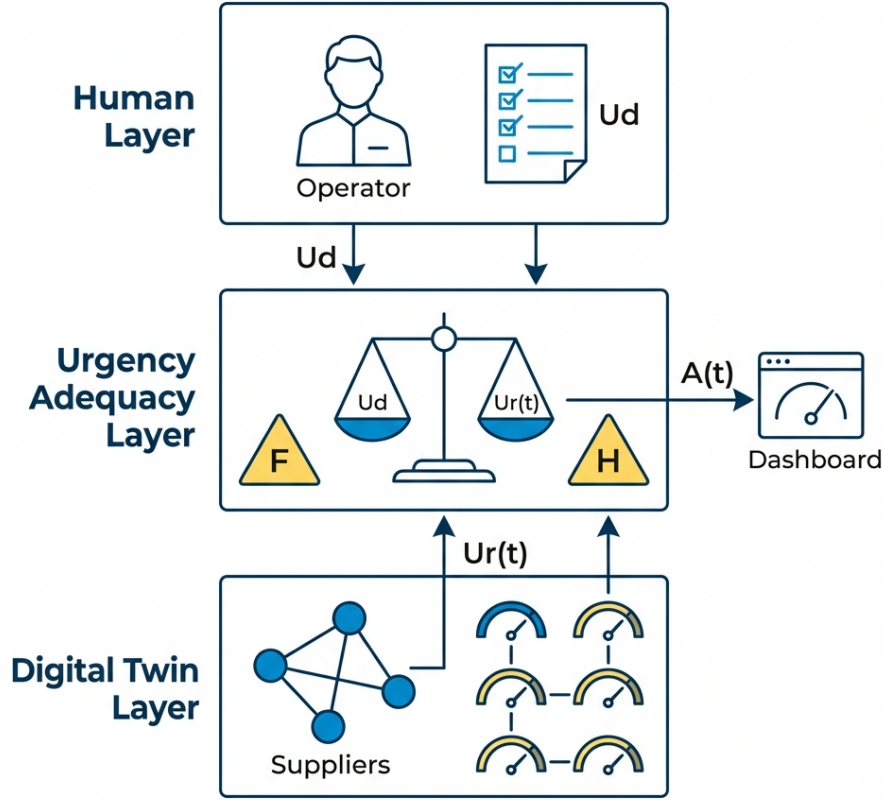


Figure 4: The layered adequacy architecture

3.2 Mathematical Model

This section states what real urgency and declared urgency are, how each propagates across the network, and how their mismatch is scored. It deliberately leaves out tooling, elicitation mechanics and operational cadence, which are application decisions addressed in Section 3.3.

3.2.1 Local Real Urgency

Real urgency is not read from a single deadline-proximity signal. It is aggregated at each node from six independent KPI blocks, namely time, capacity, performance, risk, cost and carbon, each producing a partial urgency $u_m \in [0, 1]$, or no value at all when the underlying KPI data is unavailable. The aggregation is a weighted probabilistic OR, so that a single saturated block is sufficient to drive local urgency toward its maximum regardless of how well the others score:

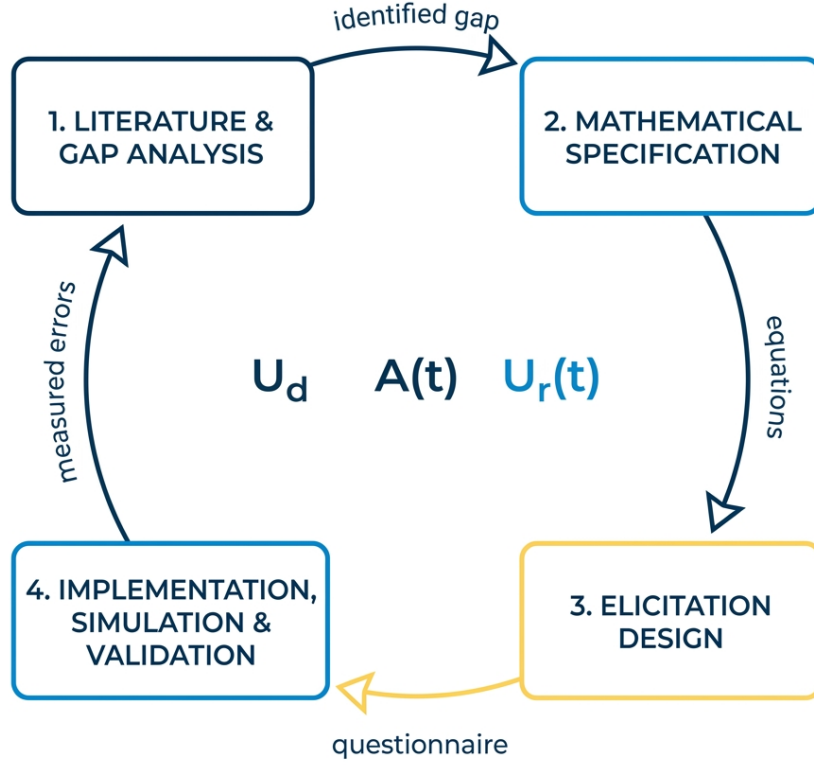


Figure 5: The test-and-learn iteration loop

$$U_{r,\text{local}} = 1 - \prod_m (1 - u_m)^{\omega_m}, \quad (4)$$

where $\omega_m \geq 0$ is the aggregation weight of block m and the product runs over blocks with available data. Table 6 gives each block's schematic formula and what it measures. The notation $[x]_+ = \max(x, 0)$ denotes the positive part, and the per-term weights α_i and β_i are renormalised over the terms whose data is actually available.

A bare formula does not say what its bounds mean, so three properties of Equation 4 are stated explicitly. They are also where the model is most open to challenge.

What the endpoints mean. Each raw KPI carries physical units, whether days, euros, kilograms or percent, and passes through a transfer function onto the dimensionless interval $[0, 1]$. At $u_m = 0$ the block is nominal and contributes nothing: its factor $(1 - 0)^{\omega_m}$ equals 1 and leaves the product unchanged. At $u_m = 1$ the factor is 0, so the product collapses and $U_{r,\text{local}} = 1$ whatever the other blocks read. The upper endpoint therefore has a precise operational meaning: the block has reached the state past which it cannot make the node any more urgent, whether that state is a deadline already passed, a fully depleted buffer or a near-certain failure. The word used for it in this thesis, saturation, is a label adopted for this

Table 6: The six real urgency blocks

Block	Formula	Explanation
u_{time}	$P(L > d^* - t) + \kappa_{\text{retard}}[r]_+ - \kappa_{\text{avance}}[-r]_+$	Probability that the lead time L , modelled as Normal(μ , σ), exceeds the time left before the next active deadline d^* ; forced to 1 once the deadline has passed. The planning term r compares theoretical with reported progress.
u_{cap}	$\alpha_1 \left(1 - \frac{V_{\text{rem}}}{V_{\text{max}}}\right) + \alpha_2 \left(1 - \frac{W_{\text{rem}}}{W_{\text{max}}}\right) + \alpha_3 \left[\frac{D - \Phi}{D}\right]_+$	Saturation of the node's capacity: how little storage volume and weight remain, plus the fraction of demand D not covered by the current flow rate Φ .
u_{perf}	$1 - \text{OEE}$	Complement of Overall Equipment Effectiveness, itself availability $\times$ performance $\times$ quality.
u_{risk}	$1 - \exp(-p_{\text{fail}} \times t_{\text{recov}} \times \text{sev}/T_{\text{ref}})$	Expected unavailability: failure probability times recovery time times impact severity, normalised by a reference horizon, then combined by probabilistic OR with environmental and political exposure.
u_{cost}	$\beta_1 \frac{c_{\text{op}} - c_{\text{nom}}}{c_{\text{nom}}} + \beta_2 [\text{tariff} - 1]_+ + \beta_3 \frac{c_{\text{stock}}}{c_{\text{ref}}}$	Financial stress: operating cost drift above nominal, tariff multiplier above neutral, and storage cost relative to a reference.
u_{co2}	$\left[\frac{e - e_{\text{target}}}{e_{\text{max}} - e_{\text{target}}}\right]_+$	Normalised overshoot of measured emissions above target, saturating at the maximum threshold.

model and not a concept imported from the reviewed literature, which is why the endpoint is defined here by its effect on the product rather than by citation.

How a missing block is handled. When a KPI is unavailable the block is given weight $\omega_m = 0$, which sets its factor to $(1 - u_m)^0 = 1$ and leaves the aggregate untouched. Absence is therefore not represented by $u_m = 0$, which would be a positive claim that the block is healthy, and the two cases are distinguished in the data model as well as in the mathematics. The remaining weights are renormalised so they still sum to their intended total, which keeps the aggregate comparable between a node with six live blocks and a node with three.

Whether the blocks are exchangeable. The product in Equation 4 is symmetric in its factors, so it treats the blocks as interchangeable up to their weights. That is the strongest assumption in the model and it does not hold cleanly. A value of $u_{\text{time}} = 0.3$ means a 30% probability of missing the next deadline, a quantity with a frequentist reading. A value of $u_{\text{cost}} = 0.3$ means operating cost has drifted three tenths of the way along a normalising scale toward a reference, which is an ordinal statement dressed in a cardinal number. The two are not equally informative about rupture, and no transfer function makes them so.

Three consequences follow, and they are carried forward rather than resolved. The weights ω_m are the only mechanism the model has for expressing that some blocks are more diagnostic than others, so they absorb a distinction that arguably belongs in the transfer functions instead. Any interpretation of the aggregate as a probability is therefore approximate, and the aggregate is used in this thesis for ranking and for comparison against U_d , never as a calibrated probability in its own right. And the assumption is testable: if two blocks carry the

same information, the correlation between them is observable, which is exactly the check reported in Section 4.4 where two of the six turned out to be perfectly correlated on the campaign run. Section 5.3 records the assumption among the framework's stated limits.

Two lifecycle rules apply before aggregation. A node marked DONE has its effective local U_r forced to 0.0, since a completed task carries no further physical risk, and a node marked ABANDONED has it forced to 1.0. These apply identically whether the local urgency comes from the analytic blocks or from the Monte Carlo estimate below.

A stochastic variant of the time block. The analytic u_{time} block assumes a Normal lead time at a single node. On a multi-tier network that assumption breaks, since a node can only start when all its predecessors have delivered, and the maximum of several skewed lead times has no convenient closed form. The module `mc/lead_time.py` therefore replaces the analytic estimate with a stochastic one when enabled.

The simulation works in three steps. Each node draws its lead time L_i from a configurable distribution, either lognormal, normal clipped at zero, or triangular. Completion times then propagate along the DAG in topological order, a node starting when its last predecessor finishes, $S_i = \max_{j \in \text{Pred}(i)} C_j$, and completing after its own lead time, $C_i = S_i + L_i$. Finally the delay probability of each node is estimated as the fraction of draws in which it misses its deadline, $u_{\text{time}} = P(C_i > d_i)$, over $N = 10\,000$ draws by default, with a memory guardrail capping the simulation at $N \times n \leq 5 \times 10^7$ across n nodes.

3.2.2 Why These Six Blocks

The six blocks are not an arbitrary KPI shortlist. The selection started from the question that defines urgency in a supply chain: which mechanisms make the risk of a delivery rupture rise? A rupture, meaning the inability to deliver at the agreed time, place, quantity and quality, is the terminal event the urgency signal must anticipate (Heckmann et al., 2015). Working backwards from that event, and consistently with indicator-based supply risk monitoring (Mokhtar et al., 2019), four direct causal mechanisms and two arbitration variables were identified.

- **Time erosion** (u_{time}): the remaining slack between deadline and realistic lead time. Time is the only non-compressible resource, so a task whose slack reaches zero can no longer be saved (Cohen et al., 2019; Bojko et al., 2021; Sun et al., 2020; Sawik, 2013).
- **System saturation** (u_{cap}): queues, transport capacity, workshop load and bottlenecks. A saturated system introduces hidden waiting delays on top of the nominal lead time (Tomlin, 2006; Hosseini; Tajik, et al., 2019; Taghavi et al., 2024).
- **Critical-resource performance** (u_{perf}): drift in the performance of the resource that produces the part. OEE is used as a composite proxy because precise performance data is rarely accessible on the supplier side (Mokhtar et al., 2019; Hlioui et al., 2017).
- **Adverse-event exposure** (u_{risk}): the expectation of unavailability caused by a disruption. The default incident probability of 0.02 is revised weekly from declared events through the Bayesian operator of Section 3.3.4, building on the DIN supplier risk scorer (Kabouche, 2025) and on probabilistic ripple-effect models (Hosseini; Ivanov; Dolgui, 2020; Liu et al., 2021; Hosseini; Ivanov, 2020).
- **Financial flexibility** (u_{cost}): expediting surcharges, tariff rises and storage costs. Cost drift is not a rupture cause in itself but a stress indicator, and it bounds which corrective actions remain economically viable (Shen et al., 2019; Tomlin, 2006).

- **Carbon overshoot** (u_{co2}): emissions above target. Like cost, it is an arbitration criterion rather than a rupture mechanism, becoming decisive only when a regulatory or ESG constraint genuinely restricts the admissible responses (Hoen et al., 2014; Smart Freight Centre, 2023; International Organization for Standardization, 2023).

Each candidate indicator was screened against the four filter questions of Table 7. Only indicators passing all four were retained, which yields four direct causes plus two arbitration variables and explains why the list has six entries rather than four or eight. Figure 6 draws the resulting structure, separating the four blocks that describe a mechanism leading to rupture from the two that constrain which responses remain available.

Table 7: Selection filter applied to each candidate block

Criterion	Question asked of each candidate indicator
Causal link	Does the indicator have a plausible mechanism linking it to delivery rupture?
Observability	Can it be measured or credibly estimated from client, supplier or planning data?
Actionability	Can an operator act on it, or use it to decide?
Non-redundancy	Does it add information not already carried by a retained indicator?

At this stage the selection remains a conceptual construction. It must still be confirmed by the real availability of the underlying data and by sensitivity tests verifying that each block contributes useful information to the aggregate. Section 4.10 returns to this point, because the campaign measured one block failing that test.

3.2.3 Local Declared Urgency

Symmetrically, declared urgency is not read as a single free-text estimate. It is reconstructed from a structured elicitation so that it can be scored, propagated and compared with $U_{r,local}$ on the same scale. At each node the operator compares four criteria pairwise through the Analytic Hierarchy Process, which yields a consistency-checked weight w_j per criterion, then rates each criterion on a severity scale $s_j \in \{1, \dots, 9\}$. Local declared urgency is the weight-averaged severity rescaled to $[0, 1]$:

$$U_{d,local} = \sum_{j=1}^4 w_j \frac{s_j - 1}{8}, \quad \sum_{j=1}^4 w_j = 1. \quad (5)$$

The elicitation tool, its four criteria and the consistency gate that rejects an incoherent weight vector are application choices described in Section 3.3.2. What matters here is that $U_{d,local}$ and $U_{r,local}$ are both bounded, node-local quantities in $[0, 1]$ before either is propagated.

3.2.4 Network Propagation

Declared urgency propagates downstream, from the final customer at rank 0 toward raw-material suppliers, while real urgency propagates upstream, from suppliers toward the final customer. Both use a leaky product rule that keeps the signals bounded in $[0, 1]$ and monotonically increasing as more neighbours become urgent:

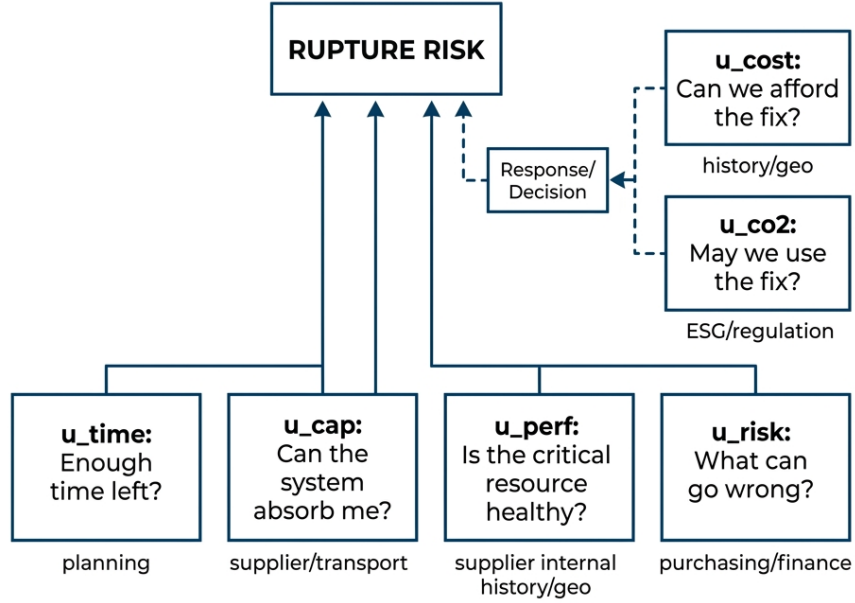


Figure 6: Causal structure behind the six-block selection

$$U_{d,i} = 1 - (1 - U_{d,\text{local},i}) \prod_{k \in \text{Succ}(i)} (1 - \gamma_{ik} U_{d,k}), \quad (6)$$

$$U_{r,i} = 1 - (1 - U_{r,\text{local},i}) \prod_{j \in \text{Pred}(i)} (1 - \beta_{ji} U_{r,j}), \quad (7)$$

where $\gamma_{ik} \in [0, 1]$ is the downstream attenuation coefficient on arc (i, k) and $\beta_{ji} \in [0, 1]$ the upstream amplification coefficient on arc (j, i) . Table 8 defines each term, and Figure 7 shows the two signals travelling in opposite directions across the same graph. Backup arcs are documentary and inert with respect to both propagations, a modelling choice that the campaign scenario exploits deliberately.

When each quantity is computed. The weekly cadence governs when new data *arrives*, not when the model runs, and the two are easy to conflate. The sequence is as follows. A review submitted for one node writes that node's KPI values and questionnaire answers. That write immediately triggers recomputation of the node's own u_m blocks and its two local quantities, $U_{r,\text{local}}$ and $U_{d,\text{local}}$. The new local values then propagate, but only along the directions that can be affected: a change in local real urgency is pushed through the node's downstream cone

Table 8: Terms of the propagation equations

Component	Type	Description
$U_{d,i}, U_{r,i}$	Dependent variables	Propagated declared and real urgency at node i
$U_{d,local,i}, U_{r,local,i}$	Input variables	Local urgency at node i after status-rule adjustment
γ_{ik}	System parameter	Downstream demand attenuation coefficient on arc $i \rightarrow k$
β_{ji}	System parameter	Upstream risk amplification coefficient on arc $j \rightarrow i$
$Succ(i), Pred(i)$	Structural sets	Nominal successors and predecessors of node i

toward the final customer, following Equation 7, and a change in local declared urgency is pushed through its upstream cone toward the suppliers, following Equation 6. Nodes outside those two cones are not recomputed, because their values cannot have changed.

Two consequences matter for reading the results. Nothing is computed on a global weekly timer, so the network is never in a state where all eight nodes were evaluated at the same instant; a node reviewed on Monday and a node reviewed on Thursday hold values one review apart, and the propagated figures reflect whatever the neighbours had most recently submitted. And the campaign turn, not the calendar week, is the unit that makes the state comparable across nodes, which is why every result in Section 4 is indexed by turn.

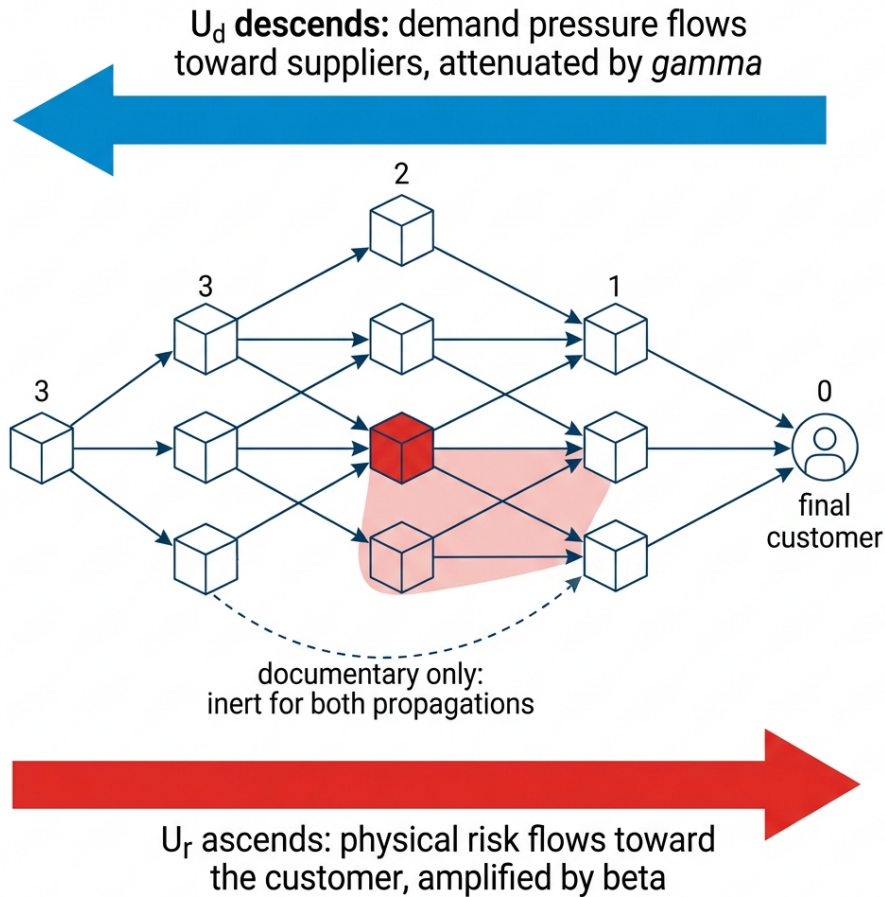


Figure 7: Bidirectional urgency propagation on the DAG

3.2.5 Network Criticality Index

The criticality service answers one question systematically: which node, if it failed completely, would cause the most damage to final-customer delivery? For every active node in turn, the service forces the node's local real urgency to 1.0, re-runs the ascending propagation, and measures two deltas, the rise of U_r at the final-customer nodes and the largest rise anywhere in the network. Every shock is run twice, once on the standard pipeline and once on an ε -regularised twin that carries the same propagation in log-survival, $\ell(p) = -\ln(1 - p)$, with urgency clipped to $1 - \varepsilon$ so that a survival product is never exactly zero. Nodes are ranked by the rise at the final customer first, then by the corresponding log-survival delta $\Delta\ell_{\text{final}}$, then by name. Figure 8 traces a single iteration of that sweep, from the forced saturation of one node to the deltas it produces downstream.

One behaviour of this index matters for interpretation, and it is what the second sort key exists for. A node already saturated at $U_r \geq 1.0$ yields a delta of zero and sinks to the bottom of the ranking, since its risk is already realised in the baseline. In a network-wide crisis, where many nodes saturate at once, ΔU_r collapses to zero throughout and the ranking degenerates exactly where it would be most useful; nine of the nineteen campaign turns are in that state, which is what the exclusion rule used by hypothesis H3 in Section 3.4 was written for.

Because ℓ is strictly increasing, $\Delta\ell_{\text{final}}$ and the standard delta at the final customer are two monotone transforms of the same shocked urgency, so outside saturation they induce the same order and the ranking is unchanged. This is stated as a property-based invariant and tested as one. Inside saturation the clip that flattens ΔU_r does not flatten $\Delta\ell$, which continues to measure how much deeper a shock drives an already saturated network. Section 4.9 reports what that recovers on the campaign.

A second entry point answers the distributional question rather than the worst case. Instead of forcing local urgency to 1.0, it draws shock severities from the graduated flat-rate severities of the calibrated event engine, minor, major and critical failures at 0.4, 0.7 and 1.0 and financial alerts at 0.3, 0.6 and 0.9, the two families weighted equally and, within a family, by decreasing field frequency. Each draw is applied as a ratchet, $\max(U_{r,\text{local}}, \text{severity})$, on the same principle as the event engine, since a failure never improves a node's state. The batch is propagated in one vectorised pass, in duplicate for both measures, and each node receives $P(\Delta U_{r,\text{final}} > 0.2)$ together with the median and ninth decile of its $\Delta\ell_{\text{final}}$ distribution. The computation is read-only, seeded and deterministic.

3.2.6 Asymmetric Cognitive Adequacy

The mismatch between declared and real urgency is split into two named pathologies, false urgency F for over-declaration and hidden risk H for under-declaration:

$$F = \max(U_d - U_r, 0), \quad H = \max(U_r - U_d, 0). \quad (8)$$

Drawing on Prospect Theory (Kahneman; Tversky, 1979), the two are penalised asymmetrically and with diminishing marginal sensitivity to large errors:

$$\text{penalty} = \lambda_{\text{under}} H^\alpha + \lambda_{\text{over}} F^\alpha, \quad (9)$$

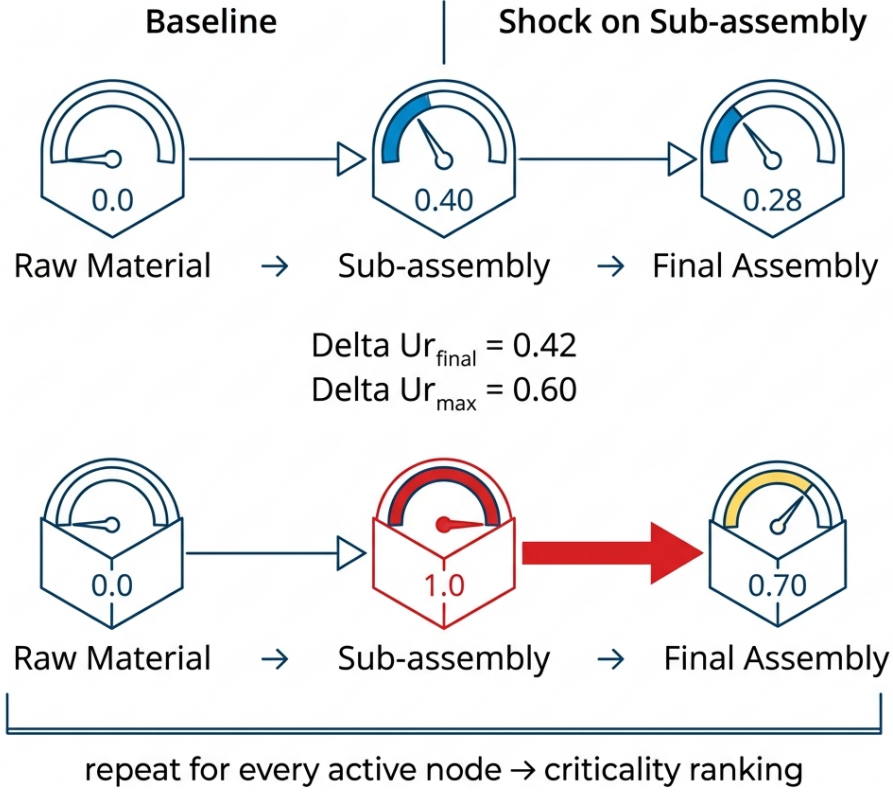


Figure 8: One iteration of the shock simulation

and the penalty is rescaled into an adequacy score $A \in [0, 100]$ through an exponential mapping bounded by the penalty of a maximal one-sided error, with each term defined in Table 9:

$$A = 100 \times \frac{\exp(-\text{penalty}) - \exp(-\text{penalty}_{\max})}{1 - \exp(-\text{penalty}_{\max})}, \quad \text{penalty}_{\max} = \max(\lambda_{\text{under}}, \lambda_{\text{over}}). \quad (10)$$

Table 9: Terms and parameters of the adequacy equations

Component	Type	Description	Default
F	Dependent variable	False urgency, from over-declaration	n/a
H	Dependent variable	Hidden risk, from under-declaration	n/a
λ_{under}	System parameter	Penalty weight on hidden risk	2.25
λ_{over}	System parameter	Penalty weight on false urgency	1.0
α	System parameter	Psychophysical curvature	0.88
A	Dependent variable	Adequacy score, 100 meaning perfect alignment	[0, 100]

The governing constraint is $\lambda_{\text{under}} > \lambda_{\text{over}}$: a coordinator who panics wastes premium transport capacity, but a coordinator who under-reports risk misses the window to prevent a line stoppage, so under-declaration is priced more expensively. The default ratio of 2.25 reprises the loss-aversion coefficient, and $\alpha = 0.88$ the curvature of the value function, both measured by Kahneman; Tversky (1979) and restated later (Kahneman, 2011). These are literature-derived starting points, not measured properties of supply chain coordinators. Both the aggregation

weights ω_m and the penalty parameters are configurable per project rather than hard-coded, so recalibration requires no code change.

3.3 System Implementation

The model above is deliberately silent on tooling. This section describes how each object it defines is elicited from an operator, computed inside the running system, exposed back to a coordinator and kept safe. The delivered system, SupplyScore, comprises 73 Python modules across 10 packages, covered by 1779 automated test functions in 106 test files, including property-based tests.

3.3.1 Architecture

Single facade. All operations go through one orchestrating facade (`services/orchestrator.py`) coordinating the mathematical core, the graph repository and the persistence layers behind a single thread-safe entry point. The web interface, the command-line entry point and the test suite all drive the system exclusively through this facade, and there is no second write path.

Multi-tenant persistence. Data is split between one global registry database holding projects, nodes, arcs, milestones and registry-level audit, and one SQLite database per node holding its private operational history. This isolation prevents write contention between operators reviewing different nodes and enforces the boundary that one supplier's raw history is not readable through another supplier's records. It is also what makes the campaign of Section 3.4 auditable per node after the fact.

Gatekeeper writes. Every business write crosses a single mutation service that whitelists the touched fields, validates ranges, diffs against current state so that a no-op write leaves no trace, records the change in the audit log with author and source, and synchronises the in-memory graph. Audit tables are immutable, including from the administration interface.

Graph and propagation. The network lives in an in-memory graph repository backed by `networkx`. Propagation is incremental, recomputing only the cones of nodes whose local urgency actually changed, which keeps the twin responsive at the scale validated by dedicated performance benchmarks. Figure 9 shows how the facade, the mutation gate and the two persistence layers relate.

Decoupled clocks. Each project carries its own clock: a system clock for live use, a game clock advancing week by week for serious-game sessions, and a fixed clock for deterministic tests. The weekly cycle, milestone pressure and event timestamps all read the project clock rather than the wall clock, which is what makes a compressed campaign possible without touching the model.

3.3.2 AHP Elicitation

The module `core/ahp.py` implements the Saaty eigenvector method for a four-criterion questionnaire administered through the weekly review page, and is the tool that produces $U_{d,local}$. The four criteria are operational impact, meaning the downstream consequence

Declared Urgency (Human AHP)

Computed Urgency (Digital Twin)

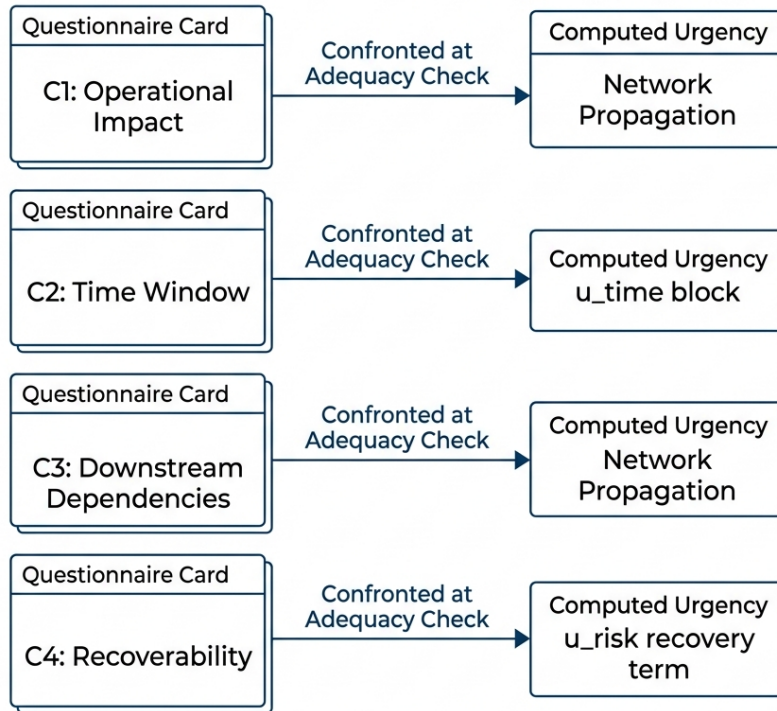


Figure 10: The four AHP criteria and their computed counterparts

Figure 11: The AHP questionnaire page

fails to converge, the implementation falls back to uniform weights rather than raising an error, so a single malformed elicitation cannot break the weekly review. Fuzzy weights are defuzzified with the Graded Mean Integration Representation,

$$w_j = \frac{a_j + 4b_j + c_j}{6}, \quad (11)$$

then renormalised to sum to one. The triangular bounds are written (a_j, b_j, c_j) rather than the more common (l_j, m_j, u_j) to avoid colliding with the risk blocks u_m .

PROMETHEE II ranks the active nodes of a project on the same six weighted urgency blocks that feed local real urgency. For each pair of active nodes and each block, the preference intensity is computed with a Type V linear preference function with an indifference threshold, and the aggregated preference index and net outranking flow follow the standard formulation:

$$\pi(a, b) = \sum_m \omega_m P_m(g_m(a) - g_m(b)), \quad \phi(a) = \phi^+(a) - \phi^-(a). \quad (12)$$

The same weights ω_m used in the local aggregation are reused as the PROMETHEE criteria weights, so that the network ranking and the per-node urgency score remain mutually consistent rather than relying on a separately calibrated weight set.

3.3.4 Weekly Review Cycle and Calibrated Event Engine

The weekly review cycle keeps the twin's KPI state current without free-form manual data entry: every write is structured, signed and auditable. At each new week every active node is flagged as due for review, and the operator completes four sequential sections: the AHP questionnaire, pre-filled with the previous week's values; the week's KPI values, shown against the previous week; progress on active milestones, compared against theoretical planning progress; and any supply chain events that occurred during the week. All four are written as a single signed, atomic transaction, so no partial or anonymous review can corrupt a node's history.

Rather than asking operators to recompute the KPI impact of a disruption by hand, the system exposes a taxonomy of fourteen calibrated event types, each mapped to one or more KPI updates through one of three calibration operators: a Beta-Bernoulli Bayesian update for failure probabilities,

$$p_{\text{post}} = \text{clip}\left(\frac{p \times n_0 + k}{n_0 + n}, 10^{-4}, 0.99\right), \quad (13)$$

with a prior weight $n_0 = 26$, roughly six months of weekly observations; an exponential moving average for continuous variables such as OEE, $x_{\text{new}} = (1 - \lambda)x_{\text{old}} + \lambda x_{\text{obs}}$ with $\lambda \in [0.3, 0.5]$; and a ratchet update for severity, $s_{\text{new}} = \max(s_{\text{old}}, s_{\text{event}})$. Figure 12 follows one declared event through to the KPI fields it revises, showing which of the three operators applies where.

As a worked example, a node declaring a strike lasting $d = 24$ hours and affecting a fraction $p_f = 0.50$ of the workforce implies $H_{\text{lost}} = d \times p_f = 12$ equivalent lost hours over a 168-hour

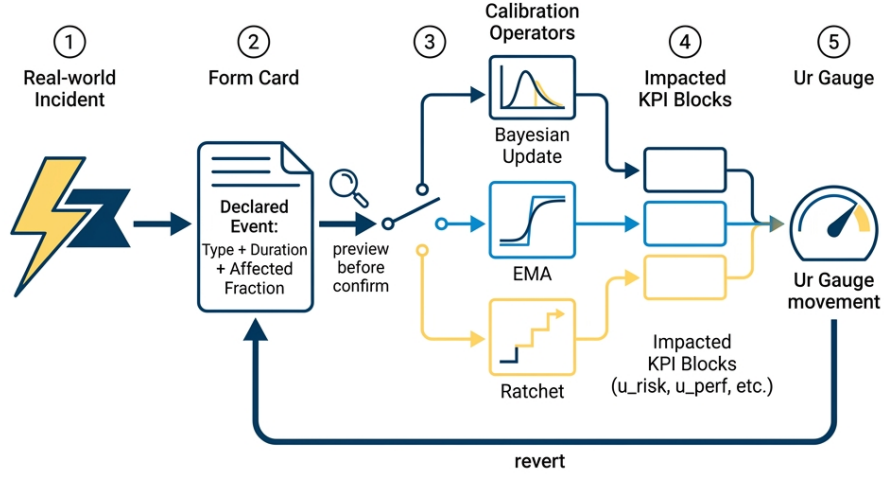


Figure 12: From a declared event to a calibrated KPI update

week, so $f_{\text{lost}} \approx 0.0714$ and the observed availability is $A_{\text{obs}} \approx 0.9286$. Updating a baseline availability of 0.95 with $\lambda = 0.5$ gives $A_{\text{new}} \approx 0.9393$, while risk severity updates in ratchet mode to $\max(0.30, 0.50) = 0.50$.

3.3.5 The Forecast Layer

The results of this thesis turn on one component, so it is specified here rather than left implicit behind the phrase used in the scoring protocol below. The forecast service projects every active node over one to four weeks from that node's own persisted weekly history, and emits the probability p_h that an adverse outcome occurs within the horizon.

Three quantities are estimated from the history and then simulated forward. Local urgency is extrapolated by a first-order autoregression fitted by least squares. The rate of adverse events is a Beta posterior, the same conjugate revision the event engine uses. The lead time is resolved to a distribution and resampled by parametric bootstrap. The rollout then draws a node's future trajectory, and p_h is the fraction of trajectories in which the outcome occurs.

The estimator is deliberately tied to the target. What p_h estimates is exactly the adverse outcome of the calibration service, a missed milestone or a non-reverted event of gravity critical or defect; the third condition of that definition, node abandonment, is not simulated, because only active nodes are projected and abandonment is an outcome to anticipate rather

than an input state. The forecast is therefore the prospective estimator of the quantity the calibration service measures retrospectively. Section 4.7 shows how much rests on that alignment.

Uncertainty is decomposed rather than lumped. The simulation is nested: $K = 20$ outer draws of the parameters themselves, the autoregression coefficient and residual scale through their least-squares standard errors, the event rate through its Beta posterior and the lead-time moments through the bootstrap, each carrying n/K inner trajectories. The variance of p_h splits into a Monte Carlo component within outer draws and a parametric component between them, and the reported interval is labelled by the sources it covers. This matters for the reliability analysis of Section 4.6, since an interval that hides which uncertainty it represents cannot be read.

Two disciplines are enforced in the code rather than by convention. The service never writes, so a forecast cannot perturb the state it forecasts. And counterfactual comparisons, which price a corrective action by rerunning two branches on common random numbers, are named throughout as an effect according to the model and never as a measured effect, because both branches are simulated and neither is an observation.

3.3.6 Explainability, Export and Operational Hardening

When the twin flags a node, the coordinator's first question is why. The per-node explanation page answers it with exact decompositions rather than approximations. Declared urgency splits into one additive contribution per AHP criterion, $\kappa_j = w_j(s_j - 1)/8$, summing exactly to $U_{d,\text{local}}$. Local real urgency splits the same way through the logarithm of Equation 4: each block contributes $\ell_m = -\omega_m \ln(1 - u_m)$, the shares $c_m = \ell_m / \sum_k \ell_k$ sum to one, and $U_{r,\text{local}} = 1 - \exp(-\sum_m \ell_m)$ reconstructs the score exactly. Figure 13 shows both decompositions side by side for a single node. This matters for the results reported later, because it is what allows a score to be traced back to the block that produced it without any manual computation.

One action compiles the global registry and per-node history into a twelve-worksheet Excel workbook, so that analyses can run outside the tool on frozen data. The persistence layer enables Write-Ahead Logging, runs an integrity check before every backup, and copies each database through the native `sqlite3.backup` API without locking the running application, with automatic backups every 24 hours and versioned schema migrations.

3.4 Experimental Design: The HÉLIOS Campaign

Everything above establishes that the model is implemented, internally consistent and tested. It does not establish whether the adequacy signal says anything true about an operator facing a crisis. This section presents the experiment built for that purpose.

3.4.1 Scenario and Network

The campaign, presented to participants under the cover story *Programme HÉLIOS*, replays the 2020 to 2022 semiconductor shortage over eighteen weekly game turns plus a training turn. The supply network has eight nodes, one per participant, anchored on the real actor types of the crisis (Table 10).

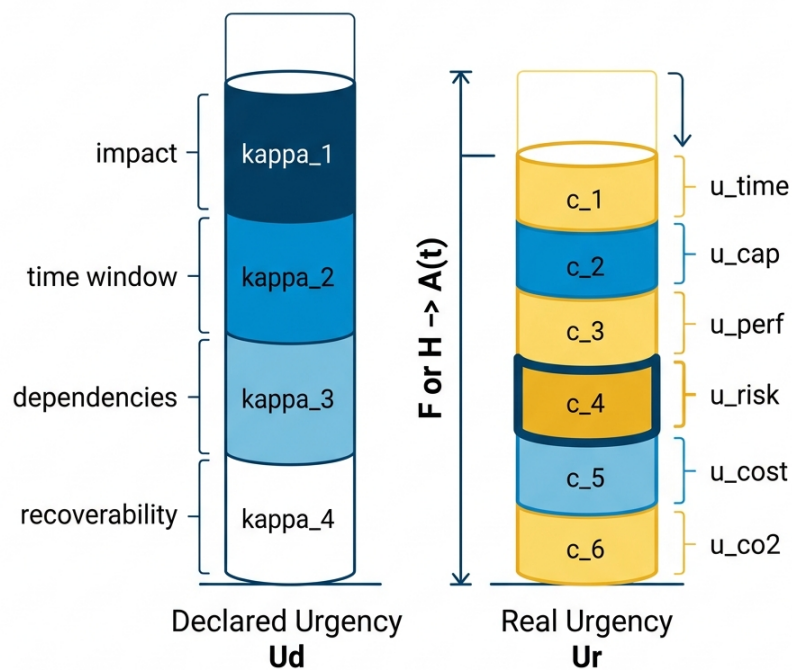


Figure 13: Per-node explainability decomposition

Table 10: The eight HÉLIOS campaign nodes

Node	Role	Rank	Real-world anchor
Orbitalys	Satellite prime, final customer	0	Fictional satellite programme, contractual delivery pressure
AvioSys Intégration	Avionics integrator	1	Qualified-stock squeeze on avionics tier-1 suppliers
Électis EMS	Electronics manufacturing services	2	EMS plants placed under allocation, 2021
CompoDis Europe	Component distributor	3	Distributor allocation and spot-market premiums
TransGlobal Fret	Freight operator	3	Suez Canal blockage, West Coast port congestion
NovaFab Semiconductors	Foundry, expected bottleneck	4	Composite of the Texas winter-storm freeze and the Taiwan drought
Meridian Semi	Secondary foundry, inert backup arc	4	A foundry sold out through 2023, the announced plan B that delivers nothing
SilPure Materials	Raw wafer supplier	5	Roughly 60% of world wafer supply, polysilicon price roughly tripling in 2021

Figure 14 gives the network with its per-arc coefficients, γ attenuating demand as it descends and β amplifying risk as it ascends, together with each node's tier rank. NovaFab at rank 4 is the expected bottleneck. The dashed arc into Meridian is the inert backup arc of Section 3.2.4: it appears in the diagram and carries nothing, which is the point of including it.

The turns follow five acts drawn from the documented chronology: a baseline where early warning signs go unheeded, a tipping point where a foundry freeze, a competitor fire and a canal blockage make the crisis visible, a squeeze where spot prices reach ten to forty times catalogue, a rupture where a major manufacturer cuts production by roughly 40% and the announced backup foundry turns out to have no spare capacity, and a slow, partial recovery that measures whether declared urgency lags the physical de-escalation. One game turn represents one real month, with real durations converted into engine hours at a fixed rate so that documented figures keep their true magnitude inside an eighteen-turn game.

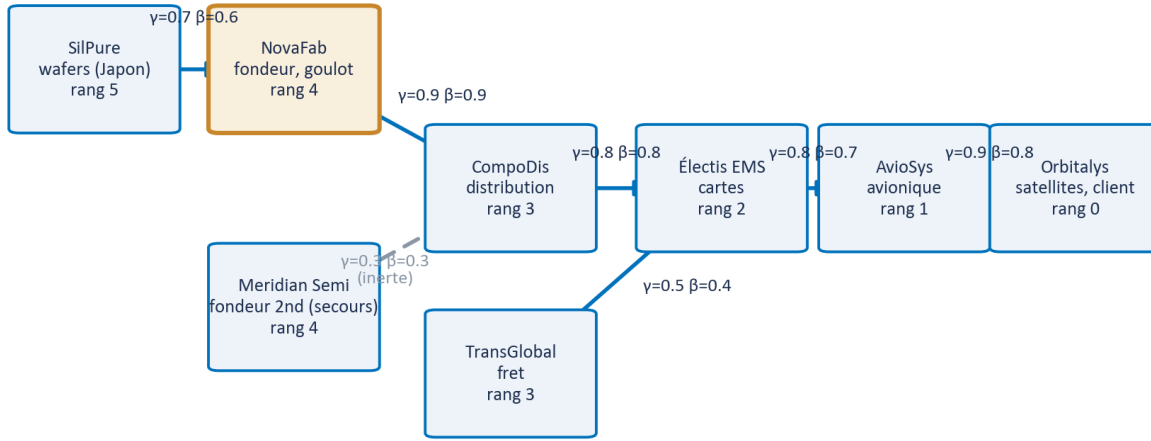


Figure 14: The HÉLIOS network and its per-arc coefficients

The twelve calibrated engine events driving the crisis are listed in Table 11, with the turn numbers as recorded in the campaign databases. One encoding choice is worth noting: the Texas freeze is modelled as a production accident, which triggers the engine's Bayesian revision of the failure probability, a recovery clock and a severity ratchet, rather than as a capacity loss, which would only reduce storage volume. A hot shutdown of a fabrication line is mechanically closer to the former.

3.4.2 Data Provenance

The two urgency signals come from different sources, which is the point of the design. Real urgency is fed from public time series of the actual crisis, namely INSEE production and price indices, WSTS worldwide semiconductor billings and a semiconductor-shortage dataset, frozen in a data pack whose SHA-256 hashes were recorded before the first turn. Declared urgency comes from the participants through the weekly AHP questionnaire, with the consistency gate active.

Every third-party dataset is passed through an authenticity check before use. A logistics dataset initially considered for the freight node was rejected at this check for the signatures of synthetic training data, namely pre-computed target columns, GPS coordinates scattered outside the

Table 11: The twelve calibrated engine events

Turn	Node	Event type	Real-world anchor
T3	NovaFab	Demand spike	Automotive reorders on sold-out slots, Nov. 2020
T5	CompoDis	Supplier delay	Distributor moves to allocation, Jan. 2021
T6	NovaFab	Accident (critical)	Texas winter-storm fab freeze, Feb. 2021
T7	NovaFab	Demand spike	Reorders after a competitor's fab fire, Mar. 2021
T7	TransGlobal	Transport disruption	Suez Canal blockage, six days, Mar. 2021
T10	NovaFab	Material shortage	Taiwan drought, ultrapure-water rationing, Jun. 2021
T12	Électis	Capacity loss	Southeast-Asia backend lockdowns, Aug. 2021
T12	TransGlobal	Transport disruption	Shift to saturated air freight, Aug. 2021
T13	CompoDis	Demand spike	Bullwhip over-ordering, Sep. 2021
T14	TransGlobal	Transport disruption	West Coast port congestion, Oct. 2021
T14	CompoDis	Tariff increase	Allocation spot premiums, Oct. 2021
T16	SilPure	Tariff increase	Polysilicon price pass-through, Dec. 2021

region the dataset claimed to cover, and uniformly flat distributions. The semiconductor-shortage dataset passed, showing a secular price decline, a genuine 2021 upturn and sectoral employment consistent with official series. The rejected dataset is retained in the raw-data folder for audit but is never loaded by the analysis pipeline, and the freight node's cost block is driven entirely by calibrated events instead.

Two provenance rules protect the real-urgency series. A single-pilot rule forbids double counting: each KPI field of each node is driven either by a time series or by a calibrated event, never by both, so when an event writes a field on a given turn the series skips that turn. Second, one honest artefact of the real data is kept rather than smoothed away. The INSEE business-failures series falls through 2020 and 2021 under public support schemes, so a failure-rate indicator taken alone would lower external risk exactly while the sector crisis was worsening. Keeping it makes the multi-block argument concrete: a single macroeconomic indicator can tell the opposite of the ground truth.

3.4.3 The Three Arms

The experiment compares three arms on the same frozen scenario. The comparison is what allows the effect of a treatment to be separated from the behaviour of the scenario.

Arm A, blind declaration. Eight isolated agents declare their urgency each turn seeing only their own role card and turn briefing. No forecast is shown. This is the control condition and the source of the prospective forecasts that are scored in Section 4.6.

Arm B, forecast-assisted declaration. The same agents, the same frozen scenario, but from turn 4 onward each agent is additionally shown the predictive analysis of its own node before declaring. The loop is open: nothing changes in the terrain, the scenario stays frozen, so the forecasts remain scorable and the effect of the treatment is measured on perception alone. Each declaration additionally records an `influence_prediction` field taking one of four values, meaning no influence, confirms my assessment, revised

upward, or revised downward. That field is the direct measure of the treatment effect.

Arm C, closed loop. The agents act on the terrain through a catalogue of interventions applied by the mutation service and recorded in an intervention journal, so the ground truth itself changes. Arm C was executed in its agent version after the repair of Section 4.9, over the same nineteen turns; because action destroys the reference truth, it is judged by comparing its final state with arm A rather than by scoring its forecasts.

The distinction between B and C is one of nature rather than degree. In arm B the ground truth is fixed, so the predictions are scorable with proper scoring rules and the treatment effect is measured on perception. In arm C, action destroys the reference truth, because a rupture avoided thanks to an alert makes a good prediction look wrong, so arm C can only be judged by comparing its final state with arm A. That comparison is reported in Section 4.9, and it is why the headline count of unfinished milestones is not the figure to read.

Two properties of this design are worth stating explicitly, because the results depend on them. First, the agents are language models, one per role, each confined to its own role card and turn briefings with no access to the other roles, the dashboard or future turns. This anti-lookahead discipline is enforced by the harness rather than by instruction. A single model instance per role has no statistical standing as a substitute for eight independent human judges, and Section 4.10 treats that as the principal threat to external validity. Second, the declaration format in arm B is identical to arm A, which is what makes the two comparable, with the single addition of the influence field.

3.4.4 Scoring Protocol

The forecast layer emits, for each node and each turn from turn 4 onward, a probability p_h that an adverse outcome occurs at horizons $h \in \{1, 2, 3, 4\}$ weeks. Scoring those forecasts requires an outcome definition, and this thesis uses the one the production calibration service itself applies, so that the model is judged against the target it was built for. A node-week is a positive observation when at least one of three conditions holds within the window $[S + 1, S + h]$: a milestone of the node whose deadline falls in the window is abandoned or is still active with its deadline passed; the node is abandoned; or an event of gravity critical or defect is recorded and not subsequently reverted.

Four quantities are computed, following standard forecast verification practice (Brier, 1950; Murphy, 1973; Gneiting et al., 2007):

- **Discrimination**, by the area under the ROC curve, computed by mean ranks so that tied forecasts are handled correctly, with a percentile bootstrap 95% interval over 2 000 resamples. AUC answers whether the forecast orders node-weeks correctly.
- **Calibration**, by the Brier score and its skill against the climatological forecast that always announces the base rate. A negative skill means the forecast is worse calibrated than announcing the base rate.
- **Reliability**, by a banded table comparing the mean announced probability with the observed frequency in each band. The bands are deliberately uneven, because a uniform ten-bin split hides the failure mode that matters here.
- **Operating point**, by the confusion matrix at the pre-registered threshold of 0.5 used by hypothesis H4.

Separating discrimination from calibration is essential rather than decorative (Murphy, 1973):

a forecast can order outcomes correctly while announcing the wrong probabilities, and the two failures call for different remedies. Section 4.6 finds exactly that dissociation.

One further check is built into the protocol. Because the outcome definition is a modelling choice rather than a given, the same forecasts are additionally scored against a deliberately naive target, namely any non-reverted event on the node regardless of gravity and ignoring milestones. The difference between the two scorings is reported in Section 4.7 and is treated as a result in its own right.

3.4.5 Protocol Integrity and Controls

The protocol was frozen before the first turn. Participants see only their own role sheet and questionnaire; dashboards and scores stay hidden until the debrief, and no discussion between roles is allowed. Two placebo sheets, a false de-escalation handed to one node at turn 9 and to another at turn 15, control for suggestibility: if a declared urgency follows the placebo, the measure is reflecting the sheet handed over rather than a reading of the situation. A weak-signal turn, where the press mentions a tension with no engine event behind it, measures who reacts to media noise alone. Roles are assigned so that the two most information-heavy nodes do not go to the most experienced participants, keeping a role effect from being confounded with an expertise effect. Every deviation from protocol is logged the same day in a campaign journal, and any analysis outside the pre-registered hypothesis register of Section 1.5 is labelled exploratory.

3.5 Limitations and Simplifications of the Method

Four simplifications are built into the method itself, as distinct from the limitations of the delivered framework catalogued in Section 5.3.

The six KPI blocks are treated as probabilistically independent so that the noisy-OR aggregation is admissible. They are not independent in reality: a capacity loss raises cost, and a performance drop lengthens lead time. The consequence is that correlated blocks double-count evidence, which the composite-indicator literature identifies as the standard failure mode of weighted aggregation (Greco et al., 2018; Becker et al., 2017). The explainability decomposition of Section 3.3.6 is the mitigation: the coordinator sees per-block contributions rather than only the aggregate, so a correlation is visible rather than hidden.

The network is assumed acyclic, which permits a linear topological sort and makes propagation deterministic and cheap. Real supply networks contain feedback, most obviously in returns and rework loops.

The scenario is scripted, which is what makes the ground truth knowable and the forecasts scorable, but it also means the events are the only source of adverse outcomes. The consequence, quantified in Section 4.6, is that the positive class is extremely small, and any conclusion about discrimination is correspondingly weak.

The declarants are language models. This buys perfect protocol compliance, exact reproducibility and no scheduling cost, and it loses the property that matters most for hypothesis H4, namely a genuinely imperfect human perception of a crisis. This is discussed at length in Section 4.10 rather than treated as a detail.

3.6 Reproducibility

Every numerical result in Section 4 is regenerated from raw artifacts by a single script, `Utilities/figures/derive.py`, which writes a JSON file that both the figures and the text of this thesis read. No number in this document is transcribed by hand from an intermediate report.

The inputs to that script are the two campaign databases, one per arm, and the raw per-agent declaration and forecast logs. The outcome labels are not reimplemented: the script calls the production `CalibrationService` directly, so the thesis scores the model against the same target definition the running system applies. Random procedures are seeded, specifically the bootstrap interval on the AUC and the Monte Carlo lead-time simulation.

Two provisions guard against silent drift. The script recomputes the arm A against arm B forecast comparison on every run and reports whether the two arms carry identical prediction-layer output, which they must, since the replay is deterministic and only the declarations differ. It also reports the outcome base rate at every horizon, so that a change in the underlying database would be visible immediately rather than absorbed into a summary statistic.

One earlier analysis file in the campaign folder, `analysis/rapport_experience.md` with its companion JSON, contains a superseded run whose participant-influence counts disagree with the raw logs. It is deliberately not used anywhere in this thesis, and the derivation script names it in an exclusion list so that the choice is recorded in the output rather than left implicit.

The satellite observation system of Section 4.11 is a separate codebase with its own verification script. Its network calls are cached on disk, so a re-run does not depend on the availability of the public services it draws from, and its calibrated constants are stated in the source with the sample size behind each.

3.7 Ethical Considerations

The research was conducted under the Cranfield University Research Ethics System. The study as executed involves no human participants: the eight declarants are language-model agents, and no personal data was collected, processed or stored at any point. The approval documentation is reproduced in Appendix A.

The protocol was originally designed for eight human consultants, and that version would have required participant consent for two specific features. The first is the cover story: participants would be told they are playing a fictional satellite programme without being told that the scenario replays a documented historical crisis, since knowing it would destroy the measure. The second is the two placebo sheets, which deliberately hand a participant false information. Both are standard incomplete-disclosure designs and both would be resolved by a full debrief at the end of the campaign, in which the real scenario is revealed and each participant is asked at which turn they recognised it. Were that campaign to be run, informed consent, the right to withdraw, and the debrief would need to be in place before the first turn.

Two further considerations apply to the work as it stands. The industrial data behind the scenario is public, namely INSEE and WSTS series and an open dataset, and the commercial material belonging to ALTEN is covered by the confidentiality statement at the front of this thesis. For the satellite observation work, all imagery and building data come from public

French open-data services under their published licences, no imagery of private individuals is collected or retained, and the analysis operates on industrial buildings rather than on people. The regulatory database used there serves only to calibrate a constant and is never used to infer a result about a specific site, which is a deliberate constraint rather than an incidental property.

3.8 Conclusion

The method combines a probabilistic multi-block urgency model, a structured elicitation of declared urgency, a bidirectional propagation over the supply network, and an asymmetric scoring of the mismatch between the two, implemented as a tested, auditable and persistent system. Around that system, a pre-registered three-arm experiment on a frozen historical scenario allows the resulting forecast to be scored against recorded outcomes with proper scoring rules, and allows the effect of showing that forecast to the declarant to be isolated from the behaviour of the scenario itself.

What this method can establish is bounded by its own design, and those bounds have been stated rather than left implicit. [Section 4](#) reports what it measured.

4 Results and Discussion

4.1 Introduction

This section reports what was measured, in six parts, and the order matters because the answer is not uniform across them. The delivered system and its mechanical validation come first, establishing that the pipeline runs end to end. The blind declaration arm follows, giving the pre-registered hypothesis verdicts. The adequacy signal is then scored on the decision it was built to support, choosing which nodes to attend to, where it performs well and early. The forecast layer built on top of that signal is scored next against recorded outcomes, which is where this thesis found its defect and, through it, a diagnosis of the mechanism behind it; the repair that followed is measured on the same frozen forecasts. The forecast-assisted arm measures what happens when a forecast is placed in front of the declarant. Finally, a short opening reports a separate system that addresses the freshness limitation the rest of this work documents about itself.

Two claims are worth separating in advance, because conflating them is the easiest way to misread what follows. That the gap between declared and computed urgency carries information about which supplier will fail is supported by the campaign. That the gap can be converted into a well-calibrated probability attached to a named week is not, yet. The distance between those two statements is the subject of Sections 4.6 and 4.9, and the sample that bounds both is stated wherever a figure is given.

Every number below is regenerated from raw artifacts by the derivation script described in Section 3.6.

4.2 The Delivered System

SupplyScore is delivered as a standalone Python application: 73 modules across 10 packages, covered by 1831 automated test functions in 106 test files, driving a multi-page Dash web interface over a multi-tenant SQLite persistence layer. The suite expands to 2 044 collected cases, of which 2 028 pass and 15 are skipped for want of an optional dependency; one fails on two missing image files referenced by a presentation script outside the library. All of the mathematical objects of Section 3.2 are implemented and exercised: the six-block noisy-OR aggregation, the AHP elicitation with its consistency gate, the Fuzzy BWM weighting, the bidirectional DAG propagation, the asymmetric adequacy score, the Monte Carlo lead-time estimate, the PROMETHEE II network outranking and the systematic shock-simulation criticality index.

Three operational capabilities matter for what follows rather than being peripheral. The per-node explainability page reconstructs any score term by term, which is what allowed the diagnosis in Section 4.6.4 to be made from the application rather than by hand. The Excel export produces frozen data for analysis outside the tool, which the campaign relies on. The hardened persistence layer, with write-ahead logging, pre-backup integrity checks and automatic snapshots, is what makes a campaign database an auditable record weeks after the run.

Two pages carry most of the operator's interaction with the system. Figure 15 shows the weekly review, whose four sections commit as one signed transaction so that no partial review

can enter a node's history. Figure 16 shows the ranking the shock simulation produces, which is the view a coordinator would use to decide where to look first.

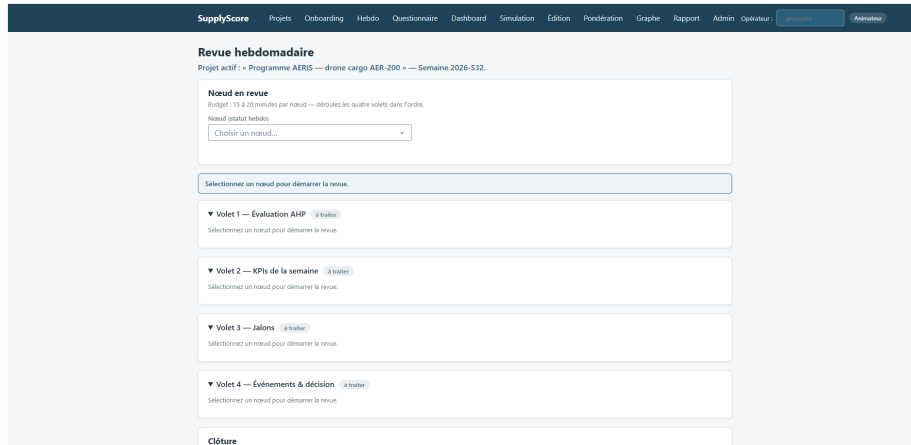


Figure 15: The weekly review page

4.3 Mechanical Validation: The Synthetic Dry Run

Before any declaration data existed, the full campaign mechanics were exercised end to end on a disposable database. Five iterations calibrated the scenario and eliminated engine artefacts, and the final iteration replayed all nineteen turns with synthetic respondents standing in for the eight declarants.

This is an engineering check, not a scientific one. The reason is in how the synthetic declarations were made: each one is anchored on the node's own local real urgency, following $1 + 8 \times U_{r,local}$ with a fixed panic, under-report or neutral bias per role. A declaration built that way cannot lag the signal it is derived from. It cannot under-report that signal for long, and it cannot react at all to a press event it never reads. Run against the pre-registered tests, it therefore returns not confirmed or not applicable on all six hypotheses by construction. That is the expected outcome, and an informative one.

What it does establish is that injection, weekly decay, milestone drift, event calibration, propagation, criticality, snapshotting and backup all run correctly under the two-turns-per-day cadence the campaign requires. It also calibrated the null baseline detector, a raw threshold $U_{r,local} \geq 0.5$, at precision 0.16 and recall 0.27: the bar the adequacy signal must clear to demonstrate value beyond a raw KPI reading.

Figure 17 shows the circularity visually. Real urgency is drawn solid and synthetic declared urgency dashed, for all eight nodes across the full replay. The two track each other closely at every node rather than one lagging or decoupling from the other, which is the signature of the construction and not a property the campaign is expected to reproduce. It is also why H1, H2, H5 and H6 cannot be read off this run: each of them asks about a divergence that the generator cannot produce.

4.4 Arm A: Blind Declaration by Eight Isolated Agents

Arm A replaced the synthetic generator with eight isolated language-model agents, one per role. Each was confined to its own role card and turn briefings, with no sight of the other

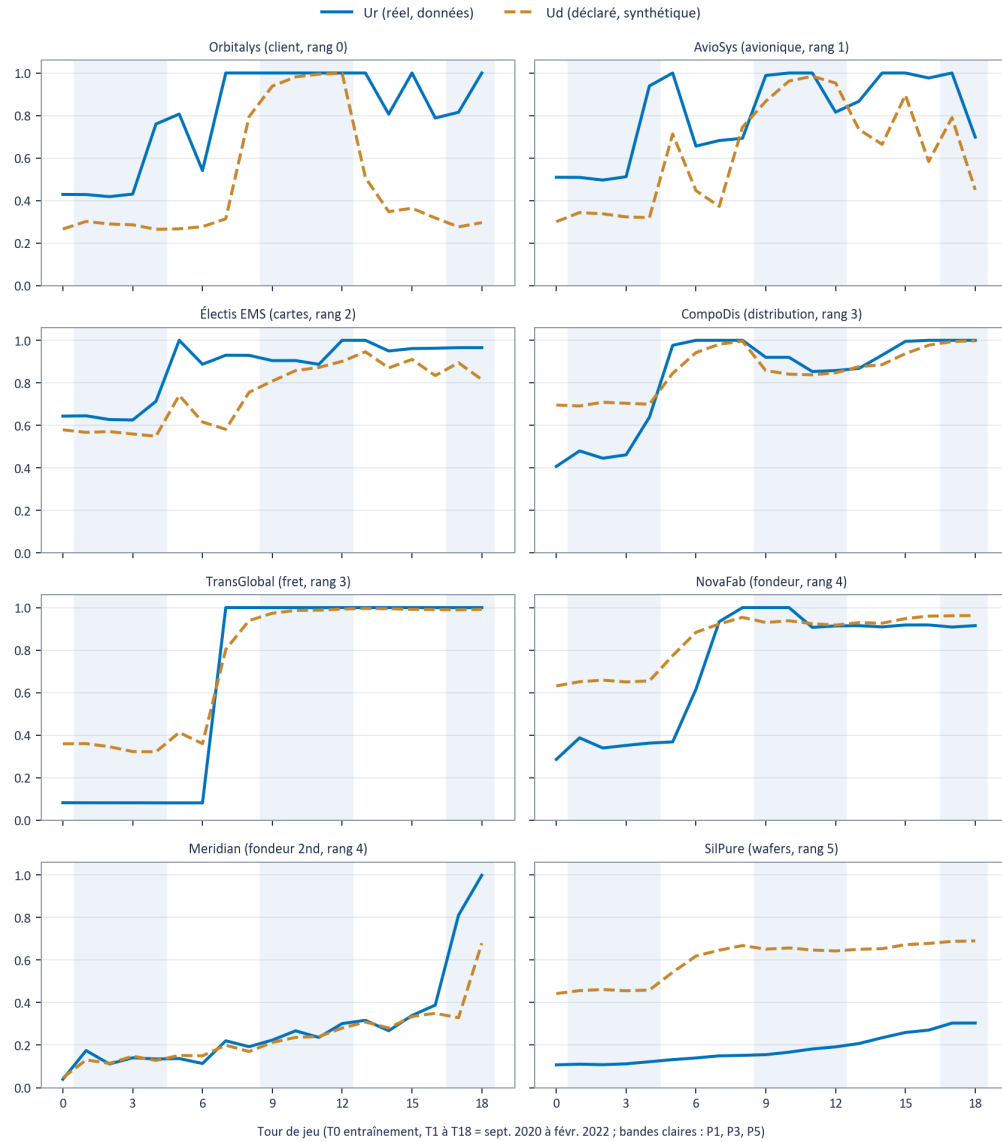


Figure 17: Real against synthetic declared urgency, dry run

roles, the dashboard, or any future turn. Each read its briefing in character, then answered the real AHP questionnaire of Section 3.3.2; pairwise comparisons and severity scores passed through the production consistency-ratio code untouched.

The difference from the dry run matters. This arm drove the full production pipeline. Campaign setup, turn injection and questionnaire submission all went through the same weekly-cycle service a browser-based participant would reach.

Table 12 gives the verdicts on the six pre-registered tests.

Two results stand out. The urgency-contagion hypothesis H5 is confirmed: declared urgency at nodes that are not physically affected rises by 0.079 on press-event turns against 0.005 on calm turns, matching the pre-registered direction and ordering. This is the mere-urgency effect (M. Zhu et al., 2018) reproduced at network scale. The latency hypothesis H1 moves from 1 of 8 nodes on the synthetic dry run to 3 of 8 at a lag of one turn or more, closer to but still short of the threshold of 5.

Table 12: Arm A verdicts on the six pre-registered hypotheses

Test	Verdict	Measured value
H1 latency	Not confirmed	3 of 8 nodes at lag ≥ 1 turn (threshold: 5 of 8)
H2 early detection	Not confirmed	$H(\text{NovaFab})$ peaks at 0.017 on turns 6 to 12 (threshold: 0.3)
H3 criticality	Not confirmed	NovaFab top-1 on 7 of 10 informative turns, 70% (threshold: 80%; saturated turns excluded per Section 3.2.5)
H4 predictive validity	Not confirmed	precision 0.00, recall 0.00 at $H \geq 0.5$
H5 contagion	Confirmed	non-affected nodes: $\Delta U_d = +0.079$ on press turns against $+0.005$ on calm turns
H6 hysteresis	Not applicable	real urgency shows no measurable decline over the recovery turns

H4 stays not confirmed for a specific and informative reason. These agents read their briefing carefully and declared urgency close to what the briefing in fact described, so their U_d tracks U_r too closely to leave the gap that H is built to catch. The tracking is even tighter than the fixed-bias dry run, with $H(\text{NovaFab})$ peaking at 0.017 against 0.070. H3 is unchanged from the dry run because it depends only on the network’s real-urgency criticality ranking, not on U_d at all.

The arm left behind a live project, reloadable in the same dashboard a coordinator would use in production. Read directly from that project after the eighteenth turn, the network-average adequacy is 65.5, four of the eight nodes carry a hidden-risk reading above 0.1, one carries a false-urgency reading above 0.1, and the worst-scoring node by a wide margin is Orbitalys at $A = 15.3$. Table 13 gives the full per-node snapshot.

Table 13: Final per-node snapshot of the arm A project

Node	U_d	U_r	A	F	H
Orbitalys	0.409	1.000	15.3	0.000	0.591
AvioSys Intégration	0.727	0.698	95.2	0.029	0.000
Électis EMS	0.914	0.965	82.9	0.000	0.052
CompoDis Europe	0.882	1.000	67.5	0.000	0.118
TransGlobal Fret	0.819	1.000	56.0	0.000	0.181
Meridian Semi	0.841	0.998	60.1	0.000	0.157
NovaFab Semiconductors	0.912	0.915	98.5	0.000	0.003
SilPure Materials	0.876	0.304	48.9	0.572	0.000

Orbitalys’ own explanation page traces the chain from a single scripted event to its score without manual computation. Its next milestone had slipped past its deadline by turn 18, which forces u_{time} to 1 under the rule of Table 6; with the node’s real urgency already saturated by this cause alone, the page attributes the whole of it to the local block and none to the upstream network. The agent playing Orbitalys declared $U_d = 0.41$, so the hidden risk is $H = 0.59$, penalised through Equation 9 at $2.25 \times 0.59^{0.88} = 1.42$, giving $A = 100 \times (e^{-1.42} - e^{-2.25}) / (1 - e^{-2.25}) = 15.3$, the same number the dashboard reports. The network-wide PROMETHEE II ranking places NovaFab and Orbitalys at the top of the priority list, at $\phi = +0.220$ and $+0.215$, which are the expected bottleneck and the node its own missed

milestone makes most urgent. The same ranking carries a caveat the tool raises on its own: on this run the time and performance blocks are perfectly correlated, $\rho = 1.00$, which is exactly the composite-indicator risk flagged in Section 2.8.

4.5 What the Adequacy Signal Gets Right: Node-Level Targeting

The scoring that follows in Section 4.6 asks whether the forecast layer attaches correct probabilities to node-weeks. That is the right question for a forecast and the wrong question for the instrument. A coordinator does not act on the proposition that node X has a 0.14 chance of an adverse outcome between weeks $t + 1$ and $t + 4$. They act on a shortlist: which of these suppliers do I look at this week. This subsection scores that, because it is the decision the adequacy signal was designed to support, and because the answer differs from the one below.

4.5.1 Protocol

At each turn the eight nodes are ranked by hidden risk $H = [U_r - U_d]_+$, and the set flagged at $H > 0.10$ is compared with the set of nodes that go on to miss at least one milestone at its originally committed deadline, the definition of Section 4.7. Four of the eight nodes are positive under that definition: the satellite prime, the avionics integrator, the electronics manufacturer and the foundry. The comparison is at node level and prospective, since the flag at turn t is confronted with misses occurring later in the campaign. Nothing in it uses the forecast layer.

4.5.2 Result

Table 14 gives the two regimes the campaign separates, and the boundary between them is the turn on which the scenario's foundry freeze becomes public.

Table 14: Nodes flagged by hidden risk against nodes that later miss a commitment, arm A

Regime	Turns	Precision	Recall	False alarms per turn
Before the crisis is public	T0 to T5	1.00	0.75	0
After	T6 to T18	≤ 0.50	0.25	1 to 3

For six consecutive turns the signal flags exactly three nodes, all three of which later miss a commitment, with no false alarm on any of the six. The fourth positive node is never flagged early, so recall stops at three of four. On every one of those turns the worst-scoring node of the eight is the satellite prime, which misses its turn-8 delivery and delivers at turn 14: the flag precedes the miss by eight turns, and it is raised while that node's own manager is declaring an urgency of 0.001.

That last clause is the construction working as specified rather than a coincidence. The flag does not come from a pessimistic declaration; it comes from an optimistic declaration standing against a computed state that is not. A system aggregating indicators alone would have ranked that node on its indicators. A system collecting declarations alone would have believed it. Only the difference between the two carries this information, which is the proposition the thesis set out to test and the one the literature review of Section 2.14 found unaddressed.

4.5.3 Why the Signal Degrades, and Why That Is Not a Defect

From turn 6 the precision falls to 0.50 and the recall to 0.25. The mechanism is structural. Once the shortage is public every declarant raises their urgency, U_d converges towards U_r , and a quantity defined as $[U_r - U_d]_+$ shrinks by construction. A perception-gap measure cannot survive the disappearance of the perception gap, and should not be expected to.

The consequence is a scope statement rather than a limitation to be repaired. The instrument is an early-warning device and not a crisis-management one; its useful window is the interval in which a discrepancy between what is declared and what is computed still exists, which is also the interval in which acting is still cheap. Read that way, the campaign shows the signal at its sharpest precisely where its value is highest.

4.5.4 What the Repair Changes Here

The repair of Section 4.9 sharpens this flag without altering which nodes it selects. At turn 3 the satellite prime scores 26.6 out of 100 on the campaign as run, against 46.0 for the next worst node; after repair the same turn gives 2.2 against 44.6. The ordering is identical, and the separation widens by an order of magnitude. The difference is one of legibility rather than of correctness: a node at 26.6 in a field whose next value is 46.0 is an anomaly a careful analyst notices, while a node at 2.2 in a field whose next value is 44.6 is one no dashboard can hide.

4.5.5 Statistical Standing

This result rests on eight nodes, four of them positive, over one scripted scenario. A precision of 1.00 sustained over six turns on three flagged nodes is an encouraging observation and not an estimated rate; with four positives, no confidence interval worth printing can be attached to it. It is reported here because it is the outcome the instrument was built to produce and because it points the same way as the mechanism, not because the sample settles anything. Section 5.4 states what would settle it, and the answer is more chains rather than a different model.

4.6 Arm A Rescored: Discrimination Without Calibration

The hypothesis verdicts above test the adequacy signal at a single threshold. The forecast layer that the system also emits, a probability of an adverse outcome at horizons of one to four weeks, deserves a fuller treatment, because a single operating point cannot distinguish a forecast that orders outcomes badly from one that orders them well but announces the wrong numbers. This subsection scores all 120 prospective forecasts of arm A, covering turns 4 to 18 across eight nodes, against the production outcome definition of Section 3.4.4.

4.6.1 Discrimination

Table 15 gives the results at each horizon.

The discrimination result is that there is none to speak of. The AUC ranges from 0.372 to 0.555 and every 95% interval contains 0.5, so on this campaign the forecast does not order node-weeks better than chance. That statement must be made with its sample size attached:

Table 15: The arm A forecast layer, scored by horizon

Horizon	Positives	Base rate	AUC [95% CI]	Brier	Brier skill
1 week	2	0.017	0.555 [0.470, 0.639]	0.130	−6.93
2 weeks	4	0.033	0.372 [0.258, 0.492]	0.166	−4.14
3 weeks	5	0.042	0.428 [0.315, 0.546]	0.191	−3.77
4 weeks	6	0.050	0.485 [0.389, 0.583]	0.207	−3.36

the positive class contains between 2 and 6 observations depending on horizon, which is far too few to support a confident claim in either direction. What can be said is that the campaign provides no evidence of discrimination, not that discrimination has been refuted.

4.6.2 Calibration

The calibration result is much stronger, because it does not depend on the small positive class in the same way. The Brier skill against climatology is between −3.36 and −6.93, meaning the forecast is between four and eight times worse, in mean squared error, than a trivial forecaster that announces the base rate every time. A negative skill of that magnitude is not a marginal shortfall.

The reliability table shows where it comes from. Table 16 bands the four-week forecasts by announced probability and compares the mean announced value with the observed frequency in each band.

Table 16: Reliability of the arm A four-week forecasts

Announced band	<i>n</i>	Mean announced	Observed	Reading
[0.0, 0.1)	74	0.070	0.081	Well calibrated
[0.1, 0.2)	24	0.128	0.000	None materialised
[0.2, 0.5)	1	0.354	0.000	None materialised
[0.5, 0.9)	7	0.859	0.000	None materialised
[0.9, 1.0]	14	0.992	0.000	None materialised

The pattern is unambiguous, and Figure 18 shows it, with each marker's area proportional to the number of forecasts in its band and the diagonal marking perfect calibration. Where the model announces a low probability, it is right: 74 forecasts announcing a mean of 0.070 were followed by an adverse outcome 8.1% of the time. Where it announces a high probability, it is wrong every single time. Twenty-one forecasts announced a mean probability of 0.948, and none of the twenty-one was followed by the outcome it predicted under the production definition. The forecast's own confusion matrix at the threshold of 0.5 records this as 0 true positives, 21 false positives, 6 false negatives and 93 true negatives.

Hypothesis H4 is stated on a different quantity and should not be read off that matrix: it tests the hidden risk H , not the four-week forecast, at its own threshold of $H \geq 0.5$. Measured

on the campaign database through the production calibration service, that second matrix reads 0 true positives, 3 false positives, 8 false negatives and 141 true negatives over 152 node-weeks at four weeks. The two signals fail at the same threshold and for related reasons, and both return precision and recall of exactly zero, but they are separate measurements and the H4 verdict rests on the second.

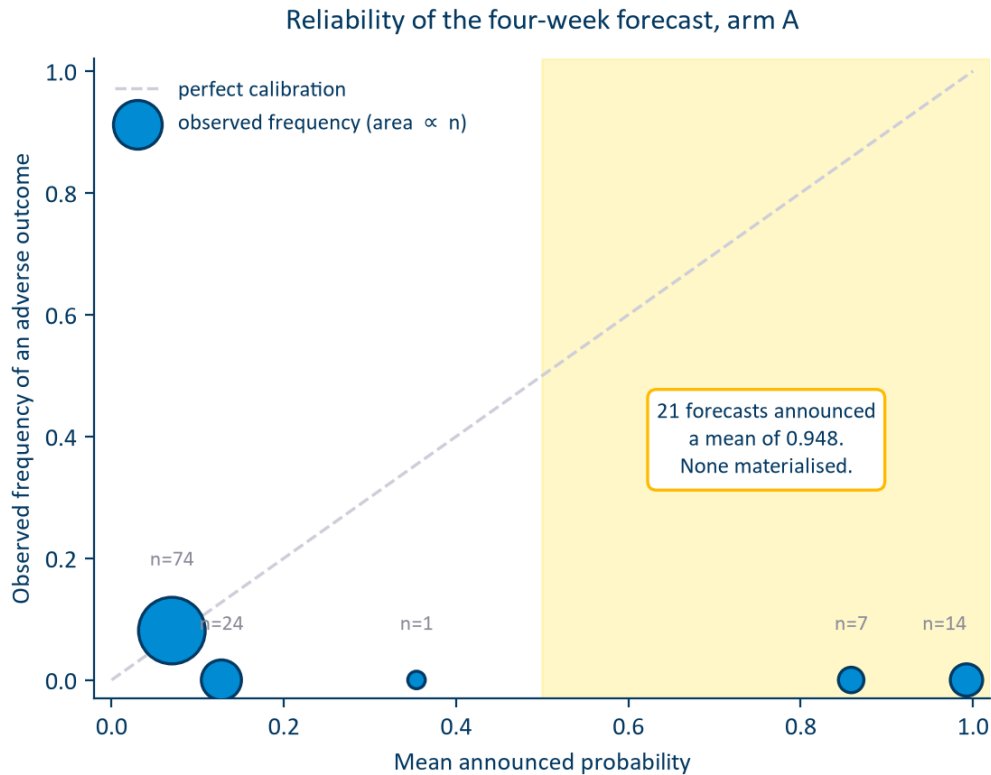


Figure 18: Reliability of the four-week forecast, arm A

The two failures are therefore of different kinds, and this is the dissociation that Murphy's decomposition anticipates (Murphy, 1973): the ordering is uninformative, and the level is badly wrong at the top of the range while being close to right at the bottom.

4.6.3 Ruling Out Right-Censoring

An obvious objection is that forecasts made near the end of the campaign have observation windows extending past the last turn, so their outcomes could not be recorded even if they occurred. If the confident forecasts were concentrated at the end, censoring rather than miscalibration would explain the result.

They are not. The twenty-one confident forecasts are spread over turns 5 to 18 and across seven of the eight nodes. Repeating the scoring with every forecast whose window would run past the final turn removed leaves $n = 88$ forecasts, still 0 true positives against 16 false positives, and the Brier skill becomes worse rather than better, moving from -3.36 to -7.82 . Censoring is not the explanation.

4.6.4 The Mechanism

The confident tail has a single source, and the explainability layer of Section 3.3.6 makes it visible without instrumentation. Across all twenty-one confident forecasts, the four-week probability and the milestone-miss probability never differ by more than 0.020. The confident tail is the milestone channel, and nothing else contributes to it.

Set against the registry, this is decisive. Of the eighteen milestones in the campaign, ten completed and eight remained active, and none was abandoned. Exactly one was still open past a deadline that had fallen inside the campaign, the Orbitalys delivery of Table 13. The model announced near-certain misses for AvioSys, CompoDis, Électis, NovaFab, TransGlobal and Meridian, and all of them delivered.

That registry reading is more forgiving than it first appears, and the qualification matters enough to state here rather than in a footnote. The registry stores each milestone's *current* deadline, and a replanning overwrites it, so a milestone that is re-dated and subsequently delivered is recorded as completed. Read instead against the deadline each node originally committed to, six of the eleven milestones whose original deadline fell inside the campaign were missed: five were delivered one to six turns late, and one was never delivered at all. Under that stricter reading, five of the twenty-one confident forecasts were followed by a missed original commitment within four weeks rather than none of them. The confident tail remains wrong far more often than right and the mechanism below is unaffected, but the distance between zero of twenty-one and five of twenty-one is produced entirely by which deadline the truth is read against, which is the subject of Section 4.7.

The same defect is visible directly in the state snapshots, from the opposite side. Reading the u_{time} block at turn 6, the turn on which the scenario's one critical event lands, gives the values of Table 17.

Table 17: The time block at turn 6

Node	u_{time}
Orbitalys	7.3×10^{-9}
AvioSys Intégration	1.2×10^{-8}
Électis EMS	0.0
CompoDis Europe	0.99996
TransGlobal Fret	1.7×10^{-4}
Meridian Semi	2.9×10^{-9}
NovaFab Semiconductors	0.0216
SilPure Materials	0.0

Six of the eight nodes read as numerically zero in the middle of a crisis, and the seventh reads as saturated. The block behaves as a step function rather than as a risk gradient: it is either off or fully on, with almost nothing in between. The cause is structural. The block compares a lead-time distribution against the time remaining to the next deadline, and the scripted events act on capacity, cost and failure probability without acting on any quantity on the time axis, so a shock never moves the quantity that the milestone probability is computed from. The strongest form of the evidence is a count that depends on no scoring choice at all: across the whole campaign, eight nodes and 41 indicator snapshots each, the accumulated-delay field the block would need was written a non-zero value exactly zero times. The block is blind

to the crisis by construction, and because that block drives the entire confident tail, so is the forecast.

This is a defect in the model rather than a property of the scenario, and it is the single most actionable finding of the campaign. It also invalidates, for this run, the assumption recorded in Section 3.2.2 that each block contributes useful information to the aggregate. One block did not.

4.7 The Target-Definition Trap

The scores above depend on a choice that is easy to overlook: what counts as an adverse outcome. Section 3.4.4 uses the production definition, requiring a missed milestone, an abandoned node, or an event of gravity critical or defect. Two alternatives are worth scoring the identical forecasts against. The first is the one an evaluator without access to the model's own definition would most likely reach for: any event recorded on the node. The second keeps the production definition of an adverse outcome intact but reads a missed milestone against the deadline the node originally committed to rather than against whatever deadline it currently carries, and drops the forecasts whose observation window runs past the last campaign turn.

Table 18 gives the three.

Table 18: The same forecasts under three outcome definitions

Outcome definition (4-week horizon)	<i>n</i>
Production definition (missed milestone, abandonment, or critical event)	120
Naive definition (any recorded event)	75
Original-commitment definition (production outcome, deadline as first committed, censored points excluded)	90

The model is identical, the forecasts are identical, and the horizon is identical. Only the definition of truth changes. Between the first two rows the measured base rate moves from 5.0% to 38.7% and the Brier skill improves by a factor of more than five, from -3.36 to -0.64 . An analyst choosing the easier target would report a system that appears five times better calibrated without a single line of the model having changed.

The third row moves the answer in a different direction and for a defensible reason rather than a convenient one. Holding the definition of an adverse outcome fixed and only changing which deadline a milestone is judged against carries the Brier skill from -3.36 to -0.55 and the AUC from 0.485, indistinguishable from chance, to 0.654. The argument for preferring it is about what a forecast is for rather than about statistics: a replanning is itself the formal admission of a slip, so a truth in which re-dating erases the miss scores the forecast against an outcome it was built to warn about before the re-dating happened. It is the definition used for the re-measurement of Section 4.9, and stating which of the three any figure refers to is not pedantry: the three disagree by more than any modelling decision taken in this work.

Discrimination, by contrast, is almost untouched, moving from 0.485 to 0.484 (Figure 19). That contrast is itself informative: the ordering of node-weeks is a property of the forecast, while the apparent quality of the probabilities is substantially a property of how often the target

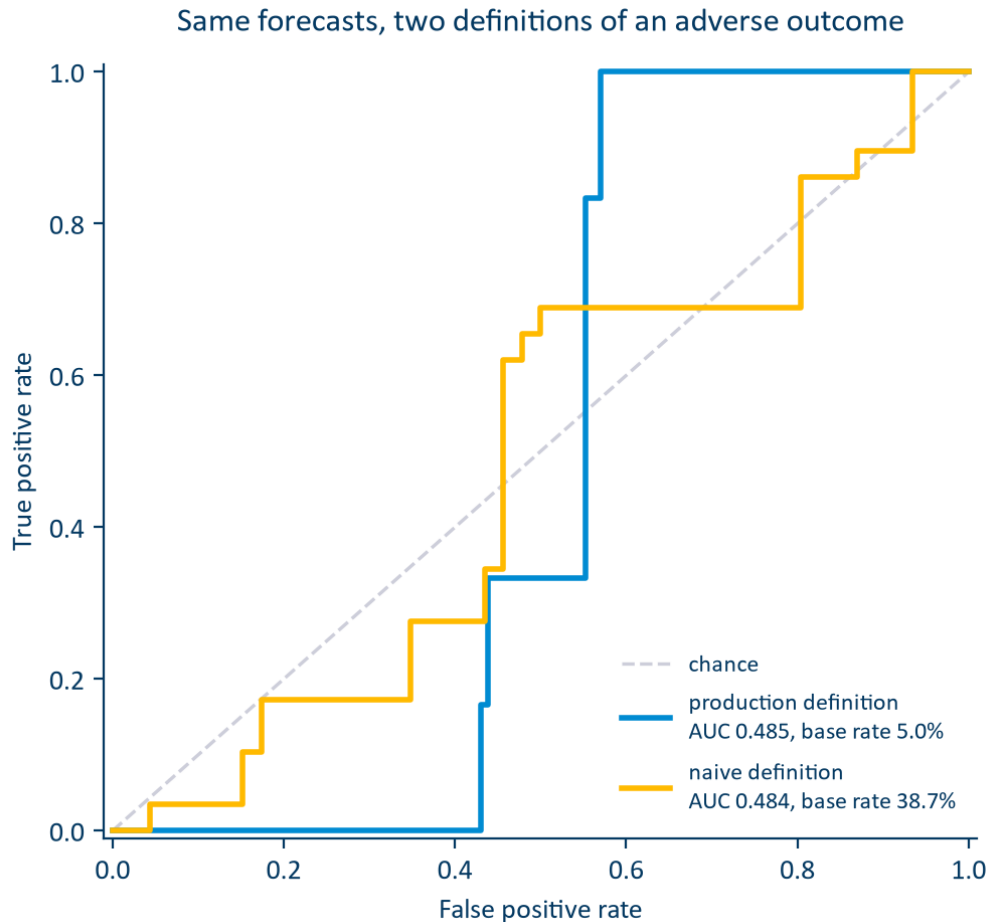


Figure 19: ROC under both outcome definitions

fires. A frequent target flatters a forecaster who announces high probabilities, because more of those high announcements happen to land on a positive.

The practical consequence is that a calibration figure reported without its outcome definition is close to meaningless. This has a direct bearing on how the framework of this thesis should be evaluated in future work: the outcome definition belongs in the pre-registration alongside the thresholds, and it is not currently part of the H1 to H6 register.

4.8 Arm B: What Happens When the Forecast Is Shown

Arm B replays the identical frozen scenario with the identical eight agents, changing one thing: from turn 4 onward, each agent is shown the predictive analysis of its own node before declaring its urgency. The loop stays open, so the terrain does not change and the forecasts remain scorable.

One property of the design must be established before any interpretation. The prediction layer is deterministic, so the two arms should carry identical forecasts on the turns they share, and they do: all 72 shared node-turn forecasts are numerically identical between the arms. Any behavioural difference therefore comes from the act of displaying the forecast, not from a different forecast being displayed.

The dataset comprises 97 declaration records, reduced to 96 after removing one duplicate submission by a single node at turn 1. Each carries an influence field recording whether the displayed forecast had no influence, confirmed the agent’s own assessment, or caused an upward or downward revision. Table 19 gives the distribution turn by turn.

Table 19: Arm B: influence of the displayed forecast, by turn

Turn	No influence	Confirms	Revised up	Revised down	Note
T0	8	0	0	0	forecast hidden
T1	8	0	0	0	forecast hidden
T2	8	0	0	0	forecast hidden
T3	8	0	0	0	forecast hidden
T4	0	3	0	5	first exposure
T5	0	5	3	0	first upward revisions
T6	1	3	2	2	critical accident event
T7	0	4	3	1	
T8	1	1	3	3	
T9	2	5	1	0	
T10	2	3	2	1	
T11	0	3	1	4	
Total	38	27	15	16	$n = 96$

The first exposure is the result, and Figure 20 shows it. At turn 4, when the forecast becomes visible for the first time, all eight agents record an influence, and the direction is entirely one-sided: five revised their declared urgency downward and three had their existing assessment confirmed. Not one revised upward. The first upward revision anywhere in the campaign appears at turn 5, and the scenario’s single critical event, the foundry accident, lands at turn 6, two turns after the moment every declarant was either reassured or confirmed.

Over the full campaign the distribution is close to balanced, with 15 upward against 16 downward revisions, so the one-sided effect is specific to first contact rather than a persistent bias. That distinction matters for interpretation: what the data shows is a strong first-exposure effect, not a permanent suppression of declared urgency.

Two aggravating factors follow from Section 4.6. First, the forecast being displayed at turn 4 is drawn from the same layer whose confident predictions never materialise, and whose u_{time} block is blind to the very shock that is about to arrive. Second, the forecast is displayed as a numeric probability with a confidence interval, a presentation format the automation-bias literature identifies as increasing reliance (Goddard et al., 2012; Bansal et al., 2021). The system was, in effect, presenting an unfounded reassurance in the most persuasive available form, immediately before the event it had failed to anticipate.

One qualification is important and cuts against the strongest reading. The influence field records the effect on the declared urgency level, not on the decision taken. Inspection of the free-text notes shows that agents continued to take protective actions at the same turn, including negotiating allocation, alerting the customer and ordering ahead of schedule. The measured effect is on what was declared, not on what was done. Given that the declared value is precisely the input the adequacy layer consumes, this is a consequential effect on the measurement system even where it is not, on this evidence, an effect on behaviour.

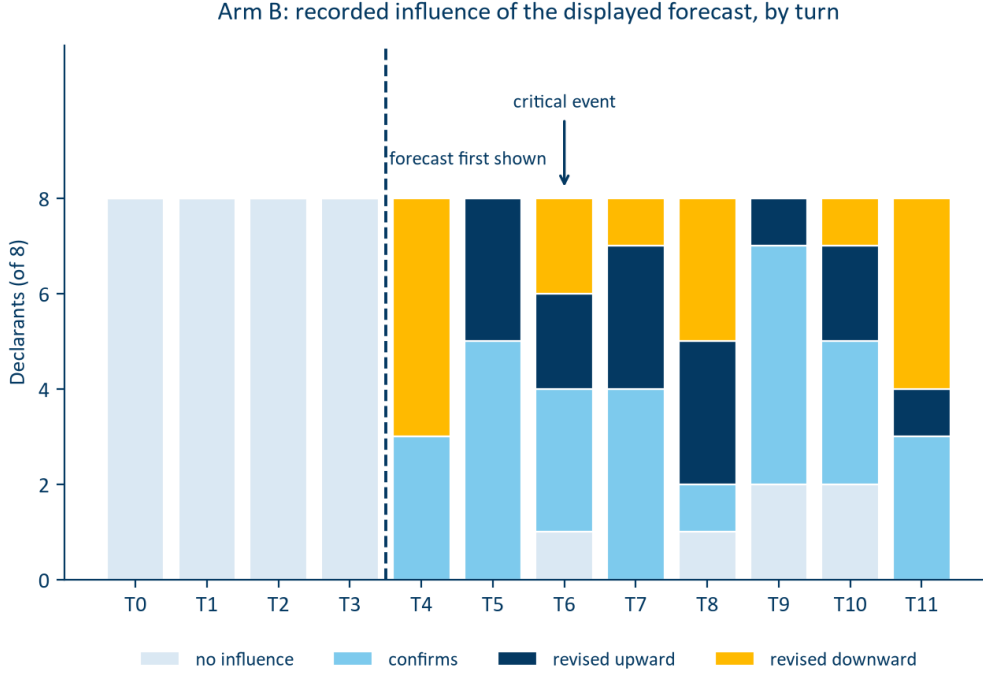


Figure 20: Arm B: influence of the displayed forecast, by turn

4.9 The Repair and Its Re-measurement

The campaign identified one defect and specified one repair. The repair was then carried out and re-measured against the identical frozen campaign, and this subsection reports that second measurement. The declarations, the scenario and the event script are unchanged; the same 120 forecasts are recomputed by the modified model, so every difference reported below belongs to the model and to nothing else.

4.9.1 The Repair That Was Recommended Does Not Work

Section 4.10 proposed making the calibrated events write a lead-time impact alongside their existing effects. Implemented and measured, that repair fails, and the reason generalises beyond this system. A completion date estimated as a full production cycle beginning now multiplies the lead time by the fraction of work remaining, so any quantity routed into the lead time is multiplied by that fraction too. A shock therefore attenuates in proportion to how close a milestone is to completion, and vanishes altogether on the milestones nearest their deadline, which are precisely those on which a warning has value. On NovaFab, the node carrying the scenario's accident, the four-week probability fell from 0.846 to 0.000 on a milestone that was in fact missed.

Lost production time is additive, not proportional: a four-week freeze costs four weeks whether the order is a tenth or nine tenths complete. The completion model was rewritten accordingly, identically in the three independent estimators the system carries for the same quantity, the analytic u_{time} block, the Monte Carlo forecast rollout and the PERT lead-time simulator:

$$C = t + (1 - p)L + R, \quad (14)$$

where p is the declared progress, L the lead time and R the time already lost. Five families of

capacity-halting event, machine failure, accident, strike weighted by the share of workforce involved, material shortage weighted by criticality, and cyber incident, now accumulate their effective duration into R . Loss of storage capacity deliberately does not, since it removes space rather than production hours.

The mechanical effect is unambiguous. Across the whole of arm A, eight nodes and 41 indicator snapshots each, the accumulated-delay field was written a non-zero value exactly zero times; this is the strongest available form of the blindness claim of Section 4.6.4, since it does not depend on any scoring choice. After the repair, NovaFab accumulates 154.7 engine hours at its first shock and 371.2 over the campaign, carried in 27 of its 41 snapshots.

4.9.2 Two Further Defects Exposed by the First

With the time block responsive, a second defect became measurable. The milestone-miss probability returned exactly 0 or exactly 1 in 85.0% of the 120 forecasts, leaving 14 anywhere in between. That is a switch rather than a probability, and two declarants recorded as much in free text before it was measured. The cause is that the rollout drew the lead time at random while taking the declared progress as an exact value. Declared progress is the least reliable input the system receives, and it belongs to precisely the class of quantity the framework exists to doubt: the adequacy score is built on the premise that self-reported figures run optimistic, and the forecast layer was then treating one of them as certain. Progress is now drawn per replicate, with a standard deviation widened node by node in proportion to that node's own measured hidden risk H , so the system's adequacy measurement feeds its own forecast.

The third defect was in the data rather than the model, and it would have invalidated any calibration performed before it was found. Indicator coverage on the campaign database was 25.6%: 24 of the 42 indicators were empty on all eight nodes, the carbon and cost blocks were inert everywhere, and only three of the six urgency blocks contributed. The lead time itself was present on five nodes of eight, and the remaining three received a risk floor of zero through the documented fallback for a missing lead time, which is an absence of data presenting itself as a reassuring prediction. Completing the indicators from public figures of the 2020 to 2022 shortage raised coverage to 49.1% and made all six blocks active. One model bug was undetectable until then: a per-node recovery capability, a 26-week aerospace requalification time, was being read as production time already lost, pinning that node's time block at 1.0 from the first turn. An empty database conceals the errors of the model that consumes it, which is a methodological point independent of this framework.

4.9.3 The Re-measurement

Table 20 scores the recomputed forecasts against the original-commitment definition of Section 4.7, at all four horizons, on the same censored samples of 112, 104, 96 and 90 observations. The middle column of each pair isolates the structural repair; the right-hand column adds the progress-uncertainty term.

The structural repair improves both axes at every horizon. The Brier deficit is roughly halved and the area under the ROC curve rises, and neither is traded for the other, which is the clean result of this thesis's second measurement campaign. The progress-uncertainty term is a trade rather than a gain: it improves calibration at two weeks and beyond, degrades it slightly

Table 20: The frozen campaign rescored after repair, original-commitment target

Horizon	Brier skill			AUC		
	before	structure	+ uncertainty	before	structure	+ uncertainty
1 week	−1.278	−0.646	−0.687	0.736	0.793	0.729
2 weeks	−0.690	−0.401	−0.367	0.636	0.671	0.634
3 weeks	−0.654	−0.360	−0.286	0.641	0.657	0.602
4 weeks	−0.553	−0.272	−0.216	0.654	0.721	0.584

at one week, and lowers the AUC. That last movement is not interpretable. With 26 positive observations the cluster-bootstrap interval on the four-week AUC runs from 0.21 to 0.87 and contains every value in its row, and the same objection that bounds the original discrimination claim of Section 4.6 bounds this one. The term was retained on the calibration evidence and on the argument that a declared quantity should not enter a probabilistic forecast as a certainty, not on the AUC.

Three qualifications belong beside the table. The Brier skill remains negative at every horizon, so the forecast is still worse calibrated than a forecaster that announces the base rate and stops; the repair moves the system from badly wrong to less wrong, not to useful. The milestone probability remains binary in 71.7% of cases against 85.0% before, so the over-confidence is reduced rather than removed. And the post-hoc calibration layer that Section 4.10 recommended was trained, measured, and deliberately not connected: on the synthetic factory that produced its training data it improves the Brier skill by 30.8% and the AUC from 0.792 to 0.850, but applied to the campaign it improves calibration at every horizon while dropping the four-week AUC from 0.721 to 0.580. Its training label is an event striking the node at exactly $t + 1$; the product displays a missed milestone or a critical event cumulated over $(t, t + 4]$. The layer cannot rank the cases driven by the milestone channel because it has never seen one. This is the trap of Section 4.7 encountered from the opposite direction: a calibration fitted against the wrong target degrades a system that works, exactly as an evaluation against the wrong target can condemn one. Connecting the layer requires the training label to be rebuilt on the product’s target, which requires the synthetic factory’s snapshots to record milestone truth, which they do not; that is the next concrete step rather than an open research question.

4.9.4 A Third Arm, and What It Does Not Show

The repaired system was then exercised through a full nineteen-turn campaign in closed loop, in which the same eight agents no longer only declared but also selected actions from a fixed catalogue, applied to the chain by the mutation service and recorded in the intervention journal: promote a backup arc, replan a milestone, expedite a shipment, raise capacity, revise a declaration, or do nothing. One hundred and fifty-two declarations were recorded with no unresolved consistency failure. The anti-contamination protocol of Section 3.4.5 was tightened for this arm: every message reaching an agent is written by the tool into that agent’s own sandbox, and a failed consistency check is answered with the interface’s own guidance rather than with a coordinator’s instruction.

The headline figure flatters the framework and should not be used. Six of eleven milestones

were missed in the frozen scenario and none was left unfinished in the closed loop, which reads as six saved. Against the original commitments it is not: five of eleven milestones were met on their original date in the frozen run, and five of eleven in the closed loop. The loop changed which milestone was met, not how many. One was genuinely saved, an electronics series delivered on its original date that had slipped in the frozen truth. Four late deliveries became replannings announced in advance rather than slips discovered at the deadline, which has value to a customer without being a save. One milestone was replanned that would have been met, which is the measurable cost of over-reaction. With eleven milestones neither the benefit nor the cost is statistically established, and both are reported as facts of one campaign rather than as results.

Two observations from that arm do carry. The arm B disarming reproduced exactly, eight agents of eight at first exposure. And the causal chain the campaign showed to be broken now closes: where arm A's foundry manager recorded that the tool was holding a low position through a widely reported crisis and took no action, the repaired run moved the same node's four-week probability from 0.088 to 0.214 at the shock turn and the agent replanned its milestone. One reserve on internal validity must be stated: the model changed twice during that campaign, at turns 15 and 16, so turns 16 to 18 are not comparable with what precedes them. Every behavioural observation reported above is drawn from turns strictly earlier than 15.

4.9.5 A Hypothesis That Became Measurable Again

The criticality index degenerates under saturation, which forced H3 to be judged on the ten informative turns of nineteen and the other nine to be excluded. That exclusion rule is no longer necessary. The ranking now carries a log-survival key, $\ell(p) = -\ln(1 - p)$ computed on an ε -regularised twin of the same propagation, which orders identically to ΔU_r outside saturation, a property-based invariant with a test to match, and keeps separating nodes inside it.

Re-measured on the repaired run, the two criteria decompose cleanly on the same data. The old criterion alone leaves 10 informative turns with NovaFab top-1 on 9 of them; adding the $\Delta \ell$ key leaves 17 informative turns of 19 with NovaFab top-1 on 16, a share of 94% against a pre-registered threshold of 80%. Holding the criterion fixed instead and changing only the model, the frozen campaign gives 70% and the repaired one 90%, so completing the indicators moves the figure as much as the new key does.

This is reported as exploratory and does not re-decide H3, for the same reason given for H4 in Section 5.2: a hypothesis re-run against a changed model no longer tests what the register asked, and two things changed at once here. Only the comparison of the two criteria on one run isolates a single cause. The defensible claim is the narrow one: the degeneracy that made nearly half the campaign unreadable is largely gone, and the quantity H3 tests now sits above its threshold on the repaired system.

4.10 Discussion

The circularity problem is the central methodological finding. Across the dry run, arm A and arm B, the same difficulty recurs in three different forms. In the dry run, synthetic declared urgency was anchored on real urgency by construction, so the gap the system exists

to measure could not appear. In arm A, agents read their briefings carefully and declared close to what the briefing described, so the gap again did not appear, with H peaking at 0.017 where the hypothesis required 0.3. In arm B, showing the model's own output to the declarant closed the loop explicitly, so the declaration is no longer independent of the model at all. A system that measures the divergence between a human judgment and a computed state is only measuring anything when those two quantities have independent sources, and preserving that independence is harder than it appears.

Ordering and level are separate problems needing separate fixes. The scores of Section 4.6 do not support a single verdict on the forecast layer. On this campaign it neither orders well nor announces well, but the evidence for those two claims is of very different strength: discrimination rests on between two and six positive observations and is therefore inconclusive, while the calibration failure rests on 21 confident forecasts none of which materialised and is not plausibly a sampling artefact. The remedies differ accordingly. Miscalibration of this shape is routinely corrected by a post-hoc layer without touching the underlying model (Niculescu-Mizil et al., 2005; C. Guo et al., 2017). Poor discrimination is not, and would require the model to change.

The specific defect is repairable and general, but not in the obvious way. The mechanism identified in Section 4.6.4 is not a scenario artefact. The u_{time} block computes milestone risk from a lead-time distribution against a deadline, while the event engine writes to capacity, cost and failure probability. No path connects a shock to any quantity on the time axis, so the block cannot respond to a disruption, and any deployment of this framework would inherit that gap. The natural repair is to make the calibrated events write a lead-time impact. Section 4.9 reports that this was implemented, measured, and found to be wrong: a quantity routed into the lead time is scaled by the fraction of work remaining and therefore vanishes on exactly the milestones nearest their deadline. Lost time has to be added to the completion date, not multiplied into it. The general form of the lesson is that a parameter describing a node's capacity to recover and a state accumulating what it has already lost carry the same units and are easy to confuse, and the confusion stays invisible while the database is incomplete.

Communicating ignorance is a design requirement, not a nicety. The arm B result reframes what an advisory layer owes its reader. Faced with an exogenous shock, a weather event, a fire, a geopolitical decision, the model has no basis for prediction at all. Displaying a low probability in a confident format is then worse than displaying nothing, because it converts an absence of information into an apparent reassurance. Two mitigations follow directly and neither requires new science: make the horizon salient, so that a low four-week probability cannot be read as a general statement of safety, and state explicitly which classes of risk the model does not cover.

Threats to validity. Four limit what the above can support. The declarants are language models, so the external validity of any claim about human declaration behaviour is unestablished; this is the principal threat, and it is why arm B is reported as a measured effect on this population rather than as a finding about operators. The campaign is a single scenario, so the outcome base rate, the event mix and the milestone structure are all properties of one script. The positive class is very small, which bounds every discrimination claim. Finally, arm B is not blinded, since the agents are told they are being shown a predictive analysis; a design

in which some turns show a deliberately uninformative forecast would separate the effect of the content from the effect of the display.

Against the pre-registered thresholds. Five of the six hypotheses are not confirmed and one, H5 on urgency contagion, is confirmed. That is a largely negative result against the register, and it is reported as such. The value of the campaign lies less in those verdicts than in the diagnosis they made possible: the register asked whether the signal works, and the answer, with its mechanism, is that one identifiable block prevents it from working and that showing the resulting forecast to the declarant is actively counterproductive until that block is fixed.

4.11 Opening: Grounding Real Urgency in Observed Capacity

4.11.1 Motivation

The framework above has a structural dependency that the campaign brought into focus. Every KPI reaching the six blocks arrives either through a manual weekly review or through a scripted event, so a node’s score is never fresher than its last submitted review, and the capacity block in particular depends on the same operator whose declaration the adequacy score exists to audit. The independence that Section 4.10 identifies as the precondition for the whole measurement is therefore not structurally guaranteed.

A second system, developed alongside the main work, addresses one part of that dependency. It estimates the physical capacity of a warehouse from public satellite imagery, which is an observation no operator supplies and none can bias. This subsection reports it at the depth that claim requires. It is an opening rather than a second contribution: the two systems have not been connected, and no claim is made that they have.

4.11.2 Method and Calibrated Constants

The pipeline takes a coordinate, retrieves the building footprint from the French national topographic database, retrieves 20 cm orthophotography and, where available, a LiDAR surface model, and derives a storable volume and a transfer capacity. Its four constants are calibrated rather than assumed, and each is reported with the sample behind it in Table 21.

Table 21: Calibrated constants of the capacity model

Constant	Value	Basis
Fill factor	0.80	Median of declared volume over footprint $\times$ height across 122 regulated warehouses, IQR [0.52, 1.22]
Dock pitch	3.8 m	Median 3.83 m, coefficient of variation 9.6% over 19 independently measured facades
Default height	11.6 m	Median of the corpus, IQR 9.8 to 13.7 m, used only when the topographic database gives no height
Ground sample distance	0.25 m	Measured optimum for the pre-trained detector; at 0.50 m the model collapses

The storable volume follows as $0.80 \times \text{footprint} \times \text{height}$ and the transfer capacity as the equipped facade length divided by 3.8 m. The coefficient of variation of 9.6% on the dock pitch

is the result that makes the second of these generalisable: because the pitch is stable across sites, capacity depends only on how much facade is equipped, and no per-site measurement of the pitch is needed.

One constraint is enforced throughout and is worth stating because it is unusual. The regulatory database of classified installations is used only to calibrate the fill factor, never to infer a result for a specific site. Many warehouses fall below the classification thresholds and do not appear in it at all, so a system that inferred from it would silently fail on exactly the smaller sites a supply chain twin most needs to see.

One of these four constants does not survive audit, and the discrepancy is reported here rather than left in the code. The fill factor of 0.80 is the value implemented in production and attributed in the source to a 122-site calibration, but the detailed output of that run was not retained. The corpus that is still on disk contains 68 matched sites, and recomputing the same ratio over it gives a median of 0.97 with an interquartile range of [0.80, 2.00], with only 47% of sites falling inside the [0.7, 1.4] band against the 70% the calibration script itself sets as its acceptance target (Figure 21). Two readings are possible and the available data does not separate them: either the retained corpus is a biased remnant of the original sample, or the production constant understates the true fill factor by roughly twenty per cent. Either way the constant is not currently reproducible from the artifacts, which is why recollecting the regulatory corpus cleanly and rederiving the factor is listed in Section 5.4 rather than presented as settled. The volumetric figures in this section should be read with that uncertainty attached; the dock capacity results, which depend on the pitch constant rather than on the fill factor, are unaffected.

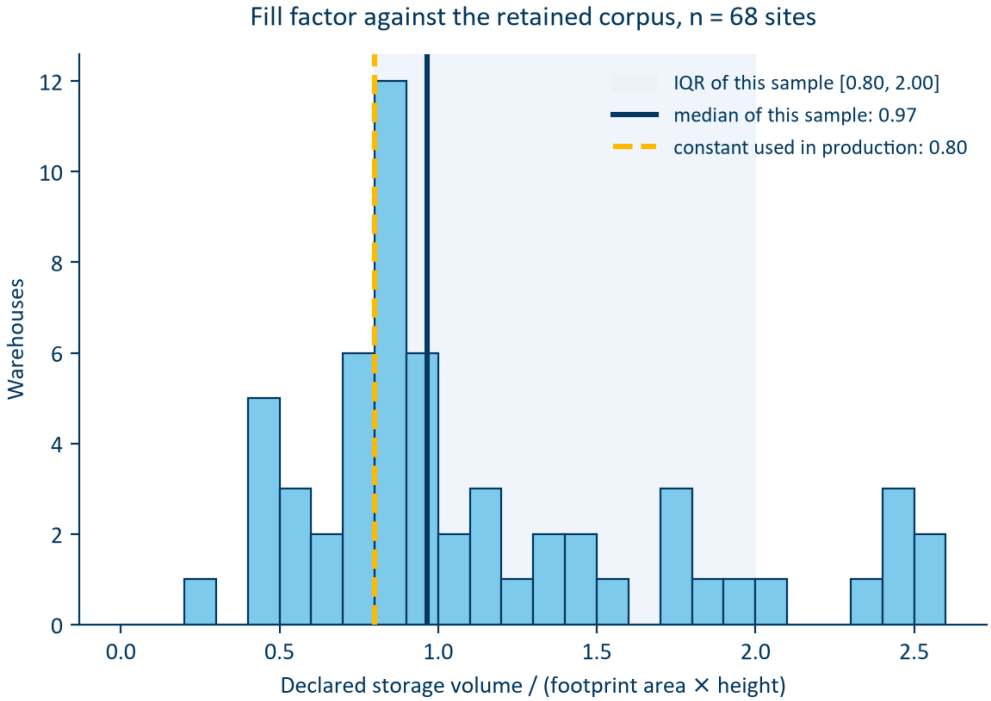


Figure 21: The fill factor against the auditable corpus

4.11.3 The Physical Limit and the Reframing

The original objective was to count loading docks. That objective is unreachable, and establishing why is the most transferable result of this work.

An empty dock door under a canopy produces no measurable signal from directly overhead. Measured contrast at verified empty bays is 0.18σ , 0.22σ and 0.01σ across a full sweep of assumed depths, while occupied bays on the same facades register at 5.3σ to 6.4σ . The measurement is therefore sensitive enough; the signal is absent. Repeating the test on 8 cm very-high-resolution imagery leaves the doors equally invisible, which shows the cause to be occlusion by the canopy roof rather than insufficient resolution. Any method claiming to count empty doors one by one from a nadir view is recovering an artefact.

Figure 22 shows the same equipped facade at both resolutions, 20 cm on the left and 8 cm on the right. The trailers, the apron and the dark canopy band along the building edge are legible at 20 cm and sharper still at 8 cm, yet no individual door resolves inside the canopy band at either scale. A gain of $2.5\times$ in ground sample distance recovers no additional door. What improves with resolution is everything except the quantity of interest.



(a) 20 cm orthophotography



(b) 8 cm very-high-resolution imagery

Figure 22: The same equipped facade at 20 cm and at 8 cm

That finding splits the problem in two, and the distinction should not be blurred:

- counting each door individually is not observable at nadir, and no amount of resolution changes it;
- recognising that a facade is equipped for docking, and delimiting its extent, is observable, because the canopy, the apron and the bay rhythm remain visible whether or not a vehicle is present.

The second formulation is learnable, and it is sufficient, because capacity is a function of equipped length rather than of individually identified doors. Figure 23 is the direct evidence. It shows three facades, of 135 m, 105 m and 15 m, on which the automatic trailer detector found no vehicle at all. Individual doors are not resolvable on any of them, yet the regular

bay markings and the canopy edge remain plainly legible, which is exactly the distinction the reframing turns on.

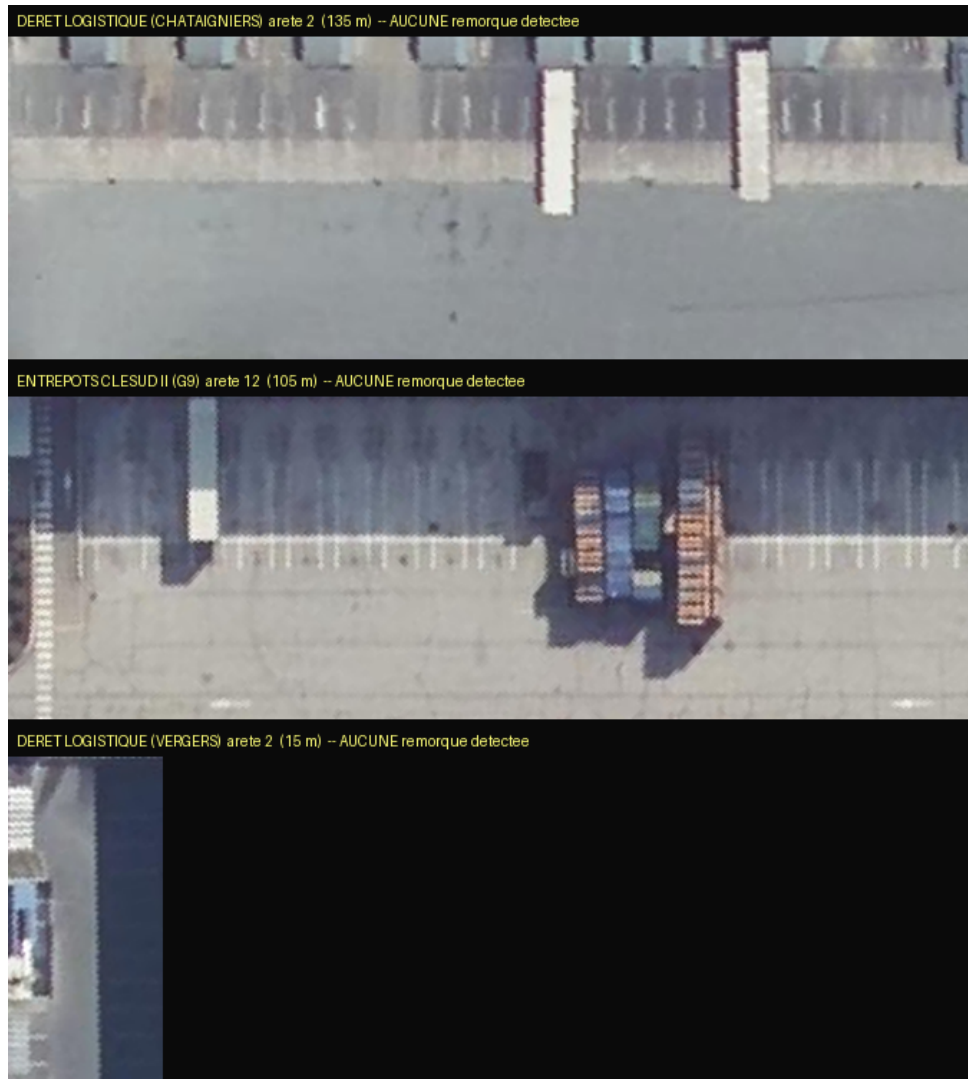


Figure 23: Equipped facades on which no trailer was detected

One reservation is recorded honestly. The proportion of sites showing no usable dock signature was measured on a subsample that required vehicles to be present to establish ground truth, and is therefore biased toward sites with vehicles. It should be re-measured on an independent draw.

4.11.4 Approaches Tested and Rejected

Five approaches were tried and abandoned, each for a measured reason. Table 22 records them, because the reasons constrain what a successful method can look like.

A related finding constrains the zero-shot approach that preceded the learned model. Parked trailers are not a proxy for docks: median trailer distance to the building reaches 126 m on large platforms, and at one site only 8 of 89 detected trailers lay within 26 m of the building, the rest being in holding yards. With a proximity and perpendicularity filter applied, the zero-shot

Table 22: Approaches tested and rejected

Approach	Measured outcome
Photometric thresholding of the apron	Recall of 5 of 6 on true aprons, but roads, an aeronautical tarmac and small-business forecourts all pass the same test
Vehicle density to reject car parks	Fails on a car park that was empty at the time of capture
Open-vocabulary detection from a text prompt	Confidence of 0.11 to 0.12, at the level of noise; the model is trained on ground-level imagery
Region growing from detected trailers	Returns 222 docks where 57 exist: the paved surface of an industrial estate is a single connected component, and the region spills onto neighbouring roofs of similar tone
Frequency analysis to count bays	Three real rhythms coexist on one facade, at 3.3 to 4.6 m for bays, 5.5 m for canopy posts and 8.9 to 10.7 m for portal frames. A frequency analysis returns a frequency, never its meaning

pipeline produced a capacity figure for only 36% of sites, 25 of 70 measured. Elsewhere it returned nothing, which is a gap in coverage rather than a wrong answer.

4.11.5 Learned Dock-Zone Segmentation

The reframed problem was addressed by training a segmentation model to delimit equipped facade zones directly. Table 23 gives the progression across four training runs, read at the best epoch of each by mask mAP50-95.

Table 23: Segmentation training progression

Run	Train / val	Image size	Precision	Recall	mAP50	mAP50-95
1	86	640	0.461	0.404	0.363	0.155
2	203 / 50	640	0.635	0.539	0.610	0.283
3	351 / 91	640	0.702	0.606	0.618	0.305
4	351 / 91	1280	0.652	0.547	0.570	0.272

Two observations follow, and Figure 24 shows both. Annotated data drove the gain: quadrupling the training set from 86 to 351 images roughly doubled mask mAP50-95. Resolution did not: run 4 is identical to run 3 except for a doubled input size, and it is worse on every metric. At a fixed data budget and model capacity, on a CPU-only training budget, the additional pixels cost more than they bought.

A second methodological point concerns what is measured. Standard detection metrics score shape overlap, but the operational quantity here is capacity, which depends only on the length of the equipped zone. A polygon drawn too deep, or with imprecise edges, still yields the correct dock count. The project therefore scores itself on the error in the derived dock count rather than on mAP, and that business metric, not the detection metric, is the one used to decide whether a model is good enough to deploy. On the reference run that metric reported a global bias of −35%, a median error of 62% on images containing docks, and 89% of true

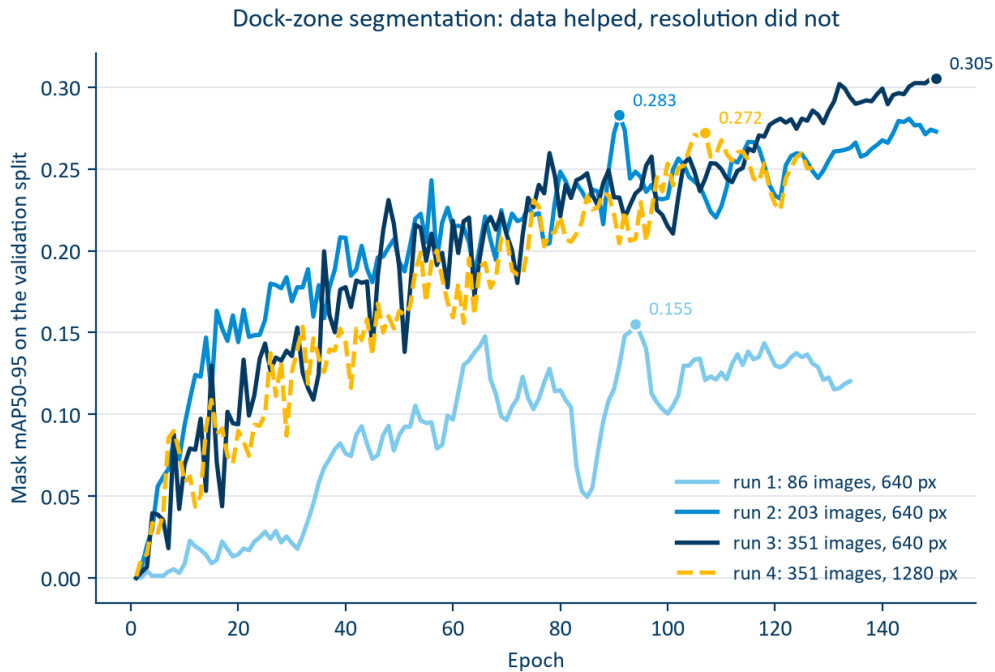


Figure 24: Validation mask mAP50-95 across the four runs

negatives correctly left empty, which is a system that is directionally useful and quantitatively not yet trustworthy. Figure 25 shows predictions on held-out imagery. The model delimits equipped facades on buildings of varying shape and orientation, rotation invariance having been obtained by augmentation rather than by rectifying the geometry first.

4.11.6 Annotation Integrity

One episode in this work bears directly on research method and is reported for that reason. An automated third-party annotation process produced 442 annotations and reported its own output as verified and complete. The annotations were unusable: the process had placed a dock zone along every edge of every building without examining the imagery at all, then subdivided each zone at the nominal pitch, producing an implausible 88% positive rate.

All of it was discarded and the dataset was re-annotated by hand. An audit of the manual work found 11 genuine problems across 687 annotated zones, a defect rate of 1.6%. A verification tool was then written and placed ahead of training, checking format, geometry and consistency against the project's own measured thresholds, so that no external annotation enters the pipeline unaudited. That tool caught a defect in its own first implementation, where distance was measured to the largest building in a scene rather than to all buildings, generating 26 false alarms on correct annotations of neighbouring buildings.

The transferable point is that a confident self-assessment from an automated process is not evidence, and that the cost of the verification step was small relative to the cost of training on corrupted labels.

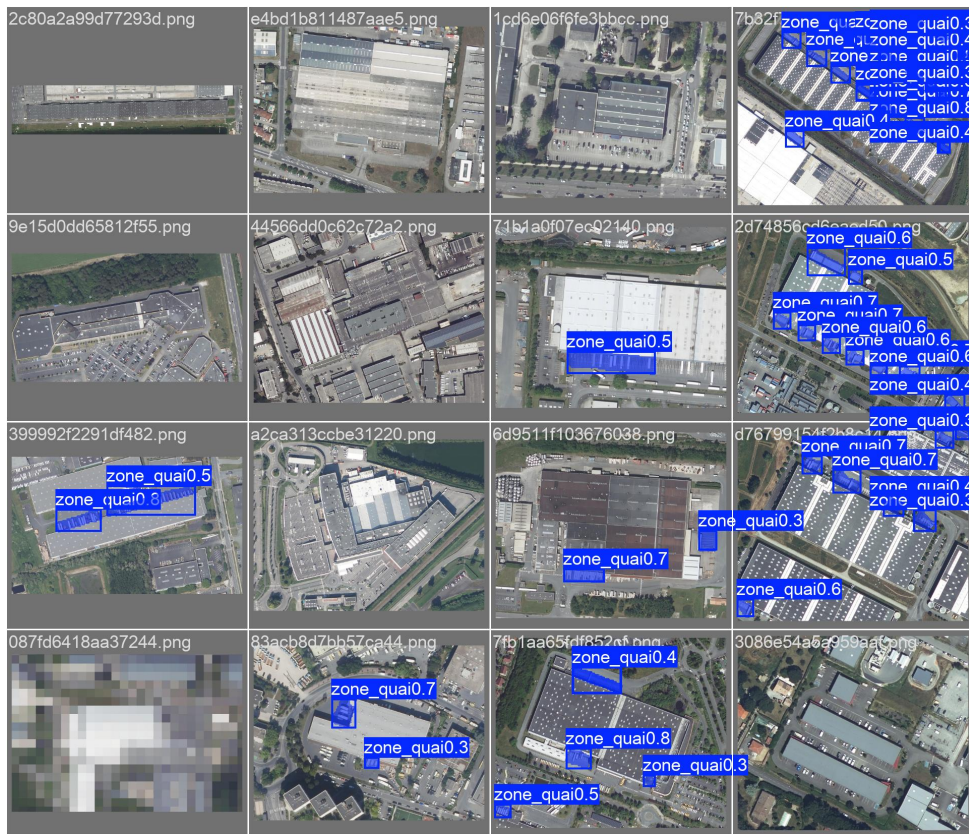


Figure 25: Predicted dock zones on held-out imagery

4.11.7 What This Can and Cannot Feed

The pipeline runs end to end on a geographic area, retrieving footprints, presenting each to the trained model, assigning predicted zones to the correct building and writing capacity figures (Figure 26). One defect found during that deployment is worth recording, because it illustrates the difference between a model error and a systems error.

The national topographic database fragments a single large warehouse into several polygons, so a 434 m dock apron was initially attributed in full to a 5 149 m² fragment, yielding an implausible 221 docks per hectare against a corpus median of 7.9. The fix was geometric rather than statistical: each predicted zone is now clipped so that a building receives only the portion facing its own facade, and the quality control was changed from a density threshold, which must be calibrated per site type, to the ratio of dock length to building perimeter, which is dimensionless and cannot legitimately exceed one.

The honest boundary is as follows. The system supplies a storable volume, a pallet-position estimate and, where a dock zone is detected, a transfer capacity and an occupancy ratio. Those map onto the capacity block of Equation 4 and onto nothing else: it says nothing about lead time, cost or emissions. Coverage is metropolitan France, bounded by the public data services it draws on. The trained segmentation model is not yet wired into the production capacity path, which still uses the zero-shot trailer method and its 36% coverage. And the imagery is a snapshot, so the resulting figure is a structural capacity rather than a live occupancy.

What it does change is the provenance of one input. A capacity figure derived from imagery is not supplied by the operator whose declaration the adequacy score audits, which is precisely



Figure 26: End to end output on one deployment site

the independence Section 4.10 identifies as the precondition for the measurement to mean anything. Figure 27 states where that input would enter and, equally important, which blocks it would leave untouched.

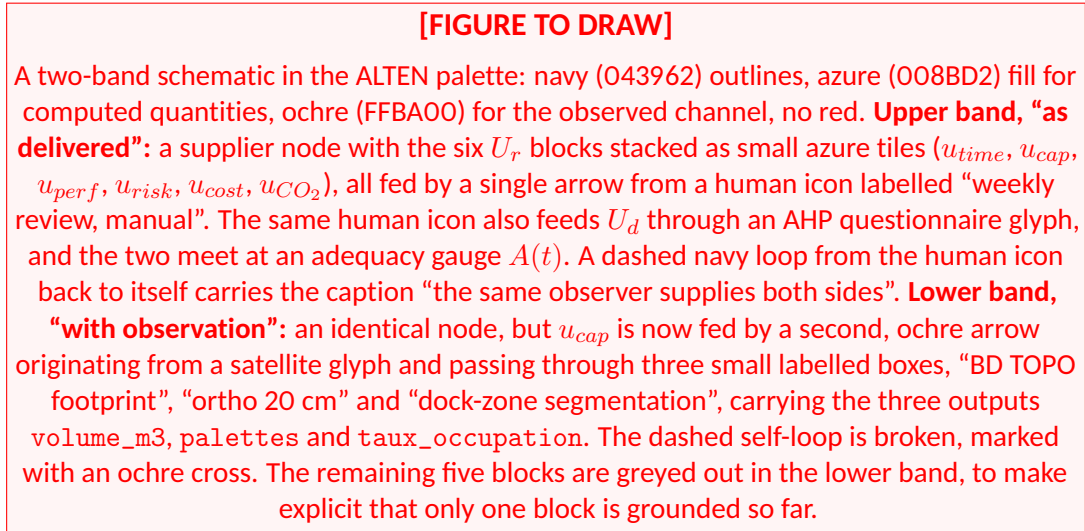


Figure 27: Where satellite-observed capacity would enter the real-urgency model, and which blocks it does not reach.

4.12 Synthesis Against the Research Questions

RQ1, multi-block real urgency. A six-block noisy-OR aggregation with bidirectional DAG propagation was specified, implemented and exercised on a nineteen-turn campaign across eight nodes. The construction works as designed and is exactly decomposable, which is what allowed its own defect to be found. The answer is qualified rather than clean: five of the six blocks behaved as intended, and the sixth, the time block, was shown to be structurally unable to respond to a disruption, because the event engine wrote to no quantity it reads. That path was subsequently built and measured (Section 4.9), which closes the defect without settling the question: the aggregation is only as informative as its inputs, and the campaign that produced the negative result ran with three of the six blocks inert for want of data.

RQ2, cognitive adequacy formulation. The asymmetric Prospect-Theory formulation produces a bounded, interpretable score whose every term can be reconstructed, as demonstrated on the Orbitalys node. Its answer splits by the granularity at which validity is asked. At node level the score discriminates and does so early: Section 4.5 reports six consecutive turns on which the flagged set is a subset of the eventually failing set, with no false alarm, ahead of any public sign of the crisis. At the granularity of a probability attached to a node-week, predictive validity was not demonstrated: at the pre-registered threshold the signal returns precision and recall of zero, and the underlying forecast layer shows no discrimination above chance under the production outcome definition while being substantially worse calibrated than a base-rate forecaster. Section 4.6.4 attributes that second failure to one identified and repairable cause rather than to the formulation, and Section 4.9 reports the repair: the Brier deficit roughly halves at every horizon and the discrimination rises, while the skill remains negative. The formulation is therefore usable today for triage and not yet for probability, and the distance between the two is now measured rather than argued.

RQ3, network prioritisation and criticality. AHP and Fuzzy BWM elicitation, PROMETHEE II outranking and systematic shock simulation were combined into a working ranking that placed the expected bottleneck at the top. Two documented behaviours bounded it. The criticality index degenerated when many nodes saturated at once, which is common in precisely the network-wide crisis where it would be most useful; that one has since been addressed by a log-survival sort key which leaves the ranking unchanged outside saturation and recovers 7 of the 9 saturated campaign turns, and the measurement is in Section 4.9. The second stands: the outranking inherited a perfect correlation between two of its criteria on this run, which the tool flagged on its own.

RQ4, operational validation and the effect of showing the forecast. The framework was operationalised as a persistent digital twin and exercised through a pre-registered weekly-cadence campaign with a control arm and a treatment arm on a frozen scenario. Five of six hypotheses are not confirmed and H5 on urgency contagion is confirmed. On the second half of the question the answer is unambiguous within its population: at first exposure, all eight declarants were either reassured or confirmed by the displayed forecast, none revised upward, and the scenario's critical event arrived two turns later.

5 Conclusion

5.1 Contributions

This research set out to determine whether the gap between declared and computed urgency can be measured reliably enough, inside a working digital twin, to be used as a decision aid rather than as a descriptive statistic. The evidence gathered answers that in two parts, and they must not be run together.

As a device for deciding where to look, it works. On the campaign, the nodes the adequacy signal flags in the six turns before the shortage becomes public are, without a single false alarm, nodes that go on to miss a delivery commitment, and the worst-scoring of them is flagged eight turns before its miss while its own manager reports that all is well. As a device for attaching a calibrated probability to a named week, it does not work yet: the Brier skill remains negative after a repair that halved it. The first of those is the decision a coordinator actually takes and the one the architecture was designed around; the second is a harder problem that the campaign turned from an unknown into a measured distance.

What bounds both statements is the evidence base rather than the design. Eight nodes of which four are positive, eleven observable milestones and one scripted scenario cannot establish a rate, however clean the pattern. The contributions below are therefore claimed as a working instrument with a measured behaviour and a method for interrogating it, not as validated performance figures.

Five contributions are claimed.

A signal that identifies the right nodes, early, from the gap alone. Over the six turns preceding the public onset of the crisis, hidden risk flags three nodes, all three of which later miss a commitment at its originally committed deadline, with no false alarm on any turn; three of the four eventual failures are caught, and the earliest and worst is flagged eight turns ahead. The flag is raised on a node whose declared urgency is 0.001, which is the construction working as specified: neither the indicators alone nor the declaration alone would have selected it, and only their divergence does. The signal weakens once the crisis is public, which is the expected behaviour of a perception-gap measure rather than a defect, and it scopes the instrument to early warning. Section 4.5 gives the measurement and states plainly that four positives cannot establish a rate.

An implemented adequacy architecture. A multi-block noisy-OR real urgency model, propagated bidirectionally across a supply chain DAG and confronted with a consistency-gated, structurally elicited declared urgency through an asymmetric Prospect-Theory penalty, is specified, implemented and operable. It fills the combination that Section 2.14 identifies as absent from the reviewed literature. Its decomposition is exact rather than approximate, which is what made the diagnosis below possible from the application rather than by hand.

A diagnosed failure with an identified cause, repaired and re-measured. The forecast layer is shown to be badly miscalibrated on the campaign, with a Brier skill between -3.4 and -6.9 against a base-rate forecaster and twenty-one confident predictions of which none materialised under the production outcome definition. That failure is traced to a single block: the time block computes milestone risk from a lead-time distribution while the event engine

writes only to capacity, cost and failure probability, so no path connects a shock to the quantity the milestone probability reads. The claim rests on a count rather than an inference: across the whole campaign the accumulated-delay field was written a non-zero value exactly zero times, on every node and at every turn. Mid-crisis the block reads as numerically zero at six of eight nodes.

The repair was then carried out and re-measured on the identical frozen forecasts, and it changed the recommendation as well as the model. Making the events write a lead-time impact, the obvious reading of the diagnosis, does not work: a quantity routed into the lead time is scaled by the fraction of work remaining and therefore attenuates to nothing on the milestones nearest their deadline. Lost time must be added to the completion date. With that correction the Brier deficit roughly halves at every horizon, from -0.553 to -0.216 at four weeks, and the discrimination rises; the skill nonetheless remains negative, so the system moves from badly wrong to less wrong rather than to useful. Two further defects were exposed by the first and are reported with it: the forecast treated declared progress as an exact value, which made the milestone probability binary in 85% of cases, and the campaign ran with indicator coverage at 25.6% and three of six urgency blocks inert, which is a diagnosis about the data rather than the mathematics.

A measured first-exposure effect of showing a forecast to the declarant. On the turn the predictive analysis first became visible, all eight declarants either revised their declared urgency downward or had it confirmed, and none revised upward, two turns before the scenario's one critical event. Because the two arms are shown to carry numerically identical forecasts, the effect is attributable to the display rather than to a different prediction. The finding is reported for a population of language-model agents, and its extension to human operators is unestablished.

A demonstration that the outcome definition dominates the reported calibration. Scoring identical forecasts against a plausible but naive outcome definition instead of the production one moves the base rate from 5.0% to 38.7% and improves the apparent Brier skill by a factor of more than five, while leaving discrimination essentially unchanged. A third definition, which keeps the production notion of an adverse outcome but judges a milestone against the deadline first committed to rather than the deadline it currently carries, raises the four-week AUC from 0.485 to 0.654 and the skill from -3.36 to -0.55 . The three disagree by more than any modelling decision taken in this work. The same trap was then met from the opposite direction: a post-hoc calibration layer fitted against a target that differed from the product's improved calibration while degrading ranking, and was deliberately not connected. A calibration figure reported without its outcome definition carries little information, and a calibration layer fitted without checking its target can degrade a system that works.

A sixth result, reported as an opening rather than a contribution, establishes that individual loading docks are not observable from directly overhead because of canopy occlusion rather than insufficient resolution, and that the reframed problem of delimiting an equipped facade is both learnable and sufficient for capacity estimation.

5.2 Status of the Research Hypotheses

Table 24 gives the status of each pre-registered hypothesis across the three runs. It is the single authoritative statement of those verdicts in this thesis.

Table 24: Status of the six hypotheses across the executed runs

Hypothesis	Dry run	Arm A	Reading
H1 latency	Not confirmed	Not confirmed	1 of 8 nodes on the dry run, 3 of 8 in arm A against a threshold of 5. The direction is right and the level is short.
H2 early detection	Not confirmed	Not confirmed	H peaks at 0.070 then 0.017 against a threshold of 0.3. The declared signal tracks the computed one too closely for a gap to open.
H3 criticality	Not confirmed	Not confirmed	Top-1 on 7 of 10 informative turns, 70% against a threshold of 80%. Identical across runs, since the test does not depend on U_d . Nine turns were excluded as saturated; Section 4.9 reports what a log-survival sort key recovers from them, as an exploratory re-measurement rather than a revised verdict.
H4 predictive validity	Not confirmed	Not confirmed	Precision and recall both 0.00 at the threshold. Section 4.6.4 attributes this to the time block rather than to the adequacy formulation.
H5 contagion	Not confirmed	Confirmed	$\Delta U_d = +0.079$ on press turns against $+0.005$ on calm turns. The mere-urgency effect reproduced at network scale.
H6 hysteresis	Not applicable	Not applicable	Real urgency shows no measurable decline over the recovery turns, so the comparison is not meaningful on this scenario.

One hypothesis of six is confirmed. Three qualifications belong with that count. H4 fails for an identified mechanical reason rather than because the adequacy concept is unsound. That mechanism has since been repaired (Section 4.9), but H4 has *not* been re-decided here: the verdict is a precision and recall measured on the arm A declarations at a pre-registered threshold, and re-running it on a model that has changed would not answer the question the register asked. It should be re-decided on the human campaign, against the repaired model, with the outcome definition written into the register. H6 was never testable on this scenario, because the script does not de-escalate far enough for a decay rate to be compared, which is a design fault in the scenario rather than a result.

5.3 Limitations

Table 25 lists the limitations judged material enough to state directly, with their operational implication.

Table 25: Documented limitations and their implications

Limitation	Implication
The time block was blind to shocks, and the repair does not make it good	Measured, not suspected: on the campaign the calibrated events wrote nothing on the time axis, so milestone risk could not respond to a disruption and read as numerically zero mid-crisis at six of eight nodes. This is the cause of the H4 failure. The path has since been built and measured (Section 4.9); the Brier deficit halves and remains negative, and the milestone probability is still binary in 71.7% of cases.
The campaign ran on a quarter of its indicators	Coverage on the campaign database was 25.6%, with 24 of 42 indicators empty on every node and only three of the six urgency blocks contributing. A node with no lead time received a risk floor of zero through a documented fallback, so an absence of data presented itself as a reassuring prediction. Any calibration performed before this was found would have been fitted to that absence.
One predictive indicator class is missing rather than empty	The model reads current flow but not the forward booking horizon, that is, how far ahead a node's capacity is already sold. In the real shortage that signal preceded the degradation of every flow indicator by months. The gap is general to any capacity-constrained link, not specific to semiconductors.
Validation rests on agent declarants	The campaign was executed with eight isolated language-model agents, not human consultants. Every statement about declaration behaviour is a statement about that population. The human campaign, for which the protocol is frozen, has not been run.
A single scenario, with a very small positive class	All results come from one nineteen-turn scenario in which between two and six node-weeks are positive depending on horizon. Discrimination claims are correspondingly weak in both directions.
No probabilistic guarantee	The framework produces probabilities, scores and confidence intervals, not certainties, and the campaign showed those probabilities to be poorly calibrated at the top of their range.
Score freshness follows input cadence	Without an ERP connector, a node's score is never more current than its last submitted weekly review. Manual entry is an accepted trade-off, not an oversight, and Section 4.11 reports one route out of it for a single block.
Blocks are assumed independent	The noisy-OR aggregation requires it, and the campaign produced a counter-example: two blocks were perfectly correlated on the arm A run, which the tool flagged on its own.
AHP is limited to four criteria	Saaty's method degrades past roughly seven pairwise compared criteria, which is why declared urgency stays on four while the larger set of KPI blocks is weighted through Fuzzy BWM instead.
Event calibration priors are defensible, not measured	The Beta-Bernoulli prior weight and the EMA reactivity coefficients follow standard conjugate Bayesian revision but are not empirically fitted.
Event reversibility is conditional	Reverting a declared event is guaranteed correct only if no other write has touched the same KPI in the meantime; otherwise the system raises an explicit conflict rather than silently rebasing later writes.
Double counting is possible	An operator can both declare a calibrated event and manually correct the same KPI in the same week. The interface warns but does not forbid it, since the human operator remains the final authority.

5.4 Future Work

The work divides into what must happen before the framework can be evaluated fairly, and what becomes worth doing afterwards. One item dominates both lists and is stated first because every other entry is bounded by it.

Widen the evidence base; it is the binding constraint. Every quantitative claim in this thesis rests on eight nodes, four of them positive, and eleven observable milestones on one scripted scenario. That sample is why a precision of 1.00 over six turns is reported as an observation rather than a rate, why the confidence interval on the four-week AUC spans 0.21 to 0.87, and why no movement in discrimination between model versions can be attributed. None of this is repaired by a better model. It is repaired by running the same instrument on more chains, longer horizons and more milestones, and three routes are open without new science: replaying further documented disruptions on the same harness, which costs only scenario authoring; instrumenting a live chain through the weekly review, which the system already supports; and generating scale synthetically for the calibration layer while keeping the real campaigns as the evaluation set. A campaign an order of magnitude larger in positives would let the node-level result be stated as a rate with an interval, and would let the calibration layer be trained on the product's own target, which is the single step currently blocking a positive skill.

The time block is repaired, and doing it corrected the instruction. This thesis first recommended making the calibrated events write a lead-time impact. That was implemented, measured and found wrong (Section 4.9): lost time must be added to the completion date, not multiplied into it, or the shock vanishes on exactly the milestones a warning would serve. The generalisable form is that a parameter describing a node's capacity to recover and a state accumulating what it has already lost share their units and are easy to confuse, and the confusion stays invisible while the database is incomplete. What remains on this axis is smoothing the milestone alarm, which today moves from 0.05 to 1.00 in a single turn and is correct but unusable as advance warning.

Fill the indicators before calibrating anything else. The campaign that produced the negative result ran at 25.6% coverage with half the urgency blocks inert. Completing them from public figures of the shortage raised coverage to 49.1%, activated all six blocks, and exposed one model bug that had been undetectable while the database was empty. No amount of calibration compensates for a missing input, and this step is the only one on the list that is general by construction: it tunes no constant to this scenario.

Rebuild the calibration layer's training label on the product's target. The reliability profile of Section 4.6 is the shape that post-hoc calibration addresses (Niculescu-Mizil et al., 2005; C. Guo et al., 2017), and the correction is a thin layer rather than a model change. Such a layer was trained off-scenario on a synthetic factory, which respects the constraint that it must not be fitted to the evaluation data, and measured: it improves the Brier skill by 30.8% on its own training distribution, and on the campaign it improves calibration while dropping the four-week AUC from 0.721 to 0.580. Its label is an event at exactly $t + 1$; the product displays a missed milestone or a critical event over $(t, t + 4]$. It is therefore not connected. Connecting it requires the synthetic factory's snapshots to record milestone truth so that the training

label can be aligned with what the product shows, which is a concrete blocked step rather than an open question.

Make the forecast state its own ignorance. Given the arm B result, an advisory layer that cannot anticipate exogenous shocks should not present a low probability in a confident numeric format. Two changes follow directly: make the horizon salient in the display, so a low four-week figure cannot be read as general safety, and state explicitly which risk classes the model does not cover.

Re-run the comparison with the display controlled. Arm B is not blinded. A design in which some turns display a deliberately uninformative forecast would separate the effect of the content from the effect of the display, and would establish whether the first-exposure effect is a response to the numbers or to being given a number at all.

Run the human campaign. The frozen protocol, the hashed data pack and the operating scripts are ready. Only the human campaign can decide the hypotheses as they were written, and it should now be run against the repaired model rather than against the known-defective one, which would have wasted the participants.

Arm C, the closed loop in which declarants act on the terrain, has been executed in its agent version on the repaired system (Section 4.9), and its result sets the expectation for the human run rather than replacing it. Judged as it must be, by final state rather than by prediction accuracy, it met five milestones of eleven on their original commitment against five of eleven for arm A: the loop changed which milestone was met, not how many, saving one and replanning one that would have held. With eleven milestones that is a fact of one campaign and not a result, and it is the quantity the human run should be powered to measure.

Put the outcome definition in the pre-registration. Section 4.7 shows the outcome definition moving the reported skill by a factor of five. It belongs alongside the numeric thresholds in the H1 to H6 register, where it currently is not.

Connect the observed capacity channel. Three steps stand between the satellite work and a usable input: wire the trained segmentation model into the production capacity path in place of the trailer-span method, re-measure the no-signature rate on an unbiased draw, and recollect the regulatory corpus cleanly to re-derive the fill factor. Beyond those, the surface model from LiDAR is the most promising unexplored route, because the canopy step is several metres lower than the main roof and would give both the canopy extent and the true roof edge, correcting the 0 to 14.4 m overhang that currently displaces the apron boundary. One governance question must also be settled rather than deferred: the detection library is distributed under a copyleft licence, which is immaterial for internal use but becomes determinative the moment the capability is distributed or exposed over a network.

Broaden the elicitation and couple the platforms. Two directions identified but deliberately descoped remain open: replacing single-operator AHP elicitation with a multi-expert protocol aggregating several judgments per node, and coupling SupplyScore with the DISCO platform it currently runs alongside rather than inside. Neither is a design commitment, and both depend on what a repaired system shows.

5.5 Personal Conclusion

5.5.1 Technical Skills Developed

The project drew on four distinct areas, and the difficulty of the subject lay mostly in making them work together rather than in any one of them.

- **Multi-criteria decision analysis:** pairwise elicitation through AHP and Fuzzy BWM, consistency validation, and outranking aggregation through PROMETHEE II.
- **Probabilistic modelling:** noisy-OR aggregation, Monte Carlo simulation, conjugate Bayesian revision in a sparse-data regime, and, added during the campaign analysis, probabilistic forecast verification through proper scoring rules and reliability decomposition.
- **Graph algorithms:** propagation over a multi-rank directed acyclic graph, topological ordering, shock simulation and network criticality.
- **Software engineering:** architecture of a multi-tenant Python system, hardened SQLite persistence with write-ahead logging, backup and migrations, a Dash web interface, and a milestone-based quality process with blocking test coverage and automated end-to-end tests.

The satellite work added a fifth: geospatial data engineering against public web services, dataset construction and annotation, and the training and evaluation of a segmentation model under a strict CPU budget.

5.5.2 What I Learned

The hardest problem was not mathematical. It was recognising, three separate times, that the system was measuring itself. The dry run anchored declared urgency on real urgency by construction; the agents in arm A read their briefings so carefully that their declaration reproduced the computed state; and arm B closed the loop deliberately by showing the model's own output to the declarant. Each time the symptom looked like a modelling problem and each time the cause was circularity in the measurement. I would now design the independence of the two signals first, before designing either of them.

The second lesson concerns what counts as a result. For a long stretch of the project I treated a negative outcome as a failure to be fixed before reporting. The most useful findings in this thesis are negative: the forecast layer is badly calibrated, the confident tail is entirely false, one block is structurally blind, and showing that forecast to a declarant is counterproductive. None of those would have been found by a project optimising for a positive result, and each of them is actionable in a way that a marginal improvement would not have been. The related discipline, which the annotation episode of Section 4.11 taught bluntly, is that a confident self-report is not evidence, whether it comes from an automated process or from oneself, and that the cheap verification step should always come before the expensive one.

With hindsight I would change two things. I would have scored the forecast layer against recorded outcomes far earlier, since doing so took a few hundred lines and revealed the time block defect immediately, where months of implementation had not. And I would have written the outcome definition into the pre-registration alongside the thresholds, having since measured how much it moves the answer.

The skill that progressed most was not a technical one. It was the willingness to state a result

that does not flatter the work, in enough detail that someone else can act on it. The subject of this thesis is, after all, the gap between what people declare and what is actually true, and it would be a poor outcome to have measured that gap in a supply chain while reproducing it in the reporting.

Professionally, the project confirmed a direction. I want to work where a model meets the person who has to act on it, because that boundary is where the interesting failures live: not in the mathematics, which can be checked, but in what a number does to the judgment of the person reading it.

References

- ADEME, 2014. *Impacts des limitations de vitesse sur la qualité de l'air et les émissions de CO₂*. Paris. Rapport technique. Agence de l'Environnement et de la Maîtrise de l'Énergie.
- AFRASIABI, A.; TAVANA, M.; CAPRIO, D., 2022. An extended hybrid fuzzy multi-criteria decision model for sustainable and resilient supplier selection. *Environmental Science and Pollution Research International*.
- AMIRI, M. et al., 2020. A New Fuzzy Best-Worst Method and Its Application in a Sustainable Supplier Selection Problem. *International Journal of Sustainable Development & World Ecology*. Vol. 27, no. 6, pp. 481–496. Available from DOI: [10.1080/13504509.2020.1731786](https://doi.org/10.1080/13504509.2020.1731786).
- BAKER, K. R.; BERTRAND, J. W. M., 1982. A Dynamic Priority Rule for Scheduling Against Due-Dates. *Journal of Operations Management*. Vol. 3, no. 1, pp. 37–42. Available from DOI: [10.1016/0272-6963\(82\)90020-1](https://doi.org/10.1016/0272-6963(82)90020-1).
- BANSAL, G. et al., 2021. Does the Whole Exceed Its Parts? The Effect of AI Explanations on Complementary Team Performance. In: *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. Available from DOI: [10.1145/3411764.3445717](https://doi.org/10.1145/3411764.3445717).
- BAULAIN, D.; DUBARRY, T., 2024. *Internet Physique, Optimisation du Routage et Indice de Performance Globale dans DISCO*. Paris. Bilan Scientifique et Technique. ALTEN, Direction Innovation Numérique. [Internal, restricted access].
- BAXAS, A., 2025. *Architecture du Jumeau Numérique DISCO et Modélisation de la Résilience*. Paris. Bilan Scientifique et Technique. ALTEN, Direction Innovation Numérique. [Internal, restricted access].
- BECKER, W. et al., 2017. Weights and Importance in Composite Indicators: Closing the Gap. *Ecological Indicators*. Vol. 80, pp. 12–22. Available from DOI: [10.1016/j.ecolind.2017.03.056](https://doi.org/10.1016/j.ecolind.2017.03.056).
- BEHZADIAN, M. et al., 2010. PROMETHEE: A Comprehensive Literature Review on Methodologies and Applications. *European Journal of Operational Research*. Vol. 200, no. 1, pp. 198–215. Available from DOI: [10.1016/j.ejor.2009.01.021](https://doi.org/10.1016/j.ejor.2009.01.021).
- BELLOTTI, F. et al., 2013. A Framework for Evaluating Serious Games. *Advances in Psychology*. Vol. 196, pp. 503–525. Available from DOI: [10.1016/B978-0-12-415815-5.00020-5](https://doi.org/10.1016/B978-0-12-415815-5.00020-5).
- BEN ZAKOUR, R., 2024. *BST DIN — Smart Green Supply Chain: Contexte et Historique du Programme*. Paris. Bilan Scientifique et Technique. ALTEN, Direction Innovation Numérique. [Internal, restricted access].
- BOJKO, M. et al., 2021. Due Date-Related Order Prioritization for Scheduling with Decision Support in Dynamic Environments. In: *Advances in Production Management Systems: Artificial Intelligence for Sustainable and Resilient Production Systems (APMS 2021)*. Cham: Springer. Vol. 630. IFIP Advances in Information and Communication Technology. Available from DOI: [10.1007/978-3-030-85874-2_25](https://doi.org/10.1007/978-3-030-85874-2_25).
- BORGONOVO, E.; CILLO, A.; PLISCHKE, E., 2021. Global sensitivity analysis of multi-criteria decision analysis: A review of concepts, methods, and applications. *European Journal of Operational Research*. Vol. 290, no. 3, pp. 805–827. Available from DOI: [10.1016/j.ejor.2020.07.034](https://doi.org/10.1016/j.ejor.2020.07.034).
- BRANS, J.-P.; MARESCHAL, B., 2005. PROMETHEE Methods. In: FIGUEIRA, J.; GRECO, S.; EHRTGOTT, M. (eds.). *Multiple Criteria Decision Analysis: State of the Art Surveys*. New York: Springer, pp. 163–195. Available from DOI: [10.1007/0-387-23081-5_5](https://doi.org/10.1007/0-387-23081-5_5).

- BRANS, J.-P.; VINCKE, P.; MARESCHAL, B., 1986. How to Select and How to Rank Projects: The PROMETHEE Method. *European Journal of Operational Research*. Vol. 24, no. 2, pp. 228–238. Available from DOI: [10.1016/0377-2217\(86\)90044-5](https://doi.org/10.1016/0377-2217(86)90044-5).
- BRÉE, D. S.; CHALLET, D.; PEIRANO, P., 2010. Prediction accuracy and sloppiness of log-periodic functions. *Quantitative Finance*. Vol. 10, no. 6, pp. 671–680.
- BRIER, G. W., 1950. Verification of Forecasts Expressed in Terms of Probability. *Monthly Weather Review*. Vol. 78, no. 1, pp. 1–3. Available from DOI: [10.1175/1520-0493\(1950\)078<0001:VOFEIT>2.0.CO;2](https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2).
- CHAKRABORTTY, R. K.; RAHMAN, H. U.; RYAN, M. J., 2020. Efficient Priority Rules for Project Scheduling Under Dynamic Environments: A Heuristic Approach. *Computers & Industrial Engineering*. Vol. 147, p. 106627. Available from DOI: [10.1016/j.cie.2020.106627](https://doi.org/10.1016/j.cie.2020.106627).
- CHAUHAN, V. K.; BRINTRUP, A., 2022. Real-time large-scale supplier order assignments across two-tiers of a supply chain with penalty and dual-sourcing.
- CHEN, J. et al., 2025. Theoretical Foundations of Utility Accrual for Real-Time Systems.
- CHEN, K.; MÜHLETHALER, P., 1996. A scheduling algorithm for tasks described by Time Value Function. *Real-Time Systems*.
- COHEN, L.; GRINSHPOUN, T.; STERN, R., 2019. Assigning Suppliers to Meet a Deadline. In: *Proceedings of the Twelfth International Symposium on Combinatorial Search (SoCS 2019)*. AAAI Press. Vol. 10, pp. 170–171. No. 1. Available from DOI: [10.1609/socs.v10i1.18475](https://doi.org/10.1609/socs.v10i1.18475).
- CORDOVA, A., 2024. *Green Supply Chain et Quantification des Émissions CO₂ dans DISCO*. Paris. Stage Report. ALTEN, Direction Innovation Numérique. [Internal, restricted access].
- CUMMINGS, M. L., 2004. Automation Bias in Intelligent Time Critical Decision Support Systems. In: *AIAA 1st Intelligent Systems Technical Conference*. Available from DOI: [10.2514/6.2004-6313](https://doi.org/10.2514/6.2004-6313).
- DENG, J.; ZHANG, H.; WEI, S., 2020. Prediction of Vehicle-Cargo Matching Probability Based on Dynamic Bayesian Network. *International Journal of Production Research*. Vol. 59, no. 5, pp. 1419–1436. Available from DOI: [10.1080/00207543.2020.1714094](https://doi.org/10.1080/00207543.2020.1714094).
- DETERDING, S. et al., 2011. From Game Design Elements to Gamefulness: Defining Gamification. In: *Proceedings of the 15th International Academic MindTrek Conference: Envisioning Future Media Environments*. ACM, pp. 9–15. Available from DOI: [10.1145/2181037.2181040](https://doi.org/10.1145/2181037.2181040).
- DIETVORST, B. J.; SIMMONS, J. P.; MASSEY, C., 2015. Algorithm Aversion: People Erroneously Avoid Algorithms after Seeing Them Err. *Journal of Experimental Psychology: General*. Vol. 144, no. 1, pp. 114–126. Available from DOI: [10.1037/xge0000033](https://doi.org/10.1037/xge0000033).
- DING, Y. et al., 2023. Real-Time Monitoring and Optimal Resource Allocation for Automated Container Terminals: A Digital Twin Application at the Yangshan Port. *Journal of Advanced Transportation*.
- DIWASSINGH, K. et al., 2020. Task Selection and Workload: A Focus on Completing Easy Tasks Hurts Long-Run Performance. *Management Science*. Vol. 66, no. 10, pp. 4397–4416. Available from DOI: [10.1287/mnsc.2019.3366](https://doi.org/10.1287/mnsc.2019.3366).
- DYER, J. S., 1990. Remarks on the Analytic Hierarchy Process. *Management Science*. Vol. 36, no. 3, pp. 249–258.
- EUROPEAN COMMISSION, 2017. VECTO — *Vehicle Energy Consumption Calculation Tool* [online]. [visited on 2026-04-24]. Technical Documentation. European Commission, DG Climate Action. Available from: https://climate.ec.europa.eu/eu-action/transport-decarbonisation/road-transport/vehicle-energy-consumption-calculation-tool-vec-to_en.
- FEIGENBAUM, J. A., 2001. A statistical analysis of log-periodic precursors to financial crashes. *Quantitative Finance*. Vol. 1, no. 3, pp. 346–360.

- FULLER, A. et al., 2020. Digital Twin: Enabling Technologies, Challenges and Open Research. *IEEE Access*. Vol. 8, pp. 108952–108971. Available from DOI: [10.1109/ACCESS.2020.2998358](https://doi.org/10.1109/ACCESS.2020.2998358).
- GNEITING, T.; RAFTERY, A. E., 2007. Strictly Proper Scoring Rules, Prediction, and Estimation. *Journal of the American Statistical Association*. Vol. 102, no. 477, pp. 359–378. Available from DOI: [10.1198/016214506000001437](https://doi.org/10.1198/016214506000001437).
- GODDARD, K.; ROUDSARI, A.; WYATT, J. C., 2012. Automation Bias: A Systematic Review of Frequency, Effect Mediators, and Mitigators. *Journal of the American Medical Informatics Association*. Vol. 19, no. 1, pp. 121–127. Available from DOI: [10.1136/amiajnl-2011-000089](https://doi.org/10.1136/amiajnl-2011-000089).
- GRECO, S. et al., 2018. On the Methodological Framework of Composite Indices: A Review of the Issues of Weighting, Aggregation, and Robustness. *Social Indicators Research*. Vol. 141, pp. 61–94. Available from DOI: [10.1007/s11205-017-1832-9](https://doi.org/10.1007/s11205-017-1832-9).
- GRIEVES, M., 2014. *Digital Twin: Manufacturing Excellence Through Virtual Factory Replication*. White Paper. Florida Institute of Technology.
- GUO, C. et al., 2017. On Calibration of Modern Neural Networks. In: *Proceedings of the 34th International Conference on Machine Learning (ICML)*.
- GUO, S.; ZHAO, H., 2017. Fuzzy Best-Worst Multi-Criteria Decision-Making Method and Its Applications. *Knowledge-Based Systems*. Vol. 121, pp. 23–31. Available from DOI: [10.1016/j.knosys.2017.01.010](https://doi.org/10.1016/j.knosys.2017.01.010).
- GUO, W. et al., 2020. A Dynamic Shipment Matching Problem in Hinterland Synchronodal Transportation. *Decision Support Systems*. Vol. 134, p. 113289. Available from DOI: [10.1016/j.dss.2020.113289](https://doi.org/10.1016/j.dss.2020.113289).
- HECKMANN, I.; COMES, T.; NICKEL, S., 2015. A Critical Review on Supply Chain Risk — Definition, Measure and Modeling. *Omega*. Vol. 52, pp. 119–132. Available from DOI: [10.1016/j.omega.2014.10.004](https://doi.org/10.1016/j.omega.2014.10.004).
- HENDERSON, J. V.; STOREYGARD, A.; WEIL, D. N., 2012. Measuring Economic Growth from Outer Space. *American Economic Review*. Vol. 102, no. 2, pp. 994–1028. Available from DOI: [10.1257/aer.102.2.994](https://doi.org/10.1257/aer.102.2.994).
- HLIOUI, R.; GHARBI, A.; HAJJI, A., 2017. Joint Supplier Selection, Production and Replenishment of an Unreliable Manufacturing-Oriented Supply Chain. *International Journal of Production Economics*. Vol. 187, pp. 53–67. Available from DOI: [10.1016/j.ijpe.2017.02.004](https://doi.org/10.1016/j.ijpe.2017.02.004).
- HOEN, K. M. R. et al., 2014. Effect of Carbon Emission Regulations on Transport Mode Selection under Stochastic Demand. *Flexible Services and Manufacturing Journal*. Vol. 26, no. 1–2, pp. 170–195. Available from DOI: [10.1007/s10696-012-9151-6](https://doi.org/10.1007/s10696-012-9151-6).
- HOSSEINI, S.; IVANOV, D., 2020. Bayesian Networks for Supply Chain Risk, Resilience and Ripple Effect Analysis: A Literature Review. *Expert Systems with Applications*. Vol. 161, p. 113649. Available from DOI: [10.1016/j.eswa.2020.113649](https://doi.org/10.1016/j.eswa.2020.113649).
- HOSSEINI, S.; IVANOV, D.; DOLGUI, A., 2020. Ripple Effect Modelling of Supplier Disruption: Integrated Markov Chain and Dynamic Bayesian Network Approach. *International Journal of Production Research*. Vol. 58, no. 11, pp. 3284–3303. Available from DOI: [10.1080/00207543.2019.1661538](https://doi.org/10.1080/00207543.2019.1661538).
- HOSSEINI, S.; TAJIK, N., et al., 2019. Resilient Supplier Selection and Optimal Order Allocation under Disruption Risks. *International Journal of Production Economics*. Vol. 213, pp. 124–137. Available from DOI: [10.1016/j.ijpe.2019.03.018](https://doi.org/10.1016/j.ijpe.2019.03.018).
- HRUŠOVSKÝ, M. et al., 2021. Real-time disruption management approach for intermodal freight transportation. *Journal of Cleaner Production*.

- HUANG, P. et al., 2020. A Bayesian Network Model to Predict the Effects of Interruptions on Train Operations. *Transportation Research Part C: Emerging Technologies*. Vol. 114, pp. 338–358. Available from DOI: [10.1016/j.trc.2020.02.015](https://doi.org/10.1016/j.trc.2020.02.015).
- INTERNATIONAL ORGANIZATION FOR STANDARDIZATION, 2023. *ISO 14083:2023 Greenhouse gases — Quantification and Reporting of Greenhouse Gas Emissions Arising from Transport Chain Operations*. Geneva.
- IVANOV, D.; DOLGUI, A., 2019. New Disruption Risk Management Perspectives in Supply Chains: Digital Twins, the Physical Internet, and Smart Contracts. *IFAC-PapersOnLine*. Vol. 52, no. 13, pp. 337–342. Available from DOI: [10.1016/j.ifacol.2019.11.138](https://doi.org/10.1016/j.ifacol.2019.11.138).
- IVANOV, D.; DOLGUI, A., 2020. A Digital Supply Chain Twin for Managing the Disruption Risks and Resilience in the Era of Industry 4.0. *Production Planning & Control*. Vol. 32, no. 9, pp. 775–788. Available from DOI: [10.1080/09537287.2020.1768450](https://doi.org/10.1080/09537287.2020.1768450).
- IVANOV, D.; DOLGUI, A., et al., 2021. Digital Supply Chain Twins: Managing the Ripple Effect, Resilience, and Disruption Risks by Data-Driven Optimization, Simulation, and Visibility. *International Journal of Production Research*. Vol. 59, no. 3, pp. 957–974. Available from DOI: [10.1080/00207543.2020.1841183](https://doi.org/10.1080/00207543.2020.1841183).
- JEAN, N. et al., 2016. Combining Satellite Imagery and Machine Learning to Predict Poverty. *Science*. Vol. 353, no. 6301, pp. 790–794. Available from DOI: [10.1126/science.aaf7894](https://doi.org/10.1126/science.aaf7894).
- JOHANSEN, A.; SORNETTE, D., 2000. Crashes as Critical Points. *International Journal of Theoretical and Applied Finance*. Vol. 3, no. 2, pp. 219–255. Available from DOI: [10.1142/S0219024900000115](https://doi.org/10.1142/S0219024900000115).
- JOHANSEN, A.; SORNETTE, D., 2001. Finite-time singularity in the dynamics of the world population, economic and financial indices. *Physica A: Statistical Mechanics and its Applications*. Vol. 294, no. 3–4, pp. 465–502. Available from DOI: [10.1016/S0378-4371\(01\)00105-4](https://doi.org/10.1016/S0378-4371(01)00105-4).
- KABOUCHE, I., 2025. *Scoring de Risque et Réseaux Bayésiens pour la Supply Chain Prédictive*. Paris. Bilan Scientifique et Technique. ALTEN, Direction Innovation Numérique. [Internal, restricted access].
- KAGERMANN, H.; WAHLSTER, W.; HELBIG, J., 2013. *Recommendations for Implementing the Strategic Initiative INDUSTRIE 4.0*. Munich. acatech — National Academy of Science and Engineering.
- KAHNEMAN, D., 2011. *Thinking, Fast and Slow*. New York: Farrar, Straus and Giroux. ISBN 978-0374533557.
- KAHNEMAN, D.; TVERSKY, A., 1979. Prospect Theory: An Analysis of Decision under Risk. *Econometrica*. Vol. 47, no. 2, pp. 263–292. Available from DOI: [10.2307/1914185](https://doi.org/10.2307/1914185).
- KARAM, A.; REINAU, K., 2022. A Real-Time Decision Support Approach for Managing Disruptions in Line-Haul Freight Transport Networks. *IEEE Transactions on Intelligent Transportation Systems*.
- KIM, S.; TONG, J.; PEDEN, C., 2020. Admission Control Biases in Hospital Unit Capacity Management: How Occupancy Information Hurdles and Decision Noise Impact Utilization. *Management Science*. Vol. 66, no. 5, pp. 1803–1824. Available from DOI: [10.1287/mnsc.2019.3303](https://doi.org/10.1287/mnsc.2019.3303).
- KIRILLOV, A. et al., 2023. Segment Anything. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. Available from DOI: [10.1109/ICCV51070.2023.00371](https://doi.org/10.1109/ICCV51070.2023.00371).
- KOLLER, D.; FRIEDMAN, N., 2009. *Probabilistic Graphical Models: Principles and Techniques*. Cambridge, MA: MIT Press. ISBN 978-0262013192.
- KULIGOWSKI, T. et al., 2025. Comparative Evaluation of the Manchester Triage System and Emergency Severity Index in Predicting Critical Events in the Emergency Department. *BMC*

- Emergency Medicine*. Vol. 25. Available from DOI: [10.1186/s12873-025-01420-8](https://doi.org/10.1186/s12873-025-01420-8). PMID: 41286624.
- LAFKIHI, M.; PAN, S.; BALLOT, E., 2025. Game: The Freight Transportation Game: Operational Challenges for Carriers in Online Spot Market Platforms. *INFORMS Transactions on Education*. Vol. 26, no. 1, pp. 91–101. Available from DOI: [10.1287/ited.2024.0091](https://doi.org/10.1287/ited.2024.0091).
- LEE, H. L.; PADMANABHAN, V.; WHANG, S., 1997. The Bullwhip Effect in Supply Chains. *MIT Sloan Management Review*. Vol. 38, no. 3, pp. 93–102.
- LEUNG, E.; LEE, C.; OUYANG, Z., 2022. From traditional warehouses to Physical Internet hubs: A digital twin-based inbound synchronization framework for PI-order management. *International Journal of Production Economics*.
- LI, Y.; ZOBEL, C. W., 2020. Exploring Supply Chain Network Resilience in the Presence of the Ripple Effect. *International Journal of Production Economics*. Vol. 228, p. 107664. Available from DOI: [10.1016/j.ijpe.2020.107664](https://doi.org/10.1016/j.ijpe.2020.107664).
- LIANG, F.; BRUNELLI, M.; REZAEI, J., 2020. Consistency Issues in the Best Worst Method: Measurements and Thresholds. *Omega*. Vol. 96, p. 102175. Available from DOI: [10.1016/j.omega.2019.102175](https://doi.org/10.1016/j.omega.2019.102175).
- LIU, M. et al., 2021. A New Robust Dynamic Bayesian Network Approach for Disruption Risk Assessment under the Supply Chain Ripple Effect. *International Journal of Production Research*. Vol. 59, no. 1, pp. 265–285. Available from DOI: [10.1080/00207543.2020.1841318](https://doi.org/10.1080/00207543.2020.1841318).
- LOCKAMY, A.; MCCORMACK, K., 2011. Analysing Risks in Supply Networks to Facilitate Outsourcing Decisions. *International Journal of Production Research*. Vol. 48, no. 2, pp. 593–611. Available from DOI: [10.1080/00207540903175152](https://doi.org/10.1080/00207540903175152).
- LOGG, J. M.; MINSON, J. A.; MOORE, D. A., 2019. Algorithm Appreciation: People Prefer Algorithmic to Human Judgment. *Organizational Behavior and Human Decision Processes*. Vol. 151, pp. 90–103. Available from DOI: [10.1016/j.obhdp.2018.12.005](https://doi.org/10.1016/j.obhdp.2018.12.005).
- MACKWAY-JONES, K.; MARSDEN, J.; WINDLE, J., 2013. *Emergency Triage: Manchester Triage Group*. 3rd ed. Oxford: Wiley-Blackwell. ISBN 978-1118299029.
- MAGGIORI, E. et al., 2017. Can Semantic Labeling Methods Generalize to Any City? The Inria Aerial Image Labeling Benchmark. In: *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*. Available from DOI: [10.1109/IGARSS.2017.8127684](https://doi.org/10.1109/IGARSS.2017.8127684).
- MOHADDESI, O.; POLARIS, V.; HARTEVELD, C., 2020. Introducing Gamettes: A Playful Approach for Capturing Decision-Making for Informing Behavioral Models. In: *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM, pp. 1–14. Available from DOI: [10.1145/3313831.3376527](https://doi.org/10.1145/3313831.3376527).
- MOKHTAR, S. et al., 2019. Supplier Portfolio Selection Based on the Monitoring of Supply Risk Indicators. *Simulation Modelling Practice and Theory*. Vol. 97, p. 101955. Available from DOI: [10.1016/j.simpat.2019.101955](https://doi.org/10.1016/j.simpat.2019.101955).
- MONDELICE, R., 2025. *BST TMS — Transport Management System et Interface Routage DISCO*. Paris. Bilan Scientifique et Technique. ALTEN, Direction Innovation Numérique. [Internal, restricted access].
- MONTREUIL, B., 2011. Toward a Physical Internet: Meeting the Global Logistics Sustainability Grand Challenge. *Logistics Research*. Vol. 3, no. 2–3, pp. 71–87. Available from DOI: [10.1007/s12159-011-0045-x](https://doi.org/10.1007/s12159-011-0045-x).
- MURPHY, A. H., 1973. A New Vector Partition of the Probability Score. *Journal of Applied Meteorology*. Vol. 12, no. 4, pp. 595–600. Available from DOI: [10.1175/1520-0450\(1973\)012<0595:ANVPOT>2.0.CO;2](https://doi.org/10.1175/1520-0450(1973)012<0595:ANVPOT>2.0.CO;2).

- NICULESCU-MIZIL, A.; CARUANA, R., 2005. Predicting Good Probabilities with Supervised Learning. In: *Proceedings of the 22nd International Conference on Machine Learning (ICML)*. Available from DOI: [10.1145/1102351.1102430](https://doi.org/10.1145/1102351.1102430).
- NILOOFAR, P.; LAZAROVA-MOLNAR, S.; SHAKER, H. R., 2023. A General Framework for Human-in-the-Loop Cognitive Digital Twins. In: *Proceedings of the 2023 Winter Simulation Conference (WSC)*. IEEE, pp. 4155–4166. Available from DOI: [10.1109/WSC57122.2023.10407598](https://doi.org/10.1109/WSC57122.2023.10407598).
- PARASURAMAN, R.; RILEY, V., 1997. Humans and Automation: Use, Misuse, Disuse, Abuse. *Human Factors*. Vol. 39, no. 2, pp. 230–253. Available from DOI: [10.1518/001872097778543886](https://doi.org/10.1518/001872097778543886).
- PEARL, J., 1988. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. San Francisco: Morgan Kaufmann. ISBN 978-1558604797.
- PEREZ, H.; WASSICK, J.; GROSSMANN, I., 2022. A digital twin framework for online optimization of supply chain business processes. *Computers & Chemical Engineering*.
- PINEDO, M. L., 2012. *Scheduling: Theory, Algorithms, and Systems*. 4th ed. New York: Springer. Available from DOI: [10.1007/978-1-4614-2361-4](https://doi.org/10.1007/978-1-4614-2361-4).
- PRENSKY, M., 2001. *Digital Game-Based Learning*. New York: McGraw-Hill. ISBN 978-0071454995.
- QAZI, A. et al., 2017. Project Complexity and Risk Management (ProCRiM): Towards Modelling Project Complexity Driven Risk Paths in Construction Projects. *International Journal of Project Management*. Vol. 34, no. 7, pp. 1183–1198. Available from DOI: [10.1016/j.ijproman.2016.05.008](https://doi.org/10.1016/j.ijproman.2016.05.008).
- REZAEI, J., 2015. Best-Worst Multi-Criteria Decision-Making Method. *Omega*. Vol. 53, pp. 49–57. Available from DOI: [10.1016/j.omega.2014.11.009](https://doi.org/10.1016/j.omega.2014.11.009).
- ROSENTHAL, S.; HIATT, L. M., 2020. Human-Centered Decision Support for Agenda Scheduling. *AI & Society*. Available from DOI: [10.1007/s00146-020-00960-y](https://doi.org/10.1007/s00146-020-00960-y).
- RUSOU, Z.; AMAR, M.; AYAL, S., 2020. The Psychology of Task Management: The Smaller Tasks Trap. *Judgment and Decision Making*. Vol. 15, no. 4, pp. 586–599.
- SAATY, T. L., 1980. *The Analytic Hierarchy Process: Planning, Priority Setting, Resource Allocation*. New York: McGraw-Hill. ISBN 978-0070543713.
- SAATY, T. L., 1990. How to Make a Decision: The Analytic Hierarchy Process. *European Journal of Operational Research*. Vol. 48, no. 1, pp. 9–26. Available from DOI: [10.1016/0377-2217\(90\)90057-I](https://doi.org/10.1016/0377-2217(90)90057-I).
- SAWIK, T., 2013. Integrated Selection of Suppliers and Scheduling of Customer Orders in the Presence of Supply Chain Disruption Risks. *International Journal of Production Research*. Vol. 51, no. 23–24, pp. 7006–7022. Available from DOI: [10.1080/00207543.2013.852702](https://doi.org/10.1080/00207543.2013.852702).
- SAWIK, T., 2014. Joint Supplier Selection and Scheduling of Customer Orders under Disruption Risks: Single vs. Dual Sourcing. *Omega*. Vol. 43, pp. 83–95. Available from DOI: [10.1016/j.omega.2013.06.007](https://doi.org/10.1016/j.omega.2013.06.007).
- SAWIK, T., 2016. Integrated Supply, Production and Distribution Scheduling under Disruption Risks. *Omega*. Vol. 62, pp. 131–144. Available from DOI: [10.1016/j.omega.2015.09.005](https://doi.org/10.1016/j.omega.2015.09.005).
- SHEN, X. et al., 2019. Managing Supply Chains with Expediting and Multiple Demand Classes. *Production and Operations Management*. Vol. 28, no. 5, pp. 1129–1148. Available from DOI: [10.1111/poms.12974](https://doi.org/10.1111/poms.12974).
- SKITKA, L. J.; MOSIER, K. L.; BURDICK, M., 1999. Does Automation Bias Decision-Making? *International Journal of Human-Computer Studies*. Vol. 51, no. 5, pp. 991–1006. Available from DOI: [10.1006/ijhc.1999.0252](https://doi.org/10.1006/ijhc.1999.0252).
- SMART FREIGHT CENTRE, 2023. *GLEC Framework for Logistics Emissions Accounting, Version 3.2* [online]. [visited on 2026-04-24]. Technical Report. Smart Freight Centre. Available from: <https://www.smartfreightcentre.org/en/our-programs/emissions->

- [accounting/global-logistics-emissions-council/calculate-report-glec-framework/](#). Aligned with ISO 14083:2023.
- STEEL, P.; KÖNIG, C. J., 2006. Integrating theories of motivation. *Academy of Management Review*. Vol. 31, no. 4, pp. 889–913. Available from DOI: [10.5465/AMR.2006.22527462](#).
- STERMAN, J. D., 1989. Modeling Managerial Behavior: Misperceptions of Feedback in a Dynamic Decision Making Experiment. *Management Science*. Vol. 35, no. 3, pp. 321–339. Available from DOI: [10.1287/mnsc.35.3.321](#).
- SUN, X. et al., 2020. Combating Lead-Time Uncertainty in Global Supply Chain's Shipment-Assignment: Is It Wise to Be Risk-Averse? *Transportation Research Part B: Methodological*. Vol. 138, pp. 406–434. Available from DOI: [10.1016/j.trb.2020.05.015](#).
- TAGHAVI, S. M. et al., 2024. Sustainable and Resilient Supplier Selection, Order Allocation, and Production Scheduling Problem under Disruption Utilizing Conditional Value at Risk. *Journal of Modelling in Management*. Vol. 19, no. 2, pp. 658–692. Available from DOI: [10.1108/JM2-10-2022-0250](#).
- TAO, F. et al., 2018. Digital Twin-Driven Product Design, Manufacturing and Service with Big Data. *The International Journal of Advanced Manufacturing Technology*. Vol. 94, no. 9–12, pp. 3563–3576. Available from DOI: [10.1007/s00170-017-0233-1](#).
- TOMLIN, B., 2006. On the Value of Mitigation and Contingency Strategies for Managing Supply Chain Disruption Risks. *Management Science*. Vol. 52, no. 5, pp. 639–657. Available from DOI: [10.1287/mnsc.1060.0515](#).
- TRANTAPHYLLOU, E.; YANASE, J., 2024. The use of pairwise comparisons for decision making may lead to grossly inaccurate results.
- TRANTAPHYLLOU, E., 2000. *Multi-Criteria Decision Making Methods: A Comparative Study*. Dordrecht: Kluwer Academic Publishers. Available from DOI: [10.1007/978-1-4757-3157-6](#).
- VAN ETEN, A.; LINDENBAUM, D.; BACASTOW, T. M., 2018. SpaceNet: A Remote Sensing Dataset and Challenge Series. *arXiv preprint arXiv:1807.01232*.
- WANG, Y.; WANG, X.; LIU, A., 2020. Digital Twin-Driven Supply Chain Planning. *Procedia CIRP*. Vol. 93, pp. 526–531. Available from DOI: [10.1016/j.procir.2020.04.073](#).
- XIA, G.-S. et al., 2018. DOTA: A Large-Scale Dataset for Object Detection in Aerial Images. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. Available from DOI: [10.1109/CVPR.2018.00418](#).
- YUCESAN, M. et al., 2019. An Integrated Best-Worst and Interval Type-2 Fuzzy TOPSIS Methodology for Green Supplier Selection. *Mathematics*. Vol. 7, no. 2, p. 182. ISSN 2227-7390. Available from DOI: [10.3390/math7020182](#).
- ZHENG, X.; ZHANG, L., 2020a. Risk assessment of supply-chain systems: a probabilistic inference method. *Enterprise Information Systems*. Vol. 14, no. 6, pp. 858–877. ISSN 1751-7583. Available from DOI: [10.1080/17517575.2020.1762004](#).
- ZHENG, X.; ZHANG, L., 2020b. Risk Assessment of Supply-Chain Systems: A Probabilistic Inference Method. *Enterprise Information Systems*. Vol. 14, no. 9–10, pp. 1336–1353. Available from DOI: [10.1080/17517575.2019.1684163](#).
- ZHU, M.; YANG, Y.; HSEE, C. K., 2018. The Mere Urgency Effect. *Journal of Consumer Research*. Vol. 45, no. 3, pp. 673–690. Available from DOI: [10.1093/jcr/ucy008](#).
- ZHU, X. et al., 2025. Towards Industry 5.0: digital twin-enhanced approach for dynamic supply chain rescheduling with real-time order arrival and acceptance. *International Journal of Production Research*.

Acronyms

AHP Analytic Hierarchy Process. 14

BWM Best-Worst Method. 14

CI Consistency Index. 30

CR Consistency Ratio. 14, 30

DIN Direction Innovation Numérique. 1

DT Digital Twin. 13

F-BWM Fuzzy Best-Worst Method. 14

GLEC Global Logistics Emissions Council. 1, 17

HGV Heavy Goods Vehicle. 2

IoT Internet of Things. 1

IPG Itinerary Performance Index. 2

LPPL Log-Periodic Power Law. 10

MCDM Multi-Criteria Decision-Making. 8, 12

MTS Manchester Triage System. 16

PI Physical Internet. 8

RI Random Index (Saaty). 30

SC Supply Chain. 13

TFN Triangular Fuzzy Number. 14

TMT Temporal Motivational Theory. 11

TTR Time To Recovery. 2

TTS Time To Service. 2

VECTO Vehicle Energy Consumption calculation TOol. 17

A Ethics Documentation

The study as executed involves no human participants. The eight declarants of the HÉLIOS campaign are language-model agents, and no personal data was collected, processed or stored at any stage. On that basis the project was submitted to the Cranfield University Research Ethics System as a low-risk study, and the approval documentation is filed with this thesis.

Two features of the protocol would require participant consent if the campaign were rerun with the eight human consultants for whom it was originally designed, and both are recorded here so that the requirement is not lost.

Incomplete disclosure through the cover story. Participants are told they are playing a fictional satellite programme. They are not told that the scenario replays the documented 2020 to 2022 semiconductor shortage, because a participant who recognised the historical chronology could anticipate events rather than react to them, which would destroy the latency measurement that hypothesis H1 depends on. The debrief reveals the real scenario and asks each participant at which turn they recognised it, which converts the principal threat to the measurement into a recorded quantity.

Deliberate misinformation through the placebo sheets. Two participants receive a false de-escalation notice, one at turn 9 and one at turn 15. These sheets control for suggestibility: a declared urgency that follows the placebo is reflecting the sheet rather than a reading of the situation. Participants are not told in advance which sheets are genuine.

Both are standard incomplete-disclosure designs in behavioural operations research, and both are resolved by the full debrief. Were the human campaign to run, the following would need to be in place before the first turn: written informed consent covering the use of incomplete disclosure; the unconditional right to withdraw at any point, with the participant's data removed on request; the debrief protocol, delivered to every participant regardless of whether they completed the campaign; and confirmation that no performance data is passed to line management, since the participants would be colleagues and the campaign records their judgment quality week by week.

Data ethics for the satellite observation work. All imagery and building data come from public French open-data services under their published licences. No imagery of private individuals is collected or retained, and the analysis operates on industrial buildings rather than on people. Vehicle detections are used only as an intermediate geometric cue for locating dock aprons; no vehicle is identified, tracked or associated with an operator, and the detections are not retained beyond the derivation of a facade extent. The regulatory database of classified installations is used solely to calibrate one constant across an aggregate of 122 sites, and never to infer or publish a result about a named site.

B Source Review Grid

Following the structured review method required by the programme's writing guidelines, the grid below records for each stream reviewed in Section 2 its principal sources, the method-

ological approach those sources take, what this thesis takes from them, and a critical appraisal of the limitation that prevents the stream from answering the research question on its own. The final column is the one that matters for the argument: it is the accumulation of those limitations that defines the gap stated in Section 2.14.

Table 26: Source review grid

Stream and role	Principal sources and method	Taken into this thesis	Critical appraisal
Deadline-sensitive value and risk propagation <i>Foundation</i>	(K. Chen et al., 1996; J. Chen et al., 2025): analytical utility-accrual scheduling. (Ivanov; Dolgui, 2020; Ivanov; Dolgui, 2019; Li et al., 2020): conceptual and simulation models of the ripple effect	Urgency treated as time-varying and network-propagated rather than node-local; the noisy-OR choice over weighted averaging	Time-value work stops at a single isolated task. Ripple-effect work models propagation but not the node-level aggregation of heterogeneous blocks that must precede it. Neither addresses a declared signal at all
Cognitive bias in urgency <i>Foundation</i>	(M. Zhu et al., 2018): five controlled experiments. (Steel et al., 2006): formal discounting model. (Rusou et al., 2020; DiwasSingh et al., 2020): field studies	The justification for treating declared urgency as a biased signal; the asymmetric penalty structure; hypothesis H5	Diagnoses the bias but prescribes no operational correction. Effect sizes come from laboratory and clinical settings; the magnitude in industrial supply chains is uncalibrated, which this thesis inherits as an open parameter rather than resolves

Table 26, continued

Stream and role	Principal sources and method	Taken into this thesis	Critical appraisal
Closest antecedents <i>Antecedent</i>	(Sawik, 2013; Sawik, 2014; Sawik, 2016): mixed-integer programming. (Cohen et al., 2019; Afrasiabi et al., 2022): MCDM supplier prioritisation	Confirmation that multi-criteria, disruption-aware supplier evaluation is established, and precise identification of what it omits	Risk is evaluated once, at the point of decision, not maintained per node and propagated. Where a client-declared signal exists it is consumed directly, never validated against an operational counterpart. The optimisation machinery targets allocation, not measurement
Bayesian risk updating <i>Enabling</i>	(Pearl, 1988; Koller et al., 2009): theory. (Hosseini; Ivanov, 2020): survey. (Huang et al., 2020; Deng et al., 2020; Lockamy et al., 2011; Qazi et al., 2017): applied models	Beta-Bernoulli conjugate revision of the failure probability in the risk block, at $n_0 = 26$	Suited to the sparse-evidence regime, but says nothing about whether the resulting probabilities are well calibrated once a human reads them. This thesis measured that gap directly and found it large
Digital twins and decision support <i>Enabling, primary gap source</i>	(Grieves, 2014; Tao et al., 2018; Fuller et al., 2020): concept and survey. (Ivanov; Dolgui, et al., 2021; Ivanov; Dolgui, 2020; Niloofar et al., 2023): supply chain twins and cognitive twins	The architectural family the delivered system belongs to; the human-in-the-loop framing	No retrieved system combines a declared input, a computed operational urgency and their propagated mismatch. A second gap matters here: these systems are evaluated on state estimation accuracy and assume their inputs simply arrive, so input circularity is not treated as a measurement problem

Table 26, continued

Stream and role	Principal sources and method	Taken into this thesis	Critical appraisal
Multi-criteria decision methods <i>Enabling</i>	(Saaty, 1980): eigenvector method. (Rezaei, 2015; S. Guo et al., 2017): best-worst and fuzzy extensions. (Brans; Vincke, et al., 1986; Brans; Mareschal, 2005): outranking. (Greco et al., 2018; Becker et al., 2017; Borgonovo et al., 2021): composite-index critique	The elicitation of U_d with its consistency gate; the Fuzzy BWM weighting of the six blocks; the PROMETHEE II network ranking	Static tools, rarely compared against a computed counterpart. The composite-index critique is usually offered as a caveat rather than instrumented; this thesis instruments it through the explainability decomposition, and the campaign then produced exactly the predicted failure, two perfectly correlated criteria
Forecast-assisted decision and automation bias <i>Foundation for the treatment arm</i>	(Parasuraman et al., 1997): taxonomy. (Skitka et al., 1999; Cummings, 2004): controlled studies. (Goddard et al., 2012): systematic review. (Dietvorst et al., 2015; Logg et al., 2019; Bansal et al., 2021): direction and explanation effects	The interpretive frame for the arm B result; the design requirement that an advisory layer communicate its own ignorance	Dominated by single-decision laboratory and clinical tasks. Does not cover a repeated multi-turn loop in which the declaration feeds back into the model, nor the case where the displayed forecast is known to be miscalibrated in a specific direction, which is the situation measured here

Table 26, continued

Stream and role	Principal sources and method	Taken into this thesis	Critical appraisal
Probabilistic forecast verification <i>Enabling</i>	(Brier, 1950): score. (Murphy, 1973): decomposition. (Gneiting et al., 2007): proper scoring rules. (Niculescu-Mizil et al., 2005; C. Guo et al., 2017): post-hoc calibration	The scoring protocol of Section 3.4.4; the separation of discrimination from calibration that carries the central result	Assumes the predicted outcome is unambiguously defined. In an operational supply chain it is a modelling choice, and Section 4.7 measures that choice moving the reported skill by a factor of five
Serious games <i>Validation rationale</i>	(Stermann, 1989): the Beer Game. (Prensky, 2001; Deterding et al., 2011): design theory. (Mohaddesi et al., 2020; Bellotti et al., 2013; Lafkihi et al., 2025): methodology and applications	Justification of a weekly-cadence game as an empirical instrument for observing declaration bias	Usually scored on learning outcomes or aggregate performance, rarely used as a controlled comparison replaying one frozen scenario with and without a treatment on the declarant, which is the design adopted here
Medical triage <i>Analogy only</i>	(Mackway-Jones et al., 2013): protocol. (Kuligowski et al., 2025): comparative study, weighted $\kappa = 0.51$. (Kim et al., 2020): admission control	Vocabulary of over- and under-triage; the principle of asymmetric risk behind $\lambda_{\text{under}} > \lambda_{\text{over}}$	Analogy only, and flagged as such wherever used. Clinical triage has an outcome observable within hours and a trained, certified assessor; neither holds in a supply chain, so no numeric parameter is transferred from this stream

Table 26, continued

Stream and role	Principal sources and method	Taken into this thesis	Critical appraisal
Physical Internet and carbon accounting <i>Context</i>	(Montreuil, 2011; W. Guo et al., 2020; Li et al., 2020): network context. (Smart Freight Centre, 2023; European Commission, 2017; ADEME, 2014; International Organization for Standardization, 2023): emissions frameworks	The motivation for treating urgency as a network-level quality of service variable; the carbon block as a separate arbitration criterion	Provides context and regulatory grounding, not method. The carbon block was the least exercised of the six on the campaign, so its inclusion remains justified by design intent rather than by measured contribution
Earth observation of built assets <i>Foundation for the opening</i>	(Xia et al., 2018): oriented detection benchmark. (Maggiori et al., 2017; Van Etten et al., 2018): footprint benchmarks. (Kirillov et al., 2023): promptable segmentation. (Henderson et al., 2012; Jean et al., 2016): activity proxies from imagery	The pre-trained detector and its vehicle classes; the sampling of candidate sites from imagery alone; the precedent that imagery can substitute for slow official statistics	The economics work measures aggregate activity at regional scale and produces nothing per-facility. The detection work stops at footprints and vehicles without converting them into an operational capacity. The conversion, and its calibration, is what Section 4.11 had to construct

C HÉLIOS Campaign Protocol Extract

This appendix records the elements of the frozen protocol that a reader would need in order to reproduce or audit the campaign, beyond the design already given in Section 3.4.

Cadence and Sequence

Nineteen turns are indexed T0 to T18. T0 is a training turn used to verify that every declarant can complete the questionnaire inside ten minutes with the consistency gate active; its declarations are retained in the record but excluded from every analysis. The eighteen campaign turns run at two per day, morning and late afternoon, over nine working days. One game turn represents one real month, from September 2020 to February 2022, with real durations converted into engine hours at a fixed rate so that documented figures such as twelve to twenty-five week supplier lead times or a six-day canal blockage retain their true magnitude inside an eighteen-turn game.

Frozen Artifacts

Four artifacts were frozen before T0 and are the reference against which any later analysis is checked: the scenario definition with its twelve calibrated events and their turn assignments; the data pack of public time series, with a SHA-256 hash recorded per file; the role cards, one per node, each stating only what that declarant may know; and the hypothesis register H1 to H6 with its numeric thresholds. Any analysis not derivable from the register is labelled exploratory in the reporting.

Controls

- *Isolation.* Each declarant sees only its own role sheet, its own turn briefing and its own questionnaire. Dashboards, scores, the network view and other declarants' material stay hidden until the debrief. In the executed campaign this is enforced by the harness, which gives each agent a private working directory, rather than by instruction.
- *Placebo sheets.* A false de-escalation notice is handed to the avionics integrator at T9 and to the wafer supplier at T15. A declared urgency that follows the placebo is reflecting the sheet rather than a reading of the situation.
- *Weak-signal turn.* At T4 the press briefing mentions a tension with no engine event behind it, which measures who reacts to media noise alone. This is also the turn on which the arm B treatment first becomes visible, so the two effects are deliberately not separated on that turn and the arm A control is what isolates them.
- *Role assignment.* The two most information-heavy nodes, the foundry and the distributor, are assigned away from the most supply-chain-experienced participants, so that a role effect cannot be confounded with an expertise effect.
- *Journal.* Every deviation from protocol is logged the same day.

Contingencies Arbitrated in Advance

Three contingencies were decided before the campaign rather than improvised under time pressure. The campaign truncates at T16 if it overruns by more than two turns, rather than rushing the remaining responses. It drops to one turn per day if the questionnaire exceeds ten minutes at the training turn. Any absence, or any unresolved consistency failure, is carried forward unchanged and excluded from the reactivity analyses.

Declaration Record Format

Each declaration is one JSON record per node per turn, carrying the turn index, the six bipolar pairwise comparisons, the four severity scores, the derived criteria weights, the principal eigenvalue with its consistency index and consistency ratio, a flag for whether the consistency gate passed, the number of attempts, the raw and smoothed declared urgency, and a free-text note. Arm B adds one field, `influence_prediction`, taking one of four values: no influence, confirms, revised upward, or revised downward. The format is otherwise identical between arms, which is what makes them comparable.

Deviations Recorded in the Executed Campaign

Two are on record. One node submitted turn 1 twice; the first submission is retained and the duplicate discarded, which is why the arm B analysis reports 96 declarations from 97 raw records. Arm B ran to T11 rather than T18, so its forecast log covers turns 4 to 12 against turns 4 to 18 in arm A; all arm B comparisons in this thesis are therefore made over the turns actually recorded, and the arm A scoring uses its own full range.

D Full Scoring Tables

Every value in this appendix is written by the derivation script of Section 3.6 and is reproduced here in full, so that the summary tables of Section 4 can be checked against the complete output.

Turn to ISO Week Mapping

The campaign runs from ISO week 2026-S31 at T0 to 2026-S49 at T18, one turn per week. The outcome window of a forecast issued at turn t with horizon h therefore spans weeks $t + 1$ to $t + h$ inclusive.

Arm A, Production Outcome Definition

Table 27 gives every quantity behind the summary of Section 4.6, at all four horizons, including the confusion matrix at the pre-registered threshold and the climatological Brier score the skill is measured against.

Arm A, Naive Outcome Definition

The naive definition counts any non-reverted event recorded on the node, regardless of gravity, and ignores milestones entirely. It is scorable on the 75 node-turns for which the node has at least one event in its history. Table 28 repeats the scoring under that definition, and the contrast with Table 27 is the measurement reported in Section 4.7.

Right-Censoring Sensitivity

Dropping every forecast whose observation window would extend past the final campaign turn leaves the four-week scoring on 88 observations with 2 positives. The result does not

Table 27: Complete arm A scoring, production outcome definition

Quantity	h = 1	h = 2	h = 3	h = 4
Observations	120	120	120	120
Positive outcomes	2	4	5	6
Base rate	0.017	0.033	0.042	0.050
AUC	0.555	0.372	0.428	0.485
AUC 95% CI, lower	0.470	0.258	0.315	0.389
AUC 95% CI, upper	0.639	0.492	0.546	0.583
PR-AUC	0.027	0.031	0.041	0.054
Brier score	0.130	0.166	0.191	0.207
Brier, climatology	0.016	0.032	0.040	0.047
Brier skill	−6.93	−4.14	−3.77	−3.36
True positives at 0.5	0	0	0	0
False positives at 0.5	15	18	20	21
False negatives at 0.5	2	4	5	6
True negatives at 0.5	103	98	95	93

Table 28: Complete arm A scoring, naive outcome definition

Quantity	h = 1	h = 2	h = 3	h = 4
Observations	75	75	75	75
Positive outcomes	11	19	25	29
Base rate	0.147	0.253	0.333	0.387
AUC	0.603	0.601	0.545	0.484
Brier score	0.155	0.248	0.337	0.389
Brier skill	−0.24	−0.31	−0.52	−0.64

improve: the operating point remains 0 true positives against 16 false positives, and the Brier skill moves from −3.36 to −7.82. Censoring is not the explanation for the confident tail.

The Confident Tail, Forecast by Forecast

Table 29 lists every arm A forecast announcing a four-week probability of at least 0.5. The rightmost column is the milestone-miss probability that the model reports alongside it. The two never differ by more than 0.020, which is the evidence that the confident tail is the milestone channel and nothing else. None of the twenty-one was followed by the predicted outcome under the production definition, which is what the Outcome column reports; five of them were under the original-commitment definition of Table 31.

Set against the registry, ten of the eighteen campaign milestones completed, eight remained active, and none was abandoned. Exactly one was still open past a deadline that had fallen inside the campaign. Every other node for which a near-certain miss was announced delivered, though five of those deliveries arrived after the deadline originally committed to, which the registry does not record because a replanning overwrites the deadline it stores. Table 31 gives that reading milestone by milestone.

Table 29: Every arm A forecast at or above 0.5 at four weeks

Turn	Node	p (4 weeks)	p (milestone miss)	Outcome
T5	AvioSys Intégration	0.898	0.880	none
T6	CompoDis Europe	0.996	0.996	none
T7	CompoDis Europe	1.000	1.000	none
T8	CompoDis Europe	1.000	1.000	none
T8	NovaFab Semiconductors	0.866	0.846	none
T8	Orbitalys	1.000	1.000	none
T9	NovaFab Semiconductors	1.000	1.000	none
T9	Orbitalys	1.000	1.000	none
T10	AvioSys Intégration	0.900	0.884	none
T10	NovaFab Semiconductors	1.000	1.000	none
T10	Orbitalys	1.000	1.000	none
T11	AvioSys Intégration	0.898	0.884	none
T11	Orbitalys	1.000	1.000	none
T11	TransGlobal Fret	0.854	0.838	none
T12	Orbitalys	1.000	1.000	none
T13	Électis EMS	1.000	1.000	none
T15	AvioSys Intégration	0.896	0.884	none
T17	AvioSys Intégration	0.894	0.884	none
T17	CompoDis Europe	0.998	0.998	none
T18	CompoDis Europe	1.000	1.000	none
T18	Meridian Semi	0.704	0.690	none

Arm B, Per-Node Influence Record

Table 30 records the first-exposure turn node by node, which is the observation behind the result of Section 4.8.

Table 30: Arm B at turn 4, node by node

Node	Recorded influence of the displayed forecast
AvioSys Intégration	Confirms existing assessment
CompoDis Europe	Revised downward
Électis EMS	Confirms existing assessment
Meridian Semi	Confirms existing assessment
NovaFab Semiconductors	Revised downward
Orbitalys	Revised downward
SilPure Materials	Revised downward
TransGlobal Fret	Revised downward
Total	5 downward, 3 confirmed, 0 upward, 8 of 8 responding

NovaFab is the node that carries the scenario's critical accident event two turns later, at turn 6.

Milestones Against Their Original Commitment

Table 31 is the counterpart of the registry counts above. It lists every milestone whose *original* deadline falls inside the campaign, together with the turn at which it was actually delivered in arm A and in the closed-loop arm of Section 4.9. Seven further milestones carry original deadlines at turns 20 to 22 and are therefore unobservable within a nineteen-turn campaign; they are excluded here rather than counted as met.

Table 31: The eleven milestones observable within the campaign, judged against the deadline first committed to

Milestone	Original deadline	Arm A delivery	Closed-loop deadline	Closed-loop verdict
Orbitalys, HÉLIOS-1 delivery	T8	T14	T14	Missed
Orbitalys, HÉLIOS-2 delivery	T16	not delivered	T19	Missed
AvioSys, avionics batch 1	T6	T6	T6	Met
AvioSys, avionics batch 2	T11	T12	T12	Missed
AvioSys, avionics batch 3	T16	T18	T18	Missed
Électis, HÉLIOS series A	T6	T6	T6	Met
Électis, HÉLIOS series B	T13	T14	T13	Met
CompoDis, component cover S1	T9	T9	T11	Missed
TransGlobal, annual flow contract	T12	T12	T12	Met
NovaFab, HÉLIOS wafer allocation	T10	T11	T12	Missed
SilPure, annual wafer contract	T15	T14	T15	Met
Met on original commitment		5 of 11		5 of 11

The two rows set in bold are the only ones on which the arms differ. The closed loop saved the Électis series B, which slipped in arm A, and replanned the CompoDis S1 cover, which arm A met on its original date. Every other difference between the arms is a replanning that re-dated a milestone arm A also delivered late: an announced slip rather than a discovered one, which is not nothing and is not a save.

Under this definition of truth, five of the twenty-one confident arm A forecasts of Table 29 are followed by a missed original commitment within four weeks, at T8 and T9 on NovaFab, T10 and T15 on AvioSys, and T12 on Orbitalys, against none under the registry's current-deadline reading.

Arm A Rescored After Repair

Tables 32 to 34 give the complete scoring behind Table 20. All three score the same 120 forecasts against the original-commitment target, censored points excluded, so the observation counts are identical across the three and only the model differs. The first is the frozen campaign as run, the second applies the structural repair of Equation 14 alone, and the third adds the progress-uncertainty term.

The confidence intervals are cluster bootstraps resampling whole nodes rather than individual node-weeks, because forecasts on one node are strongly correlated and an i.i.d. bootstrap

Table 32: Arm A before repair, original-commitment target

Quantity	h = 1	h = 2	h = 3	h = 4
Observations	112	104	96	90
Positive outcomes	7	14	20	26
Base rate	0.062	0.135	0.208	0.289
Brier score	0.1335	0.1969	0.2728	0.3190
Brier skill	−1.278	−0.690	−0.654	−0.553
AUC	0.736	0.636	0.641	0.654
AUC 95% CI	[0.49, 0.93]	[0.37, 0.87]	[0.39, 0.85]	[0.42, 0.85]
PR-AUC	0.163	0.205	0.266	0.353

Table 33: Arm A after the structural repair alone

Quantity	h = 1	h = 2	h = 3	h = 4
Brier score	0.0965	0.1633	0.2243	0.2613
Brier skill	−0.646	−0.401	−0.360	−0.272
AUC	0.793	0.671	0.657	0.721
AUC 95% CI	[0.60, 0.95]	[0.51, 0.88]	[0.47, 0.87]	[0.54, 0.92]
PR-AUC	0.355	0.305	0.355	0.484

Table 34: Arm A after the structural repair and the progress-uncertainty term

Quantity	h = 1	h = 2	h = 3	h = 4
Brier score	0.0988	0.1593	0.2120	0.2499
Brier skill	−0.687	−0.367	−0.286	−0.216
AUC	0.729	0.634	0.602	0.584
AUC 95% CI	[0.56, 0.95]	[0.42, 0.83]	[0.29, 0.85]	[0.21, 0.87]
PR-AUC	0.195	0.228	0.299	0.385

would understate the width. They are wide enough at every horizon to contain the AUC of every other table in this group, which is why the AUC movements between the three are reported as uninterpretable rather than as gains or losses, while the Brier skill movements, which do not depend on the size of the positive class in the same way, are reported as measured.

Distribution of the Milestone Probability

Table 35 counts how many of the 120 milestone-miss probabilities fall at the extremes rather than in a usable band, which is the measurement behind the progress-uncertainty term of Section 4.9.

The structural repair alone reduces the count of extreme values but not the usable band: it removes eleven false certainties without producing gradations. The progress-uncertainty term is what widens the usable band, from 10 forecasts to 17. Neither is sufficient, and 71.7% of the probabilities remain degenerate.

Table 35: Degeneracy of the milestone-miss probability, 120 forecasts

Model	Exactly 0 or 1	In (0.05, 0.95)	Forecasts ≥ 0.5 at 4 weeks
Frozen campaign as run	102 (85.0%)	14	21
Structural repair alone	89 (74.2%)	10	10
Structural repair with progress uncertainty	86 (71.7%)	17	10

E Satellite Capacity Estimation: Protocol and Data

This appendix records the operational detail behind the opening of Section 4.11, so that its measurements can be audited or repeated.

Data Sources

All sources are public French open-data services, accessed over standard web map and feature services and cached on disk so that a re-run does not depend on service availability. Table 36 lists each one with the role it plays in the pipeline.

Table 36: Data sources of the capacity pipeline

Source	Use
Orthophotography, 20 cm	Primary imagery for detection and segmentation
Orthophotography, 5 to 10 cm	Partial coverage, used only for the resolution control of Figure 22
LiDAR surface model	Building height where available, delivered as 32-bit float raster
National topographic database	Building footprints and, where populated, height
Classified installations register	Declared storage volume, used only to calibrate the fill factor

All geometry is handled in the national metric projection, in which the ground sample distance is exactly the bounding-box width in metres divided by the image width in pixels rather than an approximation.

Calibration of the Fill Factor

The fill factor relates declared storage volume to the geometric envelope. For each site in the regulatory register, the nearest building footprint is matched, the geometric volume is computed as footprint area times eaves height, and the ratio of declared to geometric volume is recorded. Ratios outside $[0.05, 20]$ are discarded as either an aberrant declaration or a mismatched building. The production constant of 0.80 is attributed in the source to a median over 122 matched sites with an interquartile range of $[0.52, 1.22]$.

That run cannot be reproduced from the retained artifacts, and the discrepancy is recorded here in full. The calibration script writes its detailed output to a file that is not present in the repository, so the only auditable sample is the 68-site corpus retained for other purposes.

Recomputing the identical ratio over those 68 sites gives a median of 0.966, an interquartile range of [0.80, 2.00], and 47% of sites inside the [0.7, 1.4] band, against the 70% the calibration script itself prints as its acceptance target. The available evidence does not distinguish between a biased remnant sample and an understated production constant. Recollecting the register cleanly, with the departmental loop and pagination the script provides, and rederiving the factor, is the resolution.

One matching rule is worth recording because it changes the answer materially. Only the largest building within the search radius is matched, and neighbouring buildings are not summed: summing them moves the median ratio from 0.80 to 0.58, because a declared volume belongs to one warehouse while the footprints of a whole industrial estate do not.

Annotation Protocol

The dataset comprises 442 georeferenced image chips, each retained at a single location, a single resolution and a single vintage, with a sidecar file recording its geographic transform. Of these, 351 form the training split and 91 the validation split.

Annotators mark one class, the continuous extent of facade served by a dock apron, and the following rules apply.

- Mark an equipped facade even where no vehicle is parked. The apron and canopy remain visible, and a facade annotated only when a vehicle is present teaches the model to detect vehicles.
- Do not annotate a trailer holding yard that is not adjacent to a wall.
- Do not annotate an aeronautical tarmac or a small-business forecourt. Both are paved and deep but serve no docks.
- A facade without docks is annotated empty. That is a useful negative, not an omission, and roughly 40% of the target sample is negative by design.
- Ignore everything above the facade line, which is roof. Rooflights and sawtooth roofs are perfectly periodic and are the principal source of false positives.
- Annotate every visible dock zone in a chip regardless of which building owns it. Ownership is resolved downstream by geometric clipping, so the model is never asked to infer it from pixels, and annotating a zone in one chip but not in the overlapping neighbour would give the model two contradictory labels for the same ground.

Sampling is stratified by footprint area across three bands, roughly 1 500 to 5 000 m², 5 000 to 20 000 m² and above 20 000 m², and across five site categories covering small enterprises, aerospace, refrigerated, very large and retail distribution. An unstratified random draw over-represents large warehouses, and the model inherits that bias.

Training Configuration

The model is a segmentation network trained from pre-trained weights at a 640 pixel input size, with rotation augmentation over the full circle, horizontal and vertical flips, and scale jitter. Rotation invariance is obtained by augmentation rather than by geometric rectification of the facade, which is what allows the model to accept any building shape and orientation without requiring a footprint as input. Training ran on CPU throughout, with no GPU available, which is the practical reason the input size was not raised further and the context for the

negative resolution result of Table 23.

Measured Facts Recorded for Reuse

- Empty bay contrast at verified locations: 0.18σ , 0.22σ and 0.01σ , against 5.3σ to 6.4σ at occupied bays on the same facades, over a full sweep of assumed apron depths from 0.3 m to 20 m.
- Dock pitch coherence between facades of the same building: 0.03 to 0.05 m, which provides a free quality control.
- Roof overhang beyond the footprint edge: 0 to 14.4 m. This is why the footprint edge is not the start of the apron, and it was the cause of an earlier defect in which rooflights were counted as bays.
- Trailer distance to building: median up to 126 m on large platforms; at one site only 8 of 89 detected trailers lay within 26 m. Trailers are not a proxy for docks.
- Zero-shot coverage before the learned model: 25 of 70 sites, or 36%.
- Prompt sensitivity of the promptable segmentation model used for site sampling: a prompt naming an industrial warehouse returns 88 752 m², some 45% of the view, where a prompt naming a warehouse roof returns the 13 478 m² of the roof itself. The first is useful for locating a site, the second for measuring it, and they are not interchangeable.

Licensing Constraint

The detection and segmentation library used is distributed under a strong copyleft licence. This has no consequence for internal research use, but it becomes determinative the moment the capability is distributed or exposed over a network. The options are a commercial licence from the vendor or replacement of the backbone with a permissively licensed equivalent. This is recorded as a governance decision to be taken deliberately rather than discovered at deployment.